Physics Grade 12 - Solved Exercises (BTBB)

Balochistan Textbook Board (BTBB)

Unit 11: Electrostatics

Exercise - Multiple Choice Questions (MCQs)

Q1. The electrostatic force between two point charges is independent of one of the following quantities:

- Magnitude of charges
- ✓ Temperature of the charges
- Distance between charges
- Medium between charges

Q2. If the distance between two equal charges is reduced to half and the magnitude of charges is also decreased to half, then the force between them will be:

- ✓ Remain same
- Decreased to half
- Increased to double
- Becomes four times

Q3. The concept of electric field was introduced by:

- Coulomb
- ✓ Faraday
- Gauss
- Ampere

Q4. The number of electrons or protons which constitutes a charge of one coulomb is:

6.25 x 10^-18

✓ 6.25 x 10^18

1.6 x 10^-19

1.6 x 10^19

Q5. For static electricity the value of relative permittivity is always:

Zero

Less than one

Equal to one

✓ Greater than one

Q6. A metallic charged sphere is placed in uniform electric field E, the electric field inside the sphere will be:

E

Less than E

Greater than E

✓ Zero

Q7. The electric intensity E at a point in a field due to a dipole depends upon distance 'r' the relationship between them is:

E proportional to r

E proportional to 1/r

E proportional to 1/r^2

✓ E proportional to 1/r^3

Q8. The dipole moment is defined as the product of:

Charge and distance

✓ Charge and displacement

Charge and force

Charge and electric field

Q9. Debye is the unit of:

Electric field

Electric charge

Lines of force

✓ Dipole moment

Q10. The electric flux is maximum when angle 'theta' between area vector and electric intensity is:

✓ 0 degree

45 degree

60 degree

90 degree

Q11. Electric flux is independent of one of the following quantities:

Charge

Medium

Field

✓ Distance

Q12. The dimension of an electric potential is same as that of:

Work

✓ Work per unit charge

Electric field per unit charge

Electric force per unit charge

Q13. Electron volt is the unit of:

✓ Energy

Charge

Current

Electric Potential

Q14. The energy of an electron which accelerates through a potential difference of 1000V is:

1.6 x 10^-22 J

1.6 x 10^-20 J

1.6 x 10^-19 J

✓ 1.6 x 10^-16 J

Q15. The unit of surface charge density is:

cm

C m^-1

C m^2

✓ C m^-2

Q16. Capacitance of a capacitor does not depend upon:

Area of plate

Distance between plates

Medium between plates

✓ Material of the plates

Q17. In the presence of dielectric material, the electric field between the plates of the capacitor will be:

Zero

Remain same

✓ Decreased

Increased

Q18. When the potential difference across the capacitor is decreased by dielectric then the capacitance of the capacitor will be:

Zero

Remain same

Decreased

✓ Increased

Q19. The unit of product of resistance and capacitance is equal to unit of:

✓ Time

Work

Potential difference

Current

Q20. A capacitor is approximately full charged after:

Two time constant

Three time constant

Four time constant

✓ Five time constant

Exercise - Short Questions

Q1. What is the effect of the medium on the electrostatic force between two point charges?

Ans. A medium other than free space reduces the electrostatic force between two point charges. In free space the force is F = (1/4 pi epsilon_o) x (q1 q2 / r^2), while in a medium of relative permittivity epsilon_r it becomes F = (1/4 pi epsilon_o epsilon_r) x (q1 q2 / r^2). Since epsilon_r is greater than one for a material medium, the force in the medium is smaller than in free space by the factor epsilon_r.

Q2. What do you know about dielectric constant or relative permittivity?

Ans. The dielectric constant or relative permittivity epsilon_r of a medium is the ratio of the permittivity epsilon of that medium to the permittivity of free space epsilon_o, that is, epsilon_r = epsilon / epsilon_o. It is a pure number with no unit. It tells how many times the electrostatic force between two charges in free space is reduced when the medium is placed between them; for free space epsilon_r = 1 and for all material media it is greater than one.

Q3. How does the electrostatic force exert on one charge by another charge?

Ans. The electrostatic force acts through the electric field: a charge sets up an electric field in the space around it, and a second charge placed in that field experiences a force F = qE. Like charges repel each other and unlike charges attract each other, and the force acts along the line joining the two charges, obeying Coulomb's law in magnitude.

Q4. Why the two lines of force do not cross each other?

Ans. Two lines of force can never cross each other because the tangent to a line of force at any point gives the direction of the electric field at that point. If two lines crossed, there would be two different tangents, and hence two different directions of the electric field, at the same point - which is impossible, since the electric field at a point has only one direction.

Q5. How can you determine the direction of electric intensity?

Ans. The direction of electric intensity at a point is the direction of the force that would act on a small positive test charge placed at that point. It is also the direction of the tangent to the line of force passing through that point: away from a positive charge and towards a negative charge.

Q6. Differentiate between point charge and test charge.

Ans. A point charge is a charged body whose size is negligible compared with the distances involved, so that it can be treated as if all its charge were concentrated at a point. A test charge is a very small positive charge used to explore an electric field; it is assumed so small that its own field does not disturb the field being measured. A point charge may be large, but a test charge must always be negligibly small.

Q7. What do you know about the dipole axis and its direction?

Ans. An electric dipole consists of two equal and opposite charges separated by a small distance. The dipole axis is the straight line passing through both charges. Its direction is taken from the negative charge towards the positive charge, and the dipole

moment vector p = qd points along this axis.

Q8. Is dipole moment vector or scalar? If vector, then where is its direction in an electric field?

Ans. Dipole moment is a vector quantity, p = qd, where d is the displacement from the negative charge to the positive charge. In an electric field a dipole experiences a torque tau = p x E which tends to align the dipole moment along the direction of the field, that is, the dipole turns until p points in the direction of E.

Q9. When the electric flux will be minimum and maximum?

Ans. Electric flux through a surface is phi = E.A cos theta, where theta is the angle between the electric field E and the area vector A. The flux is maximum when theta = 0 degree, that is, when the field is parallel to the area vector (perpendicular to the surface), giving phi = EA. It is minimum (zero) when theta = 90 degree, that is, when the field is parallel to the surface, giving phi = 0.

Q10. What do you know about the Gaussian Surface?

Ans. A Gaussian surface is an imaginary closed surface chosen to apply Gauss's law. It is selected so that the electric field is either constant in magnitude and parallel to the area vector over the surface, or perpendicular to it, which makes the flux easy to calculate. For a point charge or a spherical charge distribution a sphere is chosen; for a line charge a cylinder; and for a sheet of charge a pill-box (flat cylinder).

Q11. What is the electric field inside the hollow charged spherical

shell?

Ans. The electric field inside a hollow charged spherical shell is zero at every point. By Gauss's law, a Gaussian sphere drawn inside the shell encloses no charge, so the total flux through it is zero, and by symmetry the field must be zero everywhere inside. All the charge resides on the outer surface of the shell.

Q12. What do you know about the electric potential gradient?

Ans. The electric potential gradient is the rate of change of electric potential with distance, dV/dr. The electric field at a point is equal to the negative of the potential gradient at that point, E = -dV/dr. The negative sign shows that the field points in the direction in which the potential decreases most rapidly, and for a uniform field E = V/d.

Q13. What is the relation between work and electrical potential energy?

Ans. The work done by an external agent in moving a charge q through a potential difference V is stored as a change in the electrical potential energy of the charge, W = qV = change in potential energy. If the charge moves from infinity (zero potential) to a point at potential V, the work done is W = qV, which is the potential energy of the charge at that point.

Q14. Define one electron volt and give its numerical value.

Ans. One electron volt is the amount of energy gained or lost by an electron when it is moved across an electric potential difference of one volt. Its numerical value is 1 eV = 1.6 x 10^-19 J, since W = qV

= (1.6 x 10^-19 C)(1 V) = 1.6 x 10^-19 J.

Q15. What is the absolute electric potential?

Ans. The absolute electric potential at a point is the work done per unit positive charge in bringing a small positive test charge from infinity to that point against the electric field, V = W/q. It is called absolute because the reference point, infinity, is taken as zero potential. For a point charge q at distance r, the absolute potential is V = (1/4 pi epsilon_o) x (q/r).

Q16. What is the capacitance of a capacitor?

Ans. The capacitance of a capacitor is the ratio of the charge Q stored on either plate to the potential difference V between the plates, C = Q/V. It measures the ability of the capacitor to store charge per unit potential difference. Its SI unit is the farad (F); one farad is the capacitance of a capacitor which stores one coulomb of charge when the potential difference across it is one volt.

Q17. What is the effect of dielectric on the capacitance of a capacitor?

Ans. Inserting a dielectric between the plates of a capacitor increases its capacitance. The dielectric reduces the electric field between the plates to E = E_o/epsilon_r, so the potential difference falls to V = V_o/epsilon_r for the same charge, and since C = Q/V, the capacitance becomes C = epsilon_r C_o, that is, it increases by the factor epsilon_r, the dielectric constant.

Q18. What is the polarization of dielectric?

Ans. Polarization is the process in which the molecules of the dielectric material placed between the plates of a capacitor form dipoles: in a polar dielectric the permanent dipoles align with the field, and in a non-polar dielectric the positive and negative charge centres of each molecule are displaced. These induced dipoles set up an internal field opposing the applied field, which is why the net field between the plates is reduced.

Q19. How does dipole moment produce in dielectric material?

Ans. A dipole moment is produced in a dielectric material when the applied electric field displaces the centres of positive and negative charge within each molecule. In non-polar molecules the electron cloud is shifted relative to the nucleus, creating an induced dipole moment in the direction of the field; in polar molecules the already existing permanent dipoles rotate and align with the field. The net effect is a surface charge on the dielectric faces, called polarization charge.

Q20. How does the electric field between the plate of the capacitor reduce by dielectric?

Ans. When a dielectric is placed between the plates, its molecules become polarized and polarization charges appear on its surfaces: negative charge near the positive plate and positive charge near the negative plate. These polarization charges set up an internal electric field opposite to the original field E_o, so the net field between the plates is reduced to E = E_o/epsilon_r, where epsilon_r is the dielectric constant of the material.

Q21. What is RC-circuit and its function?

Ans. An RC-circuit is a circuit containing a resistor R and a capacitor C in series with a source of e.m.f. Its function is to control the rate of charging and discharging of the capacitor: the resistor limits the current, so the charge on the capacitor grows gradually to its maximum value (charging) or falls gradually to zero (discharging), following an exponential curve. RC-circuits are used in timing circuits, filters and camera flashes.

Q22. What is meant by time constant of the RC-circuit?

Ans. The time constant of an RC-circuit is the product of resistance and capacitance, tau = RC. It is the time in which the charge on the capacitor rises to about 63% of its maximum value during charging, or falls to about 37% of its initial value during discharging. A capacitor is approximately fully charged after about five time constants.

Q23. What is the unit of product of resistance and capacitance?

Ans. The product of resistance and capacitance, RC, has the unit of time, the second (s). Resistance has unit ohm (volt/ampere) and capacitance has unit farad (coulomb/volt), so ohm x farad = (volt/ampere) x (coulomb/volt) = coulomb/ampere = second, which is why RC is called the time constant.

Exercise - Comprehensive Questions

Q1. Define and explain charges, their kinds and properties.

Ans. Charge is the fundamental property of matter by virtue of which a body experiences a force in an electric field. There are two

kinds of charge: positive charge (as on a glass rod rubbed with silk) and negative charge (as on an ebonite rod rubbed with fur). Its properties are: (i) like charges repel and unlike charges attract each other; (ii) charge is quantized, that is, it always occurs in whole multiples of the elementary charge e = 1.6 x 10^-19 C; (iii) charge is conserved, it can neither be created nor destroyed, only transferred from one body to another; (iv) charge is additive, the total charge of a system is the algebraic sum of its charges.

Q2. State and explain Coulomb's law, also discuss the magnitude of electrostatic force of attraction or repulsion between two point charges in the absence and presence of dielectric.

Ans. Coulomb's law states that the electrostatic force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them, acting along the line joining them: F = (1/4 pi epsilon_o) x (q1 q2 / r^2) in free space, where 1/4 pi epsilon_o = 9 x 10^9 N m^2 C^-2. The force is repulsive for like charges and attractive for unlike charges. In the absence of a dielectric (free space) the force has this full value. In the presence of a dielectric medium of relative permittivity epsilon_r, the force is reduced to F = (1/4 pi epsilon_o epsilon_r) x (q1 q2 / r^2), that is, it becomes 1/epsilon_r times its free-space value, because the medium is polarized and weakens the field between the charges.

Q3. What do you know about the electric field and electric field intensity? Explain the magnitude and direction of electric field intensity.

Ans. The electric field is the region of space around a charge in which another charge experiences an electric force. Electric field intensity (or electric field strength) at a point is the force experienced per unit positive test charge placed at that point, E = F/q, with SI unit newton per coulomb (N/C). Its magnitude due to a point charge q at distance r is E = (1/4 pi epsilon_o) x (q/r^2), and due to two charges it is the vector sum of the individual fields. Its direction is the direction of the force on a positive test charge: away from a positive charge and towards a negative charge; at a point due to several charges it is the direction of the resultant force on the test charge.

Q4. What are the characteristics of the electric lines of force? Explain the electric field in terms of lines of force due to a positive charge, a negative charge and two identical positive or negative charges.

Ans. Characteristics of electric lines of force: (i) they start from a positive charge and end on a negative charge (or at infinity if the other charge is absent); (ii) they never cross each other; (iii) they are closer together where the field is stronger and farther apart where it is weaker; (iv) they leave or enter a charged surface normally; (v) they do not pass through a conductor. For a single positive charge the lines radiate straight outwards in all directions; for a single negative charge they converge straight inwards. For two identical positive charges the lines from each curve away from each other and there is a neutral point midway between them where the field is zero; for two identical negative charges the pattern is the same but with the arrows reversed, pointing inwards,

again with a neutral point midway.

Q5. What is electric dipole? Calculate electric field due to a dipole.

Ans. An electric dipole is a pair of equal and opposite charges +q and -q separated by a small distance 2a (or d). Its dipole moment is p = qd, directed from the negative to the positive charge. On the axial line at a distance r from the midpoint, the field is E = (1/4 pi epsilon_o) x (2pr/(r^2 - a^2)^2), which for r much greater than a reduces to E = (1/4 pi epsilon_o) x (2p/r^3), directed along the dipole axis. On the equatorial line (perpendicular bisector) the field is E = (1/4 pi epsilon_o) x (p/r^3), directed opposite to the dipole moment. Thus the axial field is twice the equatorial field at the same distance.

Q6. Define electric flux and discuss how does the electric flux become maximum and minimum?

Ans. Electric flux through a surface is the measure of the number of electric field lines passing through it, defined as phi = E.A cos theta, where E is the field strength, A is the area and theta is the angle between the field and the area vector (the normal to the surface). Its SI unit is N m^2 C^-1. The flux becomes maximum, phi = EA, when theta = 0 degree, that is, when the field is perpendicular to the surface (parallel to the area vector). It becomes minimum, phi = 0, when theta = 90 degree, that is, when the field is parallel to the surface, so no field lines cross it. For a closed surface the total flux equals Q/epsilon_o by Gauss's law, where Q is the enclosed charge.

Q7. State and explain Gauss's law. Also discuss the various applications of Gauss's law.

Ans. Gauss's law states that the total electric flux through any closed surface is equal to 1/epsilon_o times the total charge enclosed by the surface: phi = Q/epsilon_o. It is a consequence of Coulomb's law and is most useful when the charge distribution has symmetry. Applications: (i) field due to a point charge, E = (1/4 pi epsilon_o) x (q/r^2); (ii) field outside a charged spherical shell or solid sphere, the same as a point charge at the centre, and zero field inside a hollow shell; (iii) field due to an infinite line of charge, E = lambda/(2 pi epsilon_o r); (iv) field due to an infinite sheet of charge, E = sigma/(2 epsilon_o), independent of distance; (v) field between two oppositely charged parallel plates, E = sigma/epsilon_o.

Q8. What do you know about the electric potential? Define potential gradient and calculate potential due to a point charge.

Ans. Electric potential at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point, V = W/q, measured in volts (J/C). The potential gradient is the rate of change of potential with distance, and the electric field is its negative: E = -dV/dr, so the field points towards decreasing potential; for a uniform field E = V/d. Potential due to a point charge: the work done against the field in bringing a unit charge from infinity to a distance r from a charge q is V = (1/4 pi epsilon_o) x (q/r). It is a scalar quantity, positive for a positive charge and negative for a negative charge, and the potential due to several charges is the algebraic sum of the individual potentials.

Q9. Define capacitor and its capacitance. Also calculate the capacitance of a parallel plate capacitor.

Ans. A capacitor is a device that stores electric charge (and hence energy in its electric field); it usually consists of two conductors separated by an insulator (dielectric). Its capacitance is C = Q/V, the charge stored per unit potential difference, measured in farads. For a parallel plate capacitor of plate area A and separation d, the field between the plates is E = sigma/epsilon_o = Q/(A epsilon_o), and the potential difference is V = Ed = Qd/(A epsilon_o). Hence C = Q/V = (epsilon_o A)/d. With a dielectric of relative permittivity epsilon_r between the plates, C = (epsilon_o epsilon_r A)/d. Capacitance therefore increases with plate area and dielectric constant and decreases with plate separation.

Q10. State and explain series and parallel combinations of capacitors.

Ans. In a series combination, capacitors are joined end to end so that the same charge Q flows onto each, while the total potential difference is the sum of the individual ones, V = V1 + V2 + V3. Since V = Q/C, this gives 1/C = 1/C1 + 1/C2 + 1/C3, so the equivalent capacitance is smaller than the smallest individual capacitance. In a parallel combination, all capacitors are connected across the same potential difference V, while the total charge is the sum of the individual charges, Q = Q1 + Q2 + Q3. Since Q = CV, this gives C = C1 + C2 + C3, so the equivalent capacitance is the sum of the individual capacitances and is larger than any one of them.

Q11. What is electric polarization? Discuss the effect of dielectric on the capacitance of a capacitor.

Ans. Electric polarization is the process in which the molecules of a dielectric placed in an electric field develop dipole moments: permanent dipoles align with the field and non-polar molecules acquire induced dipoles. The aligned dipoles produce polarization charges on the dielectric surfaces which set up an opposing field, reducing the net field between the capacitor plates to E = E_o/epsilon_r. For the same stored charge the potential difference therefore falls to V = V_o/epsilon_r, and since C = Q/V, the capacitance rises to C = epsilon_r C_o. Thus a dielectric increases the capacitance by the factor epsilon_r and also allows the capacitor to store more energy and to withstand a higher voltage.

Q12. State and explain the process of charging and discharging of a capacitor.

Ans. Charging: when a capacitor is connected through a resistor to a battery, current flows and charge accumulates on the plates; the charge grows exponentially, q = Q_o(1 - e^(-t/RC)), rising quickly at first and then more slowly, reaching about 63% of its maximum in one time constant RC and becoming approximately fully charged after about five time constants, when the potential difference across the capacitor equals the battery e.m.f. and the current stops. Discharging: when the charged capacitor is disconnected from the battery and connected across a resistor, the stored charge flows back through the resistor; the charge falls exponentially, q = Q_o e^(-t/RC), dropping to about 37% of its initial value in one time

constant and to practically zero after about five time constants. In both cases the resistor controls the rate, and the energy stored (1/2)CV^2 is dissipated as heat in the resistor during discharge.

Exercise - Numerical Problems

Q1. How many electrons are contained in 1C and 3C of charges? (Book answers: 6.25 x 10^18 electrons, 1.9 x 10^19 electrons)

Charge is quantized: Q = ne, so the number of electrons is n = Q/e, where e = 1.6 x 10^-19 C.

For Q = 1 C: n = 1 / (1.6 x 10^-19) = 6.25 x 10^18 electrons.

For Q = 3 C: n = 3 / (1.6 x 10^-19) = 1.875 x 10^19 electrons, which is about 1.9 x 10^19 electrons.

1 C of charge contains 6.25 x 10^18 electrons and 3 C contains about 1.9 x 10^19 electrons. This matches the book answers.

Q2. Determine the force of repulsion between two free electrons spaced 0.5 Angstrom (1 A^0 = 10^-10 m). (Book answer: 92 nN)

Each electron carries charge e = 1.6 x 10^-19 C, and their separation is r = 0.5 Angstrom = 0.5 x 10^-10 m = 5 x 10^-11 m.

By Coulomb's law, F = k e^2 / r^2, with k = 9 x 10^9 N m^2 C^-2.

F = (9 x 10^9) x (1.6 x 10^-19)^2 / (5 x 10^-11)^2 = (9 x 10^9) x (2.56 x 10^-38) / (2.5 x 10^-21).

F = 2.304 x 10^-28 / 2.5 x 10^-21 = 9.216 x 10^-8 N = 92.16 nN.

The force of repulsion is about 92 nN. This matches the book answer.

Q3. Two coins lie 2 m apart on a table, and carry identical charge. How large is the charge on each if a coin experiences a force of 3 N? (Book answer: 3.65 x 10^-5 C)
For identical charges q at separation r = 2 m, Coulomb's law gives F = k q^2 / r^2, so q^2 = F r^2 / k.

q^2 = (3 x 2^2) / (9 x 10^9) = 12 / (9 x 10^9) = 1.333 x 10^-9 C^2.

q = sqrt(1.333 x 10^-9) = 3.65 x 10^-5 C.

The charge on each coin is 3.65 x 10^-5 C. This matches the book answer.

Q4. Two positive charges of magnitudes q1 = 5 microC and q2 = 2 microC are separated by 50 cm from each other. At what point the electric intensity due to these charges will be zero? (Book answer: 0.3 m from q1)
Let the point be at distance x from q1, on the line between the charges, so its distance from q2 is 0.5 - x. There the two fields are equal in magnitude and opposite in direction: k q1 / x^2 = k q2 / (0.5 - x)^2.

Taking square roots: sqrt(q1)/x = sqrt(q2)/(0.5 - x), that is, sqrt(5)/x = sqrt(2)/(0.5 - x).

So 2.236 x (0.5 - x) = 1.414 x, giving 1.118 - 2.236 x = 1.414 x, hence 3.650 x = 1.118.

x = 1.118 / 3.650 = 0.306 m, about 0.3 m.

The electric intensity is zero at about 0.3 m from q1 (between the two charges). This matches the book answer.

Q5. Find the force that an electron experiences in an electric field of 1000 NC^-1. If the electron is free to move, find the distance covered by it in 10 ns. The mass of electron is 9.11 x 10^-31 kg. (Book answers: 1.6 x 10^-16 N, 0.88 mm) NOTE: the book's distance answer appears to be a misprint - see the solution.
Force on the electron: F = eE = (1.6 x 10^-19 C) x (1000 N/C) = 1.6 x 10^-16 N.

Acceleration: a = F/m = (1.6 x 10^-16) / (9.11 x 10^-31) = 1.756 x 10^14 m/s^2.

Time t = 10 ns = 10 x 10^-9 s = 1 x 10^-8 s. Starting from rest, distance s = (1/2) a t^2 = 0.5 x (1.756 x 10^14) x (1 x 10^-8)^2.

s = 0.5 x (1.756 x 10^14) x (1 x 10^-16) = 0.5 x 1.756 x 10^-2 = 8.78 x 10^-3 m = 8.78 mm.

The force is 1.6 x 10^-16 N (as in the book), but the distance covered in 10 ns is 8.78 mm, not 0.88 mm: the book's 0.88 mm is smaller by a factor of ten and does not follow from its own data, so the corrected value 8.78 mm is used here and the book answer is flagged.

Q6. The diameter of a hollow metallic sphere is 10 cm and the sphere carries charge of 80 microC. Find the electric intensity (i) at a distance 60 cm from the centre of the sphere and (ii) at the surface of the sphere. (Book answers: 2 x 10^6 N/C, 2.88 x 10^8 N/C)

Radius of the sphere R = 10 cm / 2 = 5 cm = 0.05 m, and charge Q = 80 microC = 80 x 10^-6 C. Outside a charged sphere the field is that of a point charge at the centre: E = kQ/r^2.

(i) At r = 60 cm = 0.6 m: E = (9 x 10^9) x (80 x 10^-6) / (0.6)^2 = 7.2 x 10^5 / 0.36 = 2.0 x 10^6 N/C.

(ii) At the surface, r = R = 0.05 m: E = (9 x 10^9) x (80 x 10^-6) / (0.05)^2 = 7.2 x 10^5 / 0.0025 = 2.88 x 10^8 N/C.

The electric intensity is 2 x 10^6 N/C at 60 cm from the centre and 2.88 x 10^8 N/C at the surface. This matches the book answers.

Q7. An electric dipole consists of two charges +10 microC and -10 microC separated by distance 0.5 cm. Calculate the electric field intensity at a point on the axial line at a distance of 5 cm from the midpoint of the dipole. (Book answer: 7.2 x 10^6 N/C)

Dipole moment p = qd = (10 x 10^-6 C) x (0.5 x 10^-2 m) = 5 x 10^-8 C m.

On the axial line at distance r = 5 cm = 0.05 m from the midpoint, for r much greater than the charge separation, E = (1/4 pi epsilon_o) x (2p/r^3).

E = (9 x 10^9) x (2 x 5 x 10^-8) / (0.05)^3 = (9 x 10^9) x (1 x 10^-7) / (1.25 x 10^-4).

E = 900 / (1.25 x 10^-4) = 7.2 x 10^6 N/C.

The electric field intensity on the axial line is 7.2 x 10^6 N/C. This matches the book answer.

Q8. Find the electric flux through each face of a hollow cube of side 5 cm, if a charge of 6 microC is placed at its centre. (Book answer: 1.133 x 10^5 N m^2 C^-1)
By Gauss's law, the total flux through the closed cube is phi_total = Q/epsilon_o = (6 x 10^-6) / (8.85 x 10^-12).

phi_total = 6.78 x 10^5 N m^2 C^-1.

By symmetry the flux is shared equally by the six faces, so the flux through each face is phi = phi_total / 6 = (6.78 x 10^5) / 6 = 1.13 x 10^5 N m^2 C^-1.

The flux through each face is 1.133 x 10^5 N m^2 C^-1. This

matches the book answer. (The side of the cube does not affect the result.)

Q9. The potential difference between two metal plates is 120 V. The plate separation is 3 mm. Find the electric field between the plates. (Book answer: 40 kV/m)
For a uniform field between plates, E = V/d.

V = 120 V and d = 3 mm = 3 x 10^-3 m, so E = 120 / (3 x 10^-3) = 40,000 V/m.

40,000 V/m = 40 kV/m.

The electric field between the plates is 40 kV/m. This matches the book answer.

Q10. A charged particle remains stationary between the two horizontal opposite charged plates due to its weight and the electric force by the field. Find the potential difference between the plates, where the distance between the plates is 2 cm, mass of particle 4 x 10^-13 kg and charge on particle 2.4 x 10^-18 C. (Book answer: 32.6 kV)
For the particle to remain stationary, the upward electric force must balance its weight: qE = mg, so E = mg/q.

E = (4 x 10^-13 x 9.8) / (2.4 x 10^-18) = (3.92 x 10^-12) / (2.4 x 10^-18) = 1.633 x 10^6 V/m.

Potential difference V = E d = (1.633 x 10^6) x (2 x 10^-2) = 32,660 V, about 32.6 kV.

The potential difference between the plates is about 32.6 kV. This matches the book answer.

Q11. A capacitor of 5 microF is charged by a 12 V battery. Find the charge and energy stored on it. (Book answers: 60 microC, 3.6 x 10^-4 J)

Charge stored: Q = CV = (5 x 10^-6 F) x (12 V) = 60 x 10^-6 C = 60 microC.

Energy stored: W = (1/2) C V^2 = 0.5 x (5 x 10^-6) x (12)^2 = 0.5 x (5 x 10^-6) x 144.

W = 360 x 10^-6 J = 3.6 x 10^-4 J.

The charge stored is 60 microC and the energy stored is 3.6 x 10^-4 J. This matches the book answers.

Q12. Three capacitors have capacitance 2 microF, 5 microF and 7 microF. Calculate their equivalent capacitance when they are connected in (i) Series (ii) Parallel. (Book answers: 1.19 microF, 14 microF)

(i) Series: 1/C = 1/C1 + 1/C2 + 1/C3 = 1/2 + 1/5 + 1/7 = (35 + 14 + 10)/70 = 59/70 microF^-1.

So C = 70/59 microF = 1.186 microF, about 1.19 microF.

(ii) Parallel: C = C1 + C2 + C3 = 2 + 5 + 7 = 14 microF.

The equivalent capacitance is 1.19 microF in series and 14 microF in parallel. This matches the book answers.

Unit 12: Current Electricity

Exercise - Multiple Choice Questions (MCQs)

Q1. A flow of 10^7 electrons per second through a conductor produces a current of:

- ✓ 1.6 x 10^-12 A
- 1.6 x 10^12 A
- 1.6 x 10^-26 A
- 1.6 x 10^26 A

Q2. The first source of e.m.f. was discovered by:

- Ampere
- ✓ Volta
- Ohm
- Coulomb

Q3. Which one of the given cells is not rechargeable?

- ✓ Leclanche
- Nickel
- Cadmium
- Lead acid

Q4. Which one of the given sources converts light energy into electrical energy?

- Dynamo
- Generator

✓ Photocell

Thermocouple

Q5. Which one of the following material is non-ohmic?

Gold

✓ Germanium

Copper

Silver

Q6. What will be resistance of the wire when its length is doubled?

Remain Same

Half

✓ Double

Triple

Q7. If the resistivity of the conductor is 2 x 10^-6 ohm m then its conductivity is:

2 x 10^6 ohm^-1 m^-1

5 x 10^6 ohm^-1 m^-1

5 x 10^-5 ohm^-1 m^-1

✓ 5 x 10^5 ohm^-1 m^-1

Q8. Siemen is a unit of conductance and it is equal to:

ohm

✓ mho

mho . m

mho . m^-1

Q9. When resistance of the material increases by increasing the temperature then the value of co-efficient of resistivity will be:

- Constant
- Zero
- ✓ Positive
- Negative

Q10. The unit of temperature co-efficient is:

- degree C
- ✓ degree C^-1
- degree C . m
- degree C^-1 . m

Q11. In the presence of internal resistance of the source, which one of the following relations between potential difference (V) and e.m.f. (E) is correct?

- E = 0
- E = V
- ✓ E > V
- E < V

Q12. The unit of electromotive force is:

- Newton
- Joule
- Watt
- ✓ Volt

Q13. The source of e.m.f. transfers its maximum power to the

external circuit when (r = internal resistance, R = load resistance):

- r = 0
- ✓ r = R
- r < R
- r > R

Q14. Kirchhoff's 1st law is based upon law of conservation of:

- Current
- ✓ Charge
- Voltage
- Energy

Q15. The node in an electric circuit which has zero potential is known as:

- Node
- Antinode
- Negative node
- ✓ Datum node

Q16. When the direction of current is from negative to positive terminal of the battery then 'E' is taken as:

- Zero
- Normal
- ✓ Positive
- Negative

Q17. If current passes through a resistor in anticlockwise direction, then the electric potential will be:

Zero

Remain same

Raised

✓ Fall down

Q18. An electrical device which controls the current is known as:

Thermocouple

✓ Rheostat

Thermistor

Wheatstone bridge

Q19. The electrical device which is being used to compare the e.m.f. of two cells is known as:

Rheostat

Wheatstone Bridge

✓ Potentiometer

Galvanometer

Exercise - Short Questions

Q1. How does the conventional current differ from the electronic current?

Ans. Conventional current is the flow of positive charge, taken to flow from the positive terminal of a source to the negative terminal through the external circuit. Electronic current is the actual flow of free electrons in a metal, which move from the negative terminal to the positive terminal. The two are equal in magnitude but opposite in direction; by convention, circuit diagrams and Kirchhoff's laws use the conventional current direction.

Q2. How can you point out the direction of electric current?

Ans. The direction of electric current in a circuit is pointed out by the conventional direction: from the positive terminal of the source, through the external circuit, to the negative terminal. In practice it can be found with a galvanometer or ammeter (the needle deflects in the direction of the conventional current), or by the direction in which positive charge would be driven by the electric field in the conductor.

Q3. What do you know about the drift of the free electrons in a conductor?

Ans. In the absence of an electric field, free electrons in a conductor move randomly in all directions with thermal motion and there is no net current. When a potential difference is applied, the field exerts a force on the electrons, giving them a slow average motion superimposed on their random motion, in the direction opposite to the field. This slow average motion is called the drift velocity; it is very small (of the order of millimetres per second), yet it constitutes the electric current.

Q4. Which kind of current you observe in electrolyte plasma?

Ans. In an electrolyte or plasma, current is carried by both positive and negative ions moving in opposite directions: positive ions drift towards the negative electrode (cathode) and negative ions towards the positive electrode (anode). The total current is the sum of the two ion currents, and the conventional current direction is that of the positive ion flow.

Q5. What are the functions of copper and zinc plates in the source of e.m.f.?

Ans. In a simple voltaic cell, the zinc plate acts as the negative electrode (anode): zinc atoms give up electrons and go into the electrolyte as positive ions, supplying electrons to the external circuit. The copper plate acts as the positive electrode (cathode): it accepts electrons from the external circuit, where positive ions from the electrolyte gain electrons and are deposited or liberated there. The chemical reactions at the two plates maintain the potential difference that drives the current.

Q6. What do you know about the ohmic characteristics of a ohmic device?

Ans. An ohmic device is one which obeys Ohm's law: the current through it is directly proportional to the potential difference across it at constant temperature, so its V-I graph is a straight line passing through the origin and its resistance remains constant. Metals like copper, silver and gold at constant temperature are ohmic devices; their resistance does not change with the applied voltage or the current.

Q7. Which type of graph will be obtained between V and I for non-ohmic conductor?

Ans. For a non-ohmic conductor the V-I graph is not a straight line through the origin; it is a curve, because the resistance changes with the applied voltage or current. Examples are the filament of an electric bulb (resistance rises as it heats up), a semiconductor

diode (current flows easily in one direction only) and a thermistor. Such devices do not obey Ohm's law.

Q8. How does resistivity of a conductor depend upon temperature?
Ans. The resistivity of a metallic conductor increases with rise in temperature, because the vibrating ions of the metal lattice obstruct the free electrons more, so rho_t = rho_o(1 + alpha t), where alpha is the temperature coefficient of resistivity. For semiconductors and insulators, resistivity decreases with temperature, because more charge carriers are freed by thermal energy. For alloys like constantan and manganin, resistivity is almost independent of temperature.

Q9. Distinguish between positive and negative temperature coefficients of resistance.
Ans. A positive temperature coefficient means the resistance increases with rise in temperature, as in metals (for example copper, alpha about +0.0039 per degree C): heating makes the lattice vibrations larger and the electrons collide more often. A negative temperature coefficient means the resistance decreases with rise in temperature, as in semiconductors, insulators and thermistors: heating frees more charge carriers, which outweighs the increased lattice vibration.

Q10. Under what condition, the potential difference and electromotive force give the same value.
Ans. The terminal potential difference V and the e.m.f. E of a source are equal when no current is drawn from the source, that is, when

the circuit is open or the external resistance is infinite, because then there is no potential drop across the internal resistance (V = E - Ir, so with I = 0, V = E). They are also equal if the internal resistance is zero, which is the case for an ideal source.

Q11. When the source delivers its maximum power?

Ans. A source delivers maximum power to the external circuit when the load resistance R is equal to the internal resistance r of the source (maximum power transfer theorem). The power delivered is P = E^2 R/(R + r)^2, which is greatest when R = r; then the power delivered is E^2/(4r), and half of the total power is dissipated inside the source itself.

Q12. What is the working principle of a thermocouple?

Ans. A thermocouple works on the Seebeck effect: when two junctions of two dissimilar metals are kept at different temperatures, an e.m.f. is produced in the circuit, which depends on the temperature difference between the hot and cold junctions. This thermo-e.m.f. can be measured with a galvanometer and calibrated to read temperature, which is why thermocouples are used as thermometers, especially for high temperatures.

Q13. What do you know about the circuit analysis?

Ans. Circuit analysis is the method of finding the currents through, and the potential differences across, every element of an electric circuit. It uses Ohm's law together with Kirchhoff's two laws: the junction rule (sum of currents entering a junction equals the sum leaving it) and the loop rule (the algebraic sum of potential changes

around any closed loop is zero). For simple series and parallel circuits Ohm's law alone is enough; for complex networks Kirchhoff's laws are applied to a set of independent loops and junctions.

Q14. What do you know about node and loop in an electric circuit?
Ans. A node (junction) is a point in a circuit where three or more conductors meet and current can divide or combine; a node whose potential is taken as zero reference is called the datum node. A loop (mesh) is any closed conducting path in a circuit through which current can circulate. Kirchhoff's first law is applied at nodes and the second law around loops, and the number of independent loops and junctions determines how many equations are needed to solve the circuit.

Q15. What are the sign of currents when it flows toward and away from the node?
Ans. By convention, a current flowing towards a node (entering the junction) is taken as positive, and a current flowing away from the node (leaving the junction) is taken as negative, or vice versa, provided the convention is used consistently. Kirchhoff's first law then states that the algebraic sum of the currents at a node is zero: the sum of the currents entering the node equals the sum of the currents leaving it.

Q16. What are the basic principles of Kirchhoff's two laws?
Ans. Kirchhoff's first law (junction rule) is based on the conservation of charge: charge can neither accumulate at a junction nor be

created there, so the total current entering a junction equals the total current leaving it. Kirchhoff's second law (loop rule) is based on the conservation of energy: when a charge is carried once around a closed loop, the net work done on it is zero, so the algebraic sum of the e.m.f.s and the potential drops (IR) around the loop is zero.

Q17. Under what condition you use Kirchhoff's Laws for circuit analysis?

Ans. Kirchhoff's laws are used for circuit analysis when the circuit is complex and cannot be reduced by simple series-parallel combinations - that is, when it contains several sources or several loops and branches, as in a Wheatstone bridge or a multi-loop network. They are needed whenever the current through a particular resistor cannot be found from Ohm's law alone because the circuit has more than one loop or more than one source of e.m.f.

Q18. Which electrical device is being used as a potential divider?

Ans. A rheostat (variable resistance) and a potentiometer are used as potential dividers. A rheostat connected across a source with a sliding contact gives any desired fraction of the source voltage between one end and the sliding contact, and a potentiometer wire works in the same way: the potential difference across any length of the wire is proportional to that length, so tapping the wire at different points gives different voltages.

Q19. What is the difference between Rheostat and Potentiometer?

Ans. A rheostat is a variable resistance used mainly to control (vary) the current in a circuit; it is connected in series and only its resistance matters. A potentiometer is a device used to measure potential difference and e.m.f. accurately (and to compare e.m.f.s) by the null (zero-deflection) method; it draws no current from the source being measured at balance, so its reading is not affected by the internal resistance of the source. A rheostat controls current, a potentiometer measures voltage without drawing current.

Q20. What is the working principle of Wheatstone Bridge?

Ans. A Wheatstone bridge works on the null-deflection principle: four resistances P, Q, R and S are connected in a diamond, a cell across one diagonal and a galvanometer across the other. The bridge is balanced when the galvanometer shows no deflection, which happens when the potential at its two terminals is equal, giving P/Q = R/S. If three resistances are known, the fourth (unknown) resistance is found from this ratio, so the bridge measures unknown resistance accurately.

Exercise - Comprehensive Questions

Q1. Define and explain steady current, direction of current and sources of current.

Ans. Steady current is a continuous flow of charge whose magnitude remains constant with time, as supplied by a cell or battery, as opposed to a varying or alternating current. Its direction is taken by convention as the direction of flow of positive charge: from the positive terminal of the source through the external circuit to the negative terminal, while the actual electron flow in metals is

in the opposite direction. Sources of current are devices that maintain a potential difference by converting some other form of energy into electrical energy: cells and batteries (chemical energy), dynamos and generators (mechanical energy), photocells (light energy), thermocouples (heat energy) and piezoelectric sources.

Q2. State and explain Ohm's law. Also discuss its applications for ohmic and non-ohmic devices.

Ans. Ohm's law states that at constant temperature, the current through a conductor is directly proportional to the potential difference across its ends: V = IR, where R is the resistance of the conductor. For ohmic devices (metals at constant temperature), R is constant, the V-I graph is a straight line through the origin, and the law can be used directly to find any one of V, I and R from the other two, and to analyse series and parallel circuits. For non-ohmic devices (filament bulbs, diodes, thermistors, electrolytes), R changes with voltage, current or temperature, the V-I graph is curved, and Ohm's law applies only in the form R = V/I at a particular point (static resistance), not as a constant of the device.

Q3. What do you know about resistivity and conductivity of a conductor? Describe how the resistivity of a conductor depends upon the variation of its temperature.

Ans. Resistivity rho is the resistance of a one-metre cube (unit cube) of a material, defined by R = rho L/A, with SI unit ohm-metre; it is a property of the material, not of the size of the conductor. Conductivity sigma is its reciprocal, sigma = 1/rho, with unit ohm^-1 m^-1 (siemens per metre); it measures how easily a

material conducts. With temperature, the resistivity of a metal increases almost linearly, rho_t = rho_o(1 + alpha t), because lattice vibrations increase and scatter the electrons more; for semiconductors and insulators it decreases, because heating frees more charge carriers; and for alloys like manganin it stays almost constant.

Q4. Describe thermistor and its application in daily life.

Ans. A thermistor is a temperature-dependent resistor made of a semiconductor material (usually a metal oxide), whose resistance varies markedly even with a small change of temperature; most thermistors have a negative temperature coefficient, so their resistance falls as temperature rises. Because of this sensitivity they are used in daily life as temperature sensors and controllers: in thermostats, fire alarms, digital thermometers, automatic temperature control of refrigerators and air conditioners, and for temperature compensation and surge protection in electronic circuits.

Q5. What do you know about electromotive force and potential difference? Under what conditions, their values are same in an electric circuit.

Ans. Electromotive force (e.m.f.) is the work done by a source per unit charge in moving charge once around a complete circuit, that is, the energy supplied by the source per coulomb; it is the potential difference across the source terminals when no current flows. Potential difference (terminal voltage) is the work done per unit charge between two points in the external circuit, V = IR, and

equals the energy converted per coulomb in the external part. They are related by V = E - Ir, where r is the internal resistance. Their values are the same when no current is drawn (open circuit, I = 0) or when the internal resistance is negligible (r = 0). When current flows, V is less than E by the drop Ir across the internal resistance.

Q6. What is electrical power? How does the maximum power transfer take place in an electric circuit?

Ans. Electrical power is the rate at which electrical energy is converted into other forms (heat, light, mechanical work) in a circuit, P = VI = I^2 R = V^2/R, measured in watts. For a source of e.m.f. E and internal resistance r connected to a load R, the power delivered to the load is P = E^2 R/(R + r)^2. This is maximum when the load resistance equals the internal resistance, R = r (maximum power transfer theorem); then P_max = E^2/(4r). At this point half the power is delivered to the load and half is wasted as heat inside the source, so the condition gives maximum power but only 50% efficiency.

Q7. What do you know about a thermocouple and its function?

Ans. A thermocouple is an electric device made of two dissimilar metal wires joined at two junctions; it produces an e.m.f. by heat, through the Seebeck effect, when the junctions are at different temperatures. Its function is to measure temperature: the thermo-e.m.f. increases with the temperature difference between the hot junction and the cold (reference) junction, and a calibrated galvanometer or voltmeter reads the temperature. Thermocouples are used for measuring high temperatures in furnaces, engines and

laboratories, and as temperature sensors in industry.

Q8. State and explain Kirchhoff's two rules for circuit analysis.

Ans. Kirchhoff's first rule (junction rule) states that the algebraic sum of the currents at any junction is zero: the total current entering a junction equals the total current leaving it. It is based on the conservation of charge, since charge cannot accumulate at a junction. Kirchhoff's second rule (loop rule) states that the algebraic sum of the potential changes around any closed loop is zero: the sum of the e.m.f.s equals the sum of the IR drops around the loop. It is based on the conservation of energy, since a charge taken once around a loop returns to its starting point with no net gain or loss of energy. Sign convention: going through a source from negative to positive terminal, the e.m.f. is positive; going through a resistor in the direction of the current, the potential falls by IR.

Q9. State and explain node analysis and loop analysis by using Kirchhoff's two rules.

Ans. Node analysis applies Kirchhoff's first rule at the junctions of a circuit: unknown branch currents are written in terms of node potentials, and the rule that currents entering a node equal currents leaving it gives one equation per independent node (one fewer than the number of nodes, taking a datum node at zero potential). Loop analysis applies Kirchhoff's second rule around the independent closed loops (meshes) of the circuit: a current is assigned to each loop, and the rule that the algebraic sum of e.m.f.s and IR drops around a loop is zero gives one equation per loop. The two sets of equations together are solved for the

unknown currents or potentials; node analysis is convenient when there are few nodes, loop analysis when there are few loops.

Q10. Describe a rheostat and its working principle. Explain how a rheostat can be used as a variable resistance as well as a potential divider.

Ans. A rheostat is a long wire of high-resistance material (such as constantan) wound on an insulating cylinder, with a sliding contact (jockey) that can move along the wire. Its working principle is that the resistance between one end of the wire and the sliding contact is proportional to the length of wire between them, so moving the contact changes the resistance in the circuit. As a variable resistance it is connected in series with the circuit: sliding the contact changes the total resistance and hence the current. As a potential divider it is connected across the source of voltage, and the output is taken between one end and the sliding contact: the output voltage is the fraction of the source voltage corresponding to the fraction of the wire length used, V_out = (length used/total length) x V_source.

Q11. What do you know about Wheatstone bridge? Draw its circuit diagram and explain its working principle.

Ans. A Wheatstone bridge is a circuit of four resistances P, Q, R and S connected in a diamond (P and Q in one branch, R and S in the other), with a cell connected across one diagonal and a galvanometer across the other diagonal. Its circuit diagram is: P (top-left arm) and Q (top-right arm) in series across the cell, R (bottom-left arm) and S (bottom-right arm) in series across the cell,

the galvanometer joining the junction of P and R to the junction of Q and S. Its working principle is the null method: the bridge is balanced by adjusting a known resistance until the galvanometer shows zero deflection, meaning the two junctions joined by the galvanometer are at the same potential; then P/Q = R/S, so if three resistances are known, the fourth (unknown) resistance is S = (Q/P) x R.

Q12. State and explain potentiometer, its function and applications.
Ans. A potentiometer is a long uniform wire (usually 1 to 10 metres) across which a steady current is maintained by a driver cell, so that the potential falls uniformly along the wire (potential gradient k volts per metre). Its function is to measure an unknown potential difference or e.m.f. by the null method: the unknown source is balanced against a length l of the wire at which the galvanometer shows no deflection, and then E = k l. Because no current is drawn from the source at balance, the measurement is exact and unaffected by internal resistance. Applications: comparing the e.m.f.s of two cells (E1/E2 = l1/l2), measuring the internal resistance of a cell, calibrating voltmeters and ammeters, and measuring small potential differences accurately.

Exercise - Numerical Problems

Q1. How many electrons flow through a light bulb each second, if the current through the bulb is 0.8 A? (Book answer: 5 x 10^18)
Current is charge per second: I = Q/t, so the charge flowing per second is Q = I x t = 0.8 A x 1 s = 0.8 C.

The number of electrons is n = Q/e = 0.8 / (1.6 x 10^-19).

n = 5 x 10^18 electrons per second.

5 x 10^18 electrons flow through the bulb each second. This matches the book answer.

Q2. A flow of charges of 150 C through a wire in 2 hours, what is the amount of current in the wire? (Book answer: 21 mA)
Current I = Q/t. Here Q = 150 C and t = 2 hours = 2 x 3600 s = 7200 s.

I = 150 / 7200 = 0.0208 A.

0.0208 A = 20.8 mA, about 21 mA.

The current in the wire is about 21 mA. This matches the book answer.

Q3. A metal rod is 4 m long and 6 mm in diametre. Compute its resistance if the resistivity of the metal is 1.76 x 10^-8 ohm m. (Book answer: 2.5 x 10^-3 ohm)
Radius of the rod r = 6 mm / 2 = 3 mm = 3 x 10^-3 m, so its cross-sectional area is A = pi r^2 = 3.14 x (3 x 10^-3)^2 = 2.83 x 10^-5 m^2.

Resistance R = rho L / A = (1.76 x 10^-8 x 4) / (2.83 x 10^-5).

R = 7.04 x 10^-8 / 2.83 x 10^-5 = 2.49 x 10^-3 ohm, about 2.5 x 10^-3 ohm.

The resistance of the rod is about 2.5 x 10^-3 ohm. This matches the book answer.

Q4. Calculate the potential difference across the two ends of the wire of resistance 5 ohm, if the charges of 720 C pass through it per minute. (Book answer: 60 V)
Current I = Q/t = 720 C / 60 s = 12 A.

By Ohm's law, V = IR = 12 A x 5 ohm = 60 V.

The potential difference across the wire is 60 V. This matches the book answer.

Q5. A 10 m long wire of diametre 0.2 cm has a resistance 3 ohm. Find the resistivity of the material of the wire, also calculate its conductivity. (Book answers: 9.4 x 10^-7 ohm m, 1.06 x 10^6 ohm^-1 m^-1)
Radius r = 0.2 cm / 2 = 0.1 cm = 1 x 10^-3 m, so area A = pi r^2 = 3.14 x (1 x 10^-3)^2 = 3.14 x 10^-6 m^2.

Resistivity rho = R A / L = (3 x 3.14 x 10^-6) / 10 = 9.42 x 10^-7 ohm m, about 9.4 x 10^-7 ohm m.

Conductivity sigma = 1/rho = 1 / (9.42 x 10^-7) = 1.06 x 10^6 ohm^-1 m^-1.

The resistivity is about 9.4 x 10^-7 ohm m and the conductivity is 1.06 x 10^6 ohm^-1 m^-1. This matches the book answers.

Q6. The resistance of a coil is 140 ohm at 20 degree C. If current is passed through it, its temperature rises, and its resistance becomes 160 ohm at 40 degree C. Calculate the temperature co-efficient of resistance of the coil. (Book answer: 0.0071 degree C^-1)
Change in resistance = 160 - 140 = 20 ohm, over a temperature change of 40 - 20 = 20 degree C.

Temperature coefficient alpha = (change in R) / (R_o x change in T) = 20 / (140 x 20).

alpha = 20 / 2800 = 0.00714 per degree C, about 0.0071 degree C^-1.

The temperature coefficient of resistance is about 0.0071 degree C^-1. This matches the book answer.

Q7. A copper wire has resistance of 3 ohm at 0 degree C. What is its resistance at 60 degree C? The temperature coefficient of resistance of copper is 0.0039 degree C^-1. (Book answer: 3.77 ohm) NOTE: the book's answer does not follow from its own data - see the solution.
Resistance at temperature t is R_t = R_o (1 + alpha t), with R_o = 3 ohm, alpha = 0.0039 per degree C and t = 60 degree C.

R_60 = 3 x (1 + 0.0039 x 60) = 3 x (1 + 0.234) = 3 x 1.234.

R_60 = 3.702 ohm, about 3.70 ohm.

The resistance at 60 degree C is 3.70 ohm from the book's own data (3 x 1.234 = 3.702). The book's printed answer of 3.77 ohm is inconsistent with its data - it would require a coefficient of about 0.0043 per degree C - so the corrected value 3.70 ohm is used here and the book answer is flagged.

Q8. A battery has an e.m.f. of 12.5 V and an internal resistance of 2.4 x 10^-3 ohm. If the load current is 20 A, find the terminal voltage. (Book answer: 12.45 V)

Terminal voltage V = E - I r, where E = 12.5 V, I = 20 A and r = 2.4 x 10^-3 ohm.

Voltage drop across the internal resistance = I r = 20 x 2.4 x 10^-3 = 0.048 V.

V = 12.5 - 0.048 = 12.452 V, about 12.45 V.

The terminal voltage is about 12.45 V. This matches the book answer.

Q9. Determine the current through each resistor in the electric circuit as shown in Fig. 12.20, where R1 is 10 ohm, R2 is 6 ohm, R3 is 10 ohm, E1 is 6 V and E2 is 3 V. (Book answers: I1 = 0.3 A, I2 = 0, I3 = 0.3 A)

In Fig. 12.20 the current I1 flows through R1 (top-left), I2 through R2 (top-right, towards the middle junction C) and I3 downwards through R3 (middle). At junction C, Kirchhoff's first law gives I1 + I2 = I3.

For the left loop (E1, R1, R3) clockwise: 6 - 10 I1 - 10 I3 = 0, so I1 + I3 = 0.6.

For the right loop (R2, E2, R3) clockwise: 6 I2 - 3 + 10 I3 = 0, so 6 I2 + 10 I3 = 3.

From the first two equations I1 = 0.6 - I3, so I3 = (0.6 - I3) + I2, giving I2 = 2 I3 - 0.6. Substituting in the loop equation: 6(2 I3 - 0.6) + 10 I3 = 3, so 22 I3 = 6.6 and I3 = 0.3 A. Then I1 = 0.6 - 0.3 = 0.3 A and I2 = 2(0.3) - 0.6 = 0 A.

The currents are I1 = 0.3 A, I2 = 0 A and I3 = 0.3 A. This matches the book answers. (No current flows through R2 because the two loop voltages balance at junction C.)

Q10. Find the current through each resistor in the electrical network as shown in Fig. 12.21. (Book answers: I1 = 0.21 A, I2 = 0.14 A, I3 = 0.36 A)

In Fig. 12.21 a 5 V battery is connected across the terminals A and D of the network, whose resistors are: A-B = 10 ohm, B-D = 20 ohm, B-C = 10 ohm, A-C = 20 ohm and C-D = 10 ohm. Let the potentials be V_A = 5 V and V_D = 0 V, with node potentials V_B and V_C to be found.

At node B, Kirchhoff's first law: current in through A-B = currents out through B-D and B-C: (5 - V_B)/10 = (V_B - 0)/20 + (V_B - V_C)/10.

At node C: currents in through B-C and A-C = current out through C-D: (V_B - V_C)/10 + (5 - V_C)/20 = V_C/10.

Multiplying the node B equation by 20: 2(5 - V_B) = V_B + 2(V_B - V_C), so 10 = 5 V_B - 2 V_C. Multiplying the node C equation by 20: 2(V_B - V_C) + (5 - V_C) = 2 V_C, so 2 V_B + 5 = 5 V_C, giving V_C = 0.4 V_B + 1. Substituting: 10 = 5 V_B - 2(0.4 V_B + 1) = 4.2 V_B - 2, so V_B = 12/4.2 = 2.857 V and V_C = 2.143 V.

Current through A-B: I1 = (5 - 2.857)/10 = 0.214 A, about 0.21 A. Current through A-C: I2 = (5 - 2.143)/20 = 0.143 A, about 0.14 A. Total current from the battery: I3 = I1 + I2 = 0.214 + 0.143 = 0.357 A, about 0.36 A.

The currents are I1 = 0.21 A (through the 10 ohm A-B resistor), I2 = 0.14 A (through the 20 ohm A-C resistor) and total current I3 = 0.36 A. This matches the book answers.

Q11. If the e.m.f. of a battery is balanced by a length of 80 cm on a potentiometer wire, while the e.m.f. of 1.03 V of a standard cell is balanced by a length of 55 cm, then determine the e.m.f. of the battery. (Book answer: 1.5 V)

For a potentiometer, e.m.f. is proportional to the balancing length:

E/E_s = l/l_s, where E_s = 1.03 V, l = 80 cm and l_s = 55 cm.

E = 1.03 x (80/55) = 1.03 x 1.4545 = 1.498 V.

E is about 1.5 V.

The e.m.f. of the battery is about 1.5 V. This matches the book answer.

Unit 13: Electromagnetism

Exercise - Multiple Choice Questions (MCQs)

Q1. A maximum force that acts on a current carrying conductor placed in a magnetic field, when angle between the field and length of the conductor is:

- 0 degree
- 45 degree
- ✓ 90 degree
- 180 degree

Q2. Tesla in terms of base units is equal to:

- kg m^-1 A^-1
- kg m A
- kg s^-1 A^-1
- ✓ kg s^-2 A^-1

Q3. One Gauss is equal to:

- 1 T
- 10^2 T
- ✓ 10^-4 T
- 10^4 T

Q4. The unit of magnetic flux is:

- Tesla
- ✓ Weber

Gauss

Henry

Q5. The magnetic flux is maximum when angle between magnetic field and vector area is:

✓ 0 degree

30 degree

60 degree

90 degree

Q6. Magnetic flux density is defined in terms of:

Tesla

wb m^-2

N A^-1 m^-1

✓ All of them

Q7. A charged particle is moving along X-axis in a magnetic field along the Y-axis. The direction of magnetic force acting on it is:

along x-axis

along y-axis

✓ along z-axis

in xy-plane

Q8. Ampere's Law gives us the relationship between:

Force and velocity of charge

Force and magnitude field

Current and force

✓ Current and magnetic field

Q9. The value of permeability of free space is:

- 10^-7 T m A^-1
- 2 pi x 10^-7 T m A^-1
- ✓ 4 pi x 10^-7 T m A^-1
- 4 pi x 10^7 T m A^-1

Q10. The magnetic field due to a current-carrying solenoid which has 'n' number of turns per unit length is:

- ✓ B = mu_o n I
- B = mu_o n^2 I
- B = mu_o n I / l
- B = mu_o n^2 I / l

Q11. The magnetic field inside the solenoid is independent of one of the following quantities:

- Permeability
- ✓ Position vector
- Number of turns
- Flow of current

Q12. What is the magnetic force on a stationary charged particle in a uniform magnetic field?

- ✓ Zero
- F = q(v x B)
- F = qvB
- F = ILB sin theta

Q13. An electron is moving horizontally towards east. If it enters in magnetic field directed upward then the electron will be deflected in the direction of:

- East
- West
- ✓ North
- South

Q14. When the direction of motion of a charged particle is perpendicular to the direction of magnetic field, then the particle follows the path of a:

- Straight line
- helix
- ellipse
- ✓ circle

Q15. The torque due to a current carrying rectangular coil placed in a uniform magnetic field is:

- tau = IBA sin theta
- tau = IAN sin theta
- tau = IBN sin theta
- ✓ tau = NAIB sin theta

Q16. The working principle of a galvanometer is based upon:

- Momentum
- ✓ Torque
- Force
- Impulse

Q17. The current passing through the coil of a galvanometer is directly proportional to the:

- Resistance
- Conductance
- Reactance
- ✓ Angle of deflection

Q18. A shunted galvanometer is called:

- Voltmeter
- ✓ Ammeter
- Ohmmeter
- Potentiometer

Q19. A galvanometer can be converted into voltmeter by connecting it with:

- Low resistance in parallel
- Low resistance in series
- High resistance in parallel
- ✓ High resistance in series

Q20. Which one of the following quantity is not measured by Avometer?

- ✓ Charge
- Current
- Resistance
- Potential difference

Exercise - Short Questions

Q1. How can you determine the direction of the magnetic field due to a current carrying a conductor?

Ans. The direction of the magnetic field around a straight current-carrying conductor is found by the right-hand grip rule: grip the conductor with the right hand so that the thumb points in the direction of the (conventional) current; the curled fingers then point in the direction of the magnetic field lines, which form concentric circles around the conductor. For a circular coil, curl the fingers in the direction of the current and the thumb points towards the north pole face of the coil, that is, along the field inside the coil.

Q2. What is tesla and what is the relation between Tesla and Gauss?

Ans. The tesla (T) is the SI unit of magnetic flux density (magnetic field strength): one tesla is the field in which a conductor of length one metre carrying a current of one ampere, placed perpendicular to the field, experiences a force of one newton. The gauss (G) is the CGS unit of the same quantity. The relation is 1 T = 10^4 G, so 1 G = 10^-4 T.

Q3. Distinguish between Tesla and Weber.

Ans. The tesla is the unit of magnetic flux density B, that is, magnetic flux per unit area (B = phi/A), while the weber is the unit of magnetic flux phi itself, the total number of field lines passing through a surface. They are related by 1 weber = 1 tesla x 1 square metre: flux (Wb) = flux density (T) x area (m^2). Tesla measures the strength of the field at a point; weber measures the total flux

through a surface.

Q4. How can you differentiate the flow of current into the page and out of the page?

Ans. In diagrams, current flowing into the page (away from the reader) is shown by a cross (x), representing the tail of an arrow going away, and current flowing out of the page (towards the reader) is shown by a dot (.), representing the tip of an arrow coming towards the reader. The magnetic field circles around the conductor accordingly: clockwise for current into the page and anticlockwise for current out of the page, by the right-hand grip rule.

Q5. Under what condition the magnetic flux is minimum and maximum?

Ans. Magnetic flux through a surface is phi = BA cos theta, where theta is the angle between the magnetic field B and the area vector (normal to the surface). The flux is maximum, phi = BA, when theta = 0 degree, that is, when the field is perpendicular to the surface (parallel to the area vector). It is minimum (zero) when theta = 90 degree, that is, when the field is parallel to the surface, so no field lines pass through it.

Q6. Differentiate between magnetic flux and magnetic flux density.

Ans. Magnetic flux phi is the total number of magnetic field lines passing through a given surface, phi = BA cos theta, measured in webers (Wb). Magnetic flux density B is the flux per unit area perpendicular to the field, B = phi/A, measured in teslas (T) or

webers per square metre; it measures the strength of the magnetic field at a point. Flux depends on the area chosen, while flux density is a property of the field itself.

Q7. What do you know about the magnetic force on a stationary charged particle in a uniform magnetic field?

Ans. The magnetic force on a charged particle is F = q(v x B), which depends on the velocity of the particle. For a stationary particle v = 0, so the magnetic force is zero, however strong the field. A magnetic field can therefore never set a charge at rest into motion, and it can do no work on a moving charge either, since the force is always perpendicular to the velocity - it only changes the direction of motion, not the speed.

Q8. Why Ampere's law is true only for a steady current?

Ans. Ampere's circuital law, the line integral of B around a closed path equals mu_o times the current enclosed, is derived assuming the currents producing the field are steady, so that the magnetic field is static and the charge distribution does not change with time. If the current varies with time, the changing electric field between the conductors also contributes (Maxwell's displacement current), and Ampere's law in its simple form no longer holds; it must be extended to the Ampere-Maxwell law.

Q9. What do you know about the amperean path?

Ans. An amperean path (Amperian loop) is an imaginary closed path chosen to apply Ampere's circuital law. It is selected so that the magnetic field is either constant in magnitude and parallel to

the path (so that B can be taken out of the line integral) or perpendicular to it (so that part contributes nothing). For a straight conductor it is a circle around the conductor; for a solenoid it is a rectangle with one side inside the solenoid; for a toroid it is a circle through the toroid.

Q10. What are the values of the magnetic field inside and outside of a current-carrying solenoid?

Ans. Inside a long solenoid the magnetic field is uniform and strong, B = mu_o n I, where n is the number of turns per unit length and I is the current; it points along the axis of the solenoid and is independent of position inside. Outside the solenoid the magnetic field is very weak, practically zero, because the field lines spread out over a large region and the contributions cancel; an ideal infinitely long solenoid has exactly zero field outside.

Q11. State the Fleming's left-hand rule to determine the direction of force that acts on a charged particle moving perpendicular to the magnetic field.

Ans. Fleming's left-hand rule: stretch the thumb, forefinger and middle finger of the left hand mutually perpendicular to each other. If the forefinger points in the direction of the magnetic field, the middle finger in the direction of motion of the positive charge (or current), then the thumb points in the direction of the magnetic force on the charge. For a negative charge like an electron, the force is in the opposite direction to the thumb. When the particle moves perpendicular to the field, the force is perpendicular to both, and is maximum, F = qvB.

Q12. Explain the path of deflection of an electron and a proton when they enter perpendicularly in a uniform magnetic field.

Ans. When a charged particle enters a uniform magnetic field perpendicularly, the magnetic force F = qvB acts perpendicular to its velocity at every instant, providing the centripetal force, so the particle follows a circular path in the plane perpendicular to the field, with radius r = mv/(qB). An electron and a proton entering with the same velocity are deflected in opposite directions (because their charges are opposite) and in opposite senses of circulation: the proton curves one way and the electron the other. The electron's circle is much smaller, because its mass is about 1836 times smaller than the proton's, so r = mv/(qB) is much smaller for the electron.

Q13. How does an electron come under the centripetal force when its motion is perpendicular to a uniform magnetic field?

Ans. When an electron moves perpendicular to a uniform magnetic field, the magnetic force F = evB acts on it in a direction perpendicular to both its velocity and the field, at every instant. Since the force is always at right angles to the motion, it does no work and cannot change the electron's speed, but it continuously changes its direction, bending it into a circular path. A force that is always directed towards the centre of a circular path is a centripetal force, and here the magnetic force itself plays that role, giving mv^2/r = evB, so the radius is r = mv/(eB).

Q14. How can you determine e/m by velocity selector method?

Ans. In a velocity selector, crossed electric and magnetic fields are adjusted so that a charged particle passes undeflected: then the electric force equals the magnetic force, qE = qvB, giving v = E/B for the selected velocity. The particle then enters a region of uniform magnetic field only, where it follows a circle of radius r = mv/(qB). Measuring r, B and v gives e/m = v/(Br) = E/(B^2 r). In practice the particle is an electron, so this determines the charge-to-mass ratio of the electron, about 1.76 x 10^11 C/kg.

Q15. How can we increase the magnitude of torque due to a current carrying rectangular coil placed in a uniform magnetic field?

Ans. The torque on a coil is tau = NIAB sin theta, where N is the number of turns, I the current, A the area of the coil, B the field and theta the angle between the field and the normal to the coil. Its magnitude can therefore be increased by: increasing the number of turns N, increasing the current I, increasing the area A of the coil, increasing the magnetic field B, or placing the coil so that theta = 90 degree (the plane of the coil parallel to the field), which makes sin theta = 1 and gives the maximum torque.

Q16. How does radial magnetic field produce in a moving coil galvanometer?

Ans. In a moving-coil galvanometer the coil is placed between the poles of a permanent magnet that are made concave (cylindrical), and a soft-iron cylinder (core) is fixed inside the coil. The concave pole pieces and the iron core shape the field so that the magnetic field lines are always radial, that is, they point along the radius towards the core and are always in the plane of the coil, whatever

the coil's position. This makes the torque tau = NIAB independent of the angle of deflection (since the field is always parallel to the plane of the coil), so the deflection is directly proportional to the current.

Q17. Why low resistance in an ammeter is called shunt resistance? Why it is connected parallel to a galvanometer?

Ans. A shunt is a low resistance connected in parallel with a galvanometer to convert it into an ammeter. It is called a shunt because it shunts (bypasses) most of the current past the galvanometer: only a small known fraction passes through the coil, while the bulk flows through the low-resistance shunt, so the delicate coil is not damaged and the meter can measure large currents. It is connected in parallel because in parallel the same potential difference acts across the galvanometer and the shunt, so the current divides in the inverse ratio of the resistances, and the total current can be found from the galvanometer reading.

Q18. How can you convert a moving galvanometer into a voltmeter?

Ans. A moving-coil galvanometer is converted into a voltmeter by connecting a high resistance in series with it. The high resistance limits the current through the coil to its full-scale value even when a large potential difference is applied across the combination, and most of the voltage drops across the series resistance. The value needed for a range V is R = V/I_g - R_g, where I_g is the full-scale deflection current and R_g the galvanometer resistance. The voltmeter is then connected in parallel with the component whose

voltage is to be measured, and its high resistance ensures it draws negligible current.

Q19. Explain the method of measurement of resistance by using ohmmeter.

Ans. An ohmmeter is essentially a galvanometer in series with a battery and a known resistance, with a scale marked directly in ohms. To measure an unknown resistance, the resistance is connected across the ohmmeter terminals: the battery drives a current through the unknown resistance and the meter, and the deflection depends on the total resistance in the circuit. The meter is first zeroed by shorting the terminals (zero resistance gives full-scale deflection), and then the unknown resistance is connected; a larger resistance gives a smaller current and a smaller deflection, so the scale reads resistance directly, with zero at full scale and infinity at zero deflection.

Q20. Distinguish between analogue and digital Avometer.

Ans. An analogue Avometer (multimeter) uses a moving-coil galvanometer with a pointer moving over a printed scale; it measures current, voltage and resistance (Ampere, Volt, Ohm) by switching the internal circuit, and the reading is taken from the pointer position, which is continuous but less precise and must be read carefully to avoid parallax error. A digital Avometer uses electronic circuits and displays the measured value as digits on a screen; it is more accurate, easier to read, auto-ranging and free from parallax error, but it needs a battery for its electronics and cannot easily show a continuously varying value the way a moving

pointer can.

Exercise - Comprehensive Questions

Q1. What is magnetic field? Explain the magnetic field produced around a current carrying conductor.

Ans. A magnetic field is the region of space around a magnet or a moving charge in which another magnet or moving charge experiences a magnetic force. Around a straight current-carrying conductor the field forms concentric circles centred on the conductor, in planes perpendicular to it. Its direction is given by the right-hand grip rule: with the thumb along the conventional current, the curled fingers show the field direction (clockwise when the current goes into the page, anticlockwise when it comes out). Its magnitude at distance r from a long straight conductor is B = (mu_o I)/(2 pi r), so it is directly proportional to the current and inversely proportional to the distance from the conductor.

Q2. Define magnetic force and derive a relation for a magnetic force that is exerted on a current-carrying a conductor placed in uniform applied magnetic field.

Ans. Magnetic force is the force experienced by a moving charge or a current-carrying conductor in a magnetic field. Consider a conductor of length L carrying current I in a uniform field B, with the conductor at angle theta to the field. Each charge q drifting with velocity v experiences F = qvB sin theta; with n free charges per unit volume and cross-section A, the total number of charges is nAL, and the total force is F = (nAL) q v B sin theta. Since current I = nAqv, this gives F = ILB sin theta, directed perpendicular to both

the conductor and the field by Fleming's left-hand rule. It is maximum (ILB) when the conductor is perpendicular to the field and zero when parallel to it.

Q3. What do you know about Flemming left hand rule? How can you determine the direction of magnetic force by using flaming left-hand rule?

Ans. Fleming's left-hand rule is a rule for the direction of the magnetic force on a current or a moving positive charge. Hold the thumb, forefinger and middle finger of the left hand mutually at right angles: the forefinger represents the magnetic field (from north to south), the middle finger represents the current (or the motion of a positive charge), and the thumb then represents the direction of the magnetic force (motion of the conductor). To find the force on a current-carrying conductor, point the forefinger along the field and the middle finger along the current; the thumb gives the force direction. For an electron the force is opposite to the thumb direction, since the electron moves opposite to the conventional current.

Q4. Define magnetic flux, magnetic flux density and their SI units.

Ans. Magnetic flux phi is the total number of magnetic field lines passing normally through a surface, defined as phi = BA cos theta, where B is the field, A the area and theta the angle between the field and the normal to the surface. Its SI unit is the weber (Wb); one weber is the flux through one square metre held perpendicular to a field of one tesla. Magnetic flux density B is the flux per unit area perpendicular to the field, B = phi/A; it measures the strength

of the field at a point. Its SI unit is the tesla (T), equal to one weber per square metre (1 T = 1 Wb m^-2 = 1 N A^-1 m^-1).

Q5. State and explain Ampere's circuital law and calculate magnetic field due to a current carrying solenoid using Ampere's law.

Ans. Ampere's circuital law states that the line integral of the magnetic field around any closed path is equal to mu_o times the total current enclosed by the path: the sum of B.dl around the loop = mu_o I_enclosed. To find the field of a solenoid, take a rectangular Amperian loop with one side of length l inside the solenoid parallel to its axis and the other sides outside or perpendicular, where the field is zero or perpendicular to the path. The line integral reduces to B l (only the inside side contributes). The enclosed current is n l I, where n is the number of turns per unit length, so B l = mu_o n l I, giving B = mu_o n I inside the solenoid, uniform and along its axis, and approximately zero outside.

Q6. How can you determine the force that acts on a moving charged particle (proton and electron) in a uniform magnetic field?

Ans. The force on a moving charged particle in a uniform magnetic field is F = q(v x B), of magnitude F = qvB sin theta, where theta is the angle between the velocity and the field. It is perpendicular to both v and B, so it changes only the direction of motion. For a proton (positive charge) the direction is given by Fleming's left-hand rule, or by the right-hand rule for v x B. For an electron (negative charge) the force is equal in magnitude but opposite in direction to that on a proton moving the same way, since q is

negative. If the particle moves parallel to the field (theta = 0), the force is zero; if perpendicular (theta = 90 degree), it is maximum, qvB, and the particle follows a circle of radius r = mv/(qB).

Q7. Explain the torque on a current carrying rectangular coil placed in a uniform magnetic field.

Ans. When a rectangular coil of N turns, area A, carrying current I is placed in a uniform magnetic field B, the forces on its two sides perpendicular to the field are equal and opposite but act along different lines, forming a couple, while the forces on the other two sides are along the axis and cancel. The torque of the couple is tau = NIAB sin theta, where theta is the angle between the normal to the coil and the field. It is maximum, NIAB, when the plane of the coil is parallel to the field (theta = 90 degree), and zero when the plane is perpendicular to the field (theta = 0), which is the equilibrium position. This torque is the working principle of electric motors and moving-coil galvanometers.

Q8. Define and explain galvanometer, its construction and working principle.

Ans. A galvanometer is an electrical instrument used to detect small electric currents (and to show their direction). Construction: a rectangular coil of many turns of fine wire is pivoted between the concave pole pieces of a permanent magnet, with a soft-iron core inside the coil to make the field radial; a spring provides the restoring torque and a pointer attached to the coil moves over a scale. Working principle: when current passes through the coil, the magnetic field exerts a torque tau = NIAB on it (the field being

radial, the torque is independent of the deflection angle), which twists the coil against the spring until the spring torque balances it. The deflection is therefore directly proportional to the current, and the scale can be calibrated to read current.

Q9. How can you develop an ammeter and a voltmeter by using a galvanometer?

Ans. An ammeter is developed by connecting a low resistance, called a shunt, in parallel with the galvanometer: most of the current passes through the shunt and only a known small fraction through the coil, so the meter can measure large currents; the shunt value for a range I is $R_s = (I_g R_g)/(I - I_g)$, where I_g and R_g are the galvanometer's full-scale current and resistance. A voltmeter is developed by connecting a high resistance in series with the galvanometer: the high resistance limits the current to the full-scale value for the desired voltage range and drops most of the voltage; the series resistance for a range V is $R = V/I_g - R_g$. The ammeter is connected in series in a circuit and the voltmeter in parallel, and the scales are marked directly in amperes and volts.

Q10. Define and explain an Avometer, its functions and its various parts.

Ans. An Avometer (multimeter, from Ampere-Volt-Ohm meter) is a single instrument that measures current, potential difference and resistance, and often other quantities. Its functions are selected by a range switch: as an ammeter it uses shunt resistances in parallel with the moving-coil meter, as a voltmeter it uses high series resistances, and as an ohmmeter it uses an internal battery with

the meter, the scale being marked in ohms. Its various parts are: the moving-coil meter (or digital display) which shows the reading; the selector switch for function and range; the terminals (common, and separate ones for current and voltage/resistance); the internal battery for resistance measurement; and, in analogue meters, the zero-adjustment knob for the ohmmeter range. The name Avometer comes from its three basic functions: A (ampere), V (volt) and O (ohm).

Exercise - Numerical Problems

Q1. A current of 10 A carrying conductor of length 20 cm is placed in uniform magnetic field 2 T. If the length is at 60 degree to the field, then calculate the magnitude of the force acting on the conductor. (Book answer: 3.5 N)

Force on a current-carrying conductor: F = BIL sin theta, with B = 2 T, I = 10 A, L = 20 cm = 0.2 m and theta = 60 degree.

F = 2 x 10 x 0.2 x sin 60 degree = 4 x 0.866.

F = 3.464 N, about 3.5 N.

The force acting on the conductor is about 3.5 N. This matches the book answer.

Q2. A maximum magnetic force of 0.3 N exerts on a 15 cm long conductor carrying a current of 10 A in a uniform magnetic field. Find the magnitude of the magnetic field. (Book answer: 0.2 T)

Maximum force occurs when the conductor is perpendicular to the

field, so F_max = BIL.

Hence B = F/(IL) = 0.3 / (10 x 0.15).

B = 0.3 / 1.5 = 0.2 T.

The magnetic field is 0.2 T. This matches the book answer.

Q3. Calculate the value of magnetic flux density and magnetic flux within an air core solenoid of radius 2 cm and 4000 turns per unit length carrying a current of 5 A. (Book answers: 2.51 x 10^-2 T, 3.15 x 10^-5 Wb)

Magnetic flux density inside a solenoid: B = mu_o n I = (4 pi x 10^-7) x 4000 x 5.

B = 4 pi x 10^-7 x 20000 = 2.513 x 10^-2 T, about 2.51 x 10^-2 T.

Cross-sectional area A = pi r^2 = 3.14 x (0.02)^2 = 1.257 x 10^-3 m^2. Magnetic flux phi = B A = (2.513 x 10^-2) x (1.257 x 10^-3) = 3.158 x 10^-5 Wb, about 3.15 x 10^-5 Wb.

The flux density is 2.51 x 10^-2 T and the flux is 3.15 x 10^-5 Wb. This matches the book answers.

Q4. A solenoid 10 cm long has 500 turns. Find the magnetic field inside the solenoid if it carries a current of 10 A. (Book answer: 6.28 x 10^-2 T)

Number of turns per unit length n = N/L = 500 / 0.1 m = 5000 turns

per metre.

B = mu_o n I = (4 pi x 10^-7) x 5000 x 10.

B = 4 pi x 10^-7 x 50000 = 6.283 x 10^-2 T, about 6.28 x 10^-2 T.

The magnetic field inside the solenoid is 6.28 x 10^-2 T. This matches the book answer.

Q5. What will be the speed of an electron if it moves at right angle to the magnetic field of 0.2 T and it experiences a magnetic force of 2 x 10^-12 N? (Book answer: 6.25 x 10^17 ms^-1) NOTE: the book's answer is wrong by a factor of ten thousand million - see the solution.

For motion at right angles to the field, F = evB, so v = F/(eB).

v = (2 x 10^-12) / ((1.6 x 10^-19) x 0.2) = (2 x 10^-12) / (3.2 x 10^-20).

v = 6.25 x 10^7 m/s.

The speed of the electron is 6.25 x 10^7 m/s. The book's printed answer of 6.25 x 10^17 m/s is impossible (it exceeds the speed of light by a factor of about two million) and is wrong by a factor of 10^10, evidently a power-of-ten misprint; the corrected value 6.25 x 10^7 m/s is used here and the book answer is flagged.

Q6. A proton enters a magnetic field of flux density 1.5 Wb m^-2

with a velocity of 2 x 10^7 ms^-1 at an angle of 45 degree with the field. Compute the force on the proton. (Book answer: 3.4 x 10^-12 N)

Force on a moving charge: F = qvB sin theta, with q = 1.6 x 10^-19 C, v = 2 x 10^7 m/s, B = 1.5 T and theta = 45 degree.

F = (1.6 x 10^-19) x (2 x 10^7) x 1.5 x sin 45 degree = (4.8 x 10^-12) x 0.707.

F = 3.394 x 10^-12 N, about 3.4 x 10^-12 N.

The force on the proton is about 3.4 x 10^-12 N. This matches the book answer.

Q7. A velocity selector has a magnetic field of 0.3 T. If a perpendicular electric field of 10^4 V m^-1 is applied, what will be the speed of the particle when it passes through the selector? (Book answer: 3.3 x 10^4 ms^-1)

In a velocity selector the particle passes undeflected when the electric force equals the magnetic force: qE = qvB, so v = E/B.

v = 10^4 / 0.3 = 33,333 m/s, about 3.3 x 10^4 m/s.

The speed of the particle is about 3.3 x 10^4 m/s. This matches the book answer.

Q8. A rectangular coil of sides 10 cm and 6 cm having 2000 turns and carrying a current of 5 A is placed in a uniform magnetic field

of 0.5 T. Calculate the maximum torque that experiences by the coil. (Book answer: 30 Nm)
Area of the coil A = 0.1 m x 0.06 m = 0.006 m^2.

Maximum torque (when sin theta = 1) is tau_max = NIAB = 2000 x 5 x 0.006 x 0.5.

tau_max = 2000 x 5 = 10000; 10000 x 0.006 = 60; 60 x 0.5 = 30 Nm.

The maximum torque on the coil is 30 Nm. This matches the book answer.

Q9. Find the value of shunt resistance required to convert a galvanometer into ammeter of range 5 A, where the galvanometer has internal resistance 50 ohm and it gives full deflection with a current of 10 mA. (Book answer: 0.1 ohm)
Full-scale galvanometer current I_g = 10 mA = 0.01 A; the shunt must carry the rest: I_s = I - I_g = 5 - 0.01 = 4.99 A.

The potential difference across the shunt equals that across the galvanometer: I_s R_s = I_g R_g, so R_s = (I_g R_g)/(I - I_g).

R_s = (0.01 x 50) / 4.99 = 0.5 / 4.99 = 0.1002 ohm, about 0.1 ohm.

The shunt resistance required is about 0.1 ohm. This matches the book answer.

Q10. A galvanometer has an internal resistance 40 ohm and deflects full scale for 4 mA current. Calculate the high resistance that should be connected in series with galvanometer, in order to convert the galvanometer into voltmeter of range 100 volt. (Book answer: 24960 ohm)

For a voltmeter of range V, the series resistance is R = V/I_g - R_g, with V = 100 V, I_g = 4 mA = 0.004 A and R_g = 40 ohm.

V/I_g = 100 / 0.004 = 25000 ohm.

R = 25000 - 40 = 24960 ohm.

The high resistance to be connected in series is 24960 ohm. This matches the book answer.

Unit 14: Electromagnetic Induction

Exercise - Multiple Choice Questions (MCQs)

Q1. The source of induced e.m.f. is:

- Cell
- Battery
- Interaction between stationary magnet and coil
- ✓ Relative motion between magnet and coil

Q2. When the coil and a bar magnet are placed very close to each other then the value of their induced e.m.f. will be:

- Maximum
- Positive
- Negative
- ✓ Zero

Q3. The unit of induced e.m.f. is:

- Ampere
- ✓ Volt
- Watt
- Weber

Q4. The law of electromagnetic induction was introduced by:

- Maxwell
- ✓ Faraday
- Lenz

Fleming

Q5. According to Faraday's Law of electromagnetic induction, the induced e.m.f. depends upon:

Minimum magnetic flux

Maximum magnetic flux

Change in magnetic flux

✓ Rate of change of magnetic flux

Q6. Lenz's law of electromagnetic induction explains:

The production of induced e.m.f.

Magnitude of induced e.m.f.

✓ Direction of induced e.m.f.

Both magnitude and direction of induced e.m.f.

Q7. Lenz's law is related to the law of conservation of:

Momentum

✓ Energy

Mass

Charges

Q8. The mutual inductance of two coil is maximum when the coils are:

Parallel to each other

Perpendicular to each other

✓ Facing each other

Touching each other

Q9. When a current change in a coil from 4A to 6A in 0.05s and it induces an e.m.f. of 8V. The co-efficient of self-inductance is:

- 0.1 H
- ✓ 0.2 H
- 0.4 H
- 0.8 H

Q10. The unit of inductance is:

- Weber
- Tesla
- ✓ Henry
- Watt

Q11. A conducting rod of length 0.5 m moves in a magnetic field of magnitude 2 T with velocity 5 m s^-1, the e.m.f. induced in the moving rod is:

- ✓ 5 V
- 10 V
- 20 V
- 50 V

Q12. Which one of the following device stores magnetic potential energy in its magnetic field?

- Resistor
- Capacitor
- ✓ Inductor
- Thermistor

Q13. At constant inductance of 'L' henry, the energy stored in a magnetic field is:

- (1/2) C Q^2
- (1/2) m v^2
- (1/2) k x^2
- ✓ (1/2) L I^2

Q14. The working principle of an A.C. motor is:

- Self-induction
- Mutual induction
- ✓ Motional e.m.f.
- None of those

Q15. The stator and the rotor of an A.C. motor are connected by a:

- Copper Wire
- Silver wire
- Aluminum wire
- ✓ No physical connection between them

Q16. Two phase A.C. contains:

- One wire two stators
- ✓ Two wires two stators
- Three wires two stators
- Three wires three stators

Q17. A transformer works when we apply:

- ✓ A.C. voltage
- D.C. voltage

A.C. as well as D.C

Neither A.C. nor D.C

Q18. The quantity that remains constant in a transformer is:

Voltage

Current

Power

✓ Frequency

Q19. The soft iron core is used in a transformer in order to:

✓ Enhance the magnetic flux

Increase the weight

Increase the copper losses

Decrease the copper losses

Q20. Which relation is true for an ideal transformer?

Input power > output power

Input power < output power

✓ Input power = output power

Output power = 0

Q21. When a step up transformer delivers high voltage then the current will be:

Remain same

✓ Decreased

Increased

Maximum

Q22. For long distance electrical power transmission, we use:

- Smaller current and lower voltage
- Larger current and higher voltage
- Larger current and lower voltage
- ✓ Smaller current and higher voltage

Q23. Lamination is used in a transformer in order to:

- Reduce flux leakage
- Minimize hysteresis loss
- Decrease coil resistance
- ✓ Minimize eddy current

Exercise - Short Questions

Q1. What is the meaning of induced current and induced e.m.f.?

Ans. Induced current is the current that flows in a conductor or coil when the magnetic flux linked with it changes, without any battery being connected to it. Induced e.m.f. is the electromotive force (potential difference) that drives this current: it is set up in the conductor by the changing magnetic flux, and its magnitude is given by Faraday's law. The induced current lasts only while the flux keeps changing; when the flux becomes steady, the induced e.m.f. and current both fall to zero.

Q2. What factor causes of induced e.m.f.?

Ans. Induced e.m.f. is caused by a change in the magnetic flux linked with a conductor or coil. The flux can change because of: (i) relative motion between a magnet and the coil; (ii) a change in the strength of the magnetic field; (iii) a change in the area of the coil

exposed to the field; (iv) a change in the orientation of the coil with respect to the field; or (v) a change of current in a nearby coil. In every case it is the rate of change of flux, not the flux itself, that determines the induced e.m.f.

Q3. Does e.m.f. induce in a coil, when a magnet is placed near to it?

Ans. No. Merely placing a magnet near a coil does not induce an e.m.f., because the magnetic flux through the coil, though present, is not changing. An e.m.f. is induced only while the flux is changing - that is, while the magnet is being moved towards or away from the coil, or while the current producing the field is changing. Once the magnet is held still near the coil, the induced e.m.f. is zero, which is also the answer to MCQ Q2 of this exercise.

Q4. The law of electromagnetic induction is expressed as: epsilon = -N (Delta phi / Delta t). What is the meaning of negative sign?

Ans. The negative sign in Faraday's law expresses Lenz's law: the induced e.m.f. (and the induced current it drives) always acts in such a direction as to oppose the change in magnetic flux that produced it. If the flux through a coil increases, the induced current creates a field opposing the increase; if the flux decreases, the induced current creates a field supporting the original flux. The negative sign is therefore the mathematical statement of opposition, and it is also the reason electromagnetic induction obeys the law of conservation of energy.

Q5. What is the difference between Faraday's law and Lenz's law of

induction?

Ans. Faraday's law gives the magnitude of the induced e.m.f.: it states that the induced e.m.f. in a coil is directly proportional to the rate of change of magnetic flux linked with it, epsilon = -N (Delta phi / Delta t). Lenz's law gives the direction of the induced e.m.f. and current: it states that the induced current always flows in such a direction as to oppose the change of flux that produced it. In short, Faraday's law tells how large the induced e.m.f. is, and Lenz's law tells which way it acts; together they are expressed by the single equation with its negative sign.

Q6. What is the difference between induced e.m.f. and back e.m.f.?

Ans. Induced e.m.f. is the general name for any e.m.f. produced by a changing magnetic flux, as in a generator, a transformer or a coil with changing current. Back e.m.f. is a particular case: it is the e.m.f. induced in the rotating coil (armature) of a motor by its own rotation in the magnetic field, and it opposes the applied voltage that drives the motor. While an ordinary induced e.m.f. is the useful output of a generator, back e.m.f. limits the current in a motor: the running current is I = (V - epsilon_back)/R, so a motor draws a large current at start (when back e.m.f. is zero) and a smaller current when running.

Q7. What is the advantage and disadvantage of eddy current?

Ans. Eddy currents are circulating currents induced in bulk metal pieces by changing magnetic flux. Advantage: they are put to use in induction furnaces (melting metals by the heat of eddy currents), in electromagnetic brakes, in induction cookers and in the damping

of galvanometer coils (dead-beat action). Disadvantage: in transformers, motors and generators they waste energy as heat in the iron cores and reduce efficiency. To minimize this loss, cores are laminated - built up of thin insulated sheets - which break up the paths of the eddy currents and greatly reduce them.

Q8. Distinguish between self-inductance and mutual inductance.

Ans. Self-inductance L of a coil is the property by which a changing current in the coil itself induces an e.m.f. in the same coil, epsilon = -L (Delta I / Delta t); it depends on the coil's own turns, area, length and core, and its unit is the henry. Mutual inductance M between two coils is the property by which a changing current in one coil induces an e.m.f. in a nearby second coil, epsilon = -M (Delta I / Delta t); it depends on the geometry and relative position of the two coils and on their coupling, and its unit is also the henry. Self-induction opposes changes in a coil's own current; mutual induction transfers energy from one circuit to another, as in a transformer.

Q9. Define Henry and express its dimension.

Ans. One henry is the self-inductance (or mutual inductance) of a coil in which an e.m.f. of one volt is induced when the current changes at the rate of one ampere per second: 1 H = 1 V s A^-1. In terms of base units, since V = kg m^2 s^-3 A^-1, the henry is kg m^2 s^-2 A^-2. Equivalently, 1 H = 1 Wb A^-1, since inductance is also flux linkage per unit current.

Q10. How magnetic energy is stored in an inductor?

Ans. When a current is established in an inductor, the source must do work against the back e.m.f. induced by the growing current; this work is not lost but is stored in the magnetic field built up around (and in) the inductor. If the current grows from zero to I, the total work done is W = (1/2) L I^2, which is the magnetic potential energy stored in the inductor. It is returned to the circuit when the current falls: the collapsing field induces an e.m.f. that tries to keep the current flowing. This is why a circuit with an inductor cannot be switched off instantly without a spark.

Q11. Explain the motional e.m.f. due to the motion of rod of length 'l' with velocity 'v' which is perpendicular to the magnetic field B.

Ans. When a conducting rod of length l moves with velocity v perpendicular to a uniform magnetic field B, the free electrons in the rod experience a magnetic force F = evB, which pushes them to one end of the rod. Charge accumulates at the ends until the electric field set up by the separated charge balances the magnetic force, eE = evB, so E = vB. The potential difference between the ends is then epsilon = E l = vBl. More generally, if the rod makes an angle theta with the field, epsilon = vBl sin theta. This e.m.f., produced purely by motion in a field, is called motional e.m.f., and it is the basis of the generator and of the back e.m.f. of a motor.

Q12. What is the working principle of an A.C. generator?

Ans. An A.C. generator works on the principle of electromagnetic induction: when a coil rotates in a uniform magnetic field, the magnetic flux linked with it changes continuously, and an alternating e.m.f. is induced in it. The induced e.m.f. at any instant

is epsilon = NBA omega sin theta, where N is the number of turns, A the coil area, B the field, omega the angular speed and theta the angle between the field and the normal to the coil. It is maximum when the plane of the coil is parallel to the field and zero when perpendicular; the slip rings and carbon brushes carry this alternating e.m.f. to the external circuit, where it reverses direction every half rotation, giving alternating current.

Q13. What is the role of carbon brushes in an A.C. generator?

Ans. Carbon brushes are stationary conducting blocks of carbon that press lightly against the rotating slip rings of an A.C. generator. Their role is to maintain continuous electrical contact between the rotating coil and the external circuit, so that the induced current can flow out to the load without twisting wires. Carbon is used because it is a conductor, it is soft enough not to wear the slip rings, it gives a smooth contact with low friction, and it does not weld to the rings; the two brushes, one on each slip ring, deliver the alternating output to the external circuit.

Q14. Why does the bulb light become dim for a moment when we start the refrigerator or some other heavy electrical appliance at home?

Ans. When a heavy appliance like a refrigerator motor is switched on, it draws a very large starting current, because at the moment of starting the motor is not yet rotating, so there is no back e.m.f. to limit the current (I = V/R only). This large current causes a large voltage drop in the house wiring (V_drop = IR), so the voltage available at the other appliances, like a bulb, falls for a moment

and the bulb goes dim. As the motor speeds up, back e.m.f. builds up, the current falls to its normal running value, the voltage recovers, and the bulb brightens again.

Q15. What is the working principle of a transformer?

Ans. A transformer works on the principle of mutual induction: an alternating current in the primary coil produces a continuously changing magnetic flux, which is carried by the soft-iron core and linked with the secondary coil, inducing an alternating e.m.f. in it. Because the same flux links both coils, the ratio of the voltages equals the ratio of the turns: $V_s/V_p = N_s/N_p$. For an ideal transformer the input power equals the output power, so $V_p I_p = V_s I_s$. It works only on alternating current, because only a changing flux can induce an e.m.f.

Q16. What is the difference between step up and step down transformer?

Ans. A step-up transformer increases the voltage: its secondary coil has more turns than its primary ($N_s > N_p$), so $V_s > V_p$, while the secondary current is smaller than the primary current ($I_s < I_p$) for the same power. A step-down transformer decreases the voltage: its secondary coil has fewer turns than its primary ($N_s < N_p$), so $V_s < V_p$, while the secondary current is larger than the primary current. Step-up transformers are used at power stations to raise voltage for long-distance transmission, and step-down transformers near consumers to bring it down to a safe usable voltage.

Q17. Why does not transformer work on D.C.?

Ans. A transformer does not work on direct current because its action depends on a changing magnetic flux. A steady D.C. in the primary produces a steady magnetic flux in the core, and a steady flux induces no e.m.f. in the secondary (Faraday's law needs a rate of change of flux). The only moment anything is induced is at switch-on or switch-off, when the flux is changing. Moreover, the low resistance of the primary would draw a very large steady D.C., overheating and possibly burning the coil, since there is no back e.m.f. from a changing flux to limit it.

Q18. What is the meaning of an ideal transformer?

Ans. An ideal transformer is a hypothetical transformer with 100% efficiency: it has no energy losses at all, so its input power equals its output power ($V_p I_p = V_s I_s$). This requires: no copper losses (windings of zero resistance), no iron losses (no hysteresis loss and no eddy current loss in the core), no flux leakage (all the flux produced by the primary links the secondary), and no magnetic saturation. Real transformers approach this ideal but always lose some energy, typically achieving 95-99% efficiency.

Q19. Describe the power losses of a transformer?

Ans. The power losses of a transformer are: (i) Copper losses - heat ($I^2 R$) produced in the primary and secondary windings because of their resistance; they vary with the load current. (ii) Hysteresis loss - energy spent in repeatedly magnetizing and demagnetizing the iron core in each cycle of the alternating current; it depends on the core material and the frequency. (iii) Eddy current loss - heat

from eddy currents induced in the core by the changing flux. (iv) Flux leakage loss - some flux produced by the primary does not link the secondary (and vice versa), so not all the energy is transferred. Together these losses make the output power less than the input power and the efficiency less than 100%.

Q20. How can you minimize the power losses of a transformer?
Ans. Transformer losses are minimized as follows: copper losses are reduced by using thick copper wire of low resistance for the windings; hysteresis loss is reduced by using a soft magnetic material (soft iron or silicon steel) with a narrow hysteresis loop for the core; eddy current loss is reduced by laminating the core - building it from thin insulated sheets so that eddy current paths are broken up; and flux leakage is reduced by winding the primary and secondary coils close together on the same core (or one over the other) and by using a closed core shape, so that practically all the flux links both coils.

Exercise - Comprehensive Questions

Q1. What is induced e.m.f.? Explain induced e.m.f. with experimental examples.
Ans. Induced e.m.f. is the electromotive force set up in a conductor or coil when the magnetic flux linked with it changes. Experimental examples: (i) moving a bar magnet towards a coil connected to a galvanometer deflects the needle; moving it away deflects the needle the opposite way; holding it still gives no deflection - the e.m.f. exists only while the flux changes. (ii) Moving a coil towards or away from a stationary magnet gives the same result. (iii) In

Faraday's ring experiment, switching on the current in one coil induces a momentary e.m.f. in a second coil wound on the same ring, and switching off induces a momentary e.m.f. in the opposite direction. (iv) Rotating a coil in a magnetic field (the generator) induces a continuous alternating e.m.f. In every case the magnitude follows Faraday's law and the direction follows Lenz's law.

Q2. State and explain Faraday law of electromagnetic induction with examples.

Ans. Faraday's law states that the induced e.m.f. in a circuit is directly proportional to the rate of change of magnetic flux linked with it: for a coil of N turns, epsilon = -N (Delta phi / Delta t). The negative sign is Lenz's law (opposition). Examples: (i) pushing a magnet into a coil quickly gives a larger deflection of the galvanometer than pushing it slowly, because the flux changes faster. (ii) A coil of more turns gives a larger induced e.m.f. for the same flux change, since epsilon is proportional to N. (iii) In an A.C. generator, rotating the coil faster increases the rate of change of flux and hence the peak e.m.f., epsilon_0 = NBA omega. (iv) In a transformer, the alternating primary current changes the flux continuously, inducing an e.m.f. in the secondary proportional to the rate of change of that flux.

Q3. What do you know about the Lenz's law of electromagnetic induction? How can you determine the direction and conservation of energy by using Lenz's law?

Ans. Lenz's law states that the direction of an induced current is

always such that it opposes the change in magnetic flux that produced it. Direction: when a magnet's north pole approaches a coil, the near face of the coil becomes a north pole (repelling the magnet); when it recedes, the near face becomes a south pole (attracting it) - in each case the induced field opposes the motion. Conservation of energy: if the induced current helped the change instead of opposing it, the magnet would be accelerated without any work being done, creating energy from nothing. Because the induced current opposes the motion, work must be done against this opposition to move the magnet, and that mechanical work is exactly the electrical energy produced - so Lenz's law is the law of conservation of energy applied to electromagnetic induction, which is why Faraday's law carries its negative sign.

Q4. Explain motional e.m.f. and derive its mathematical relation.

Ans. Motional e.m.f. is the e.m.f. induced in a conductor by its motion through a magnetic field. Derivation: consider a rod of length l moving with velocity v perpendicular to a uniform field B. A free charge q in the rod experiences the magnetic force F = qvB, which drives positive charges to one end and negative charges to the other. Separation continues until the electric force from the accumulated charge balances the magnetic force: qE = qvB, so the electric field in the rod is E = vB. The potential difference between the ends is epsilon = E l = vBl. If the rod's velocity makes an angle theta with the field, only the perpendicular component counts, giving epsilon = vBl sin theta. This is the motional e.m.f.; for a conductor of length l it is also obtained from Faraday's law, since the area swept per second is lv and the flux cut per second is Blv.

Q5. State and explain mutual induction, co-efficient of mutual induction and its unit.

Ans. Mutual induction is the phenomenon in which a changing current in one coil induces an e.m.f. in a nearby second coil, because the changing magnetic flux produced by the first coil links the second coil. The coefficient of mutual inductance M is defined by epsilon_s = -M (Delta I_p / Delta t), where epsilon_s is the e.m.f. induced in the secondary and Delta I_p / Delta t is the rate of change of current in the primary. Equivalently, M = (flux linkage in the secondary)/(primary current) = N_s phi_s / I_p. Its SI unit is the henry (H): one henry of mutual inductance means that a current changing at one ampere per second in the primary induces one volt in the secondary. M depends on the number of turns, the geometry, the distance and orientation of the coils, and the core material.

Q6. Define and explain self-induction.

Ans. Self-induction is the phenomenon in which a changing current in a coil induces an e.m.f. in the same coil. The changing current changes the magnetic flux linked with the coil itself, and by Faraday's law an e.m.f. is induced, epsilon = -L (Delta I / Delta t), where L is the coefficient of self-inductance. By Lenz's law this induced e.m.f. opposes the change: when the current is switched on, it opposes its growth (so the current rises gradually), and when it is switched off, it opposes its fall (so the current decays gradually and a spark may appear at the switch). Self-inductance depends on the number of turns, the area and length of the coil and the permeability of its core, and is measured in henries. It is the

electrical analogue of inertia.

Q7. How does magnetic potential energy store in an inductor? Also derive its mathematical relation.

Ans. Magnetic energy is stored in an inductor because work must be done against the induced back e.m.f. to build up the current. Derivation: at an instant when the current is I and growing at rate dI/dt, the back e.m.f. is epsilon = L (dI/dt), and the power supplied by the source is P = epsilon I = L I (dI/dt). The work done in a short time dt is dW = P dt = L I dI. Integrating from current zero to I gives W = integral of L I dI from 0 to I = (1/2) L I^2. This work is stored as magnetic potential energy in the field of the inductor, U = (1/2) L I^2. When the current is reduced, this energy is returned to the circuit, which is why the current cannot fall to zero instantly.

Q8. What is A.C. generator? Explain its construction and working principle.

Ans. An A.C. generator (alternator) is a device that converts mechanical energy into alternating electrical energy by electromagnetic induction. Construction: a rectangular coil (armature) of many turns wound on a soft-iron core, mounted on an axle so it can rotate between the poles of a strong magnet (or between field coils); two slip rings attached to the ends of the coil, rotating with it; and two stationary carbon brushes pressing on the slip rings, connected to the external circuit. Working principle: as the coil rotates, the magnetic flux through it changes sinusoidally, inducing an alternating e.m.f. epsilon = NBA omega sin theta. The e.m.f. is zero when the coil is perpendicular to the field, maximum

when parallel to it, and reverses every half rotation, so the output is alternating current of frequency equal to the rotation frequency.

Q9. What is A.C. motor? Explain its construction and working principle.

Ans. An A.C. motor is a device that converts alternating electrical energy into mechanical energy. Construction: a stator (the stationary part) carrying field windings that produce a rotating magnetic field, and a rotor (the rotating part) - in an induction motor a squirrel-cage of conducting bars, in a synchronous motor a coil or permanent magnet - mounted on a shaft, with no physical electrical connection between stator and rotor. Working principle: the alternating current in the stator produces a rotating magnetic field; this changing field induces currents in the rotor conductors, and the interaction of these currents with the field produces a torque that drags the rotor around with the field. As the rotor turns, its coils cut the field and a back e.m.f. (motional e.m.f.) is induced in them, opposing the applied voltage and limiting the current; at start, when the rotor is still, there is no back e.m.f. and the motor draws a large starting current.

Q10. State and explain back e.m.f. in an A.C motor.

Ans. Back e.m.f. is the e.m.f. induced in the rotating armature (rotor) of a motor by its own rotation in the magnetic field; by Lenz's law it opposes the applied voltage. If V is the applied voltage, epsilon_b the back e.m.f. and R the resistance of the armature circuit, the current is I = (V - epsilon_b)/R. At the moment of starting, the rotor is stationary, so epsilon_b = 0 and the current

is a maximum, I = V/R, which is why a motor draws a heavy starting current and may dim the lights. As the motor speeds up, epsilon_b grows in proportion to the speed, the current falls, and at normal speed the motor draws only the small current needed to supply the load and losses. If the load increases, the motor slows slightly, epsilon_b falls, and the current rises automatically to meet the load.

Q11. Define and explain eddy current.

Ans. Eddy currents are loops of current induced in a bulk piece of conducting material when the magnetic flux through it changes. Because a solid conductor can be thought of as many closed loops, a changing flux induces circulating currents in it, like eddies in water, which dissipate energy as heat (I^2 R loss). They are produced, for example, in transformer cores, motor armatures and metal plates moving in magnetic fields, and their direction follows Lenz's law - they oppose the change that produces them. They are a disadvantage in transformers and motors, where they waste energy, so cores are laminated to break up their paths; but they are useful in induction furnaces, induction cookers, electromagnetic brakes and in damping moving-coil instruments.

Q12. What do you know about a transformer? Describe the function, working and kinds of transformer.

Ans. A transformer is a static electrical device that transfers electrical energy from one circuit to another by mutual induction, changing the voltage level without changing the frequency. Function: it steps alternating voltage up or down for transmission

and use. Working: an alternating current in the primary coil sets up a changing flux in the laminated soft-iron core, which links the secondary coil and induces an alternating e.m.f. in it; V_s/V_p = N_s/N_p, and for an ideal transformer V_p I_p = V_s I_s. Kinds: (i) step-up transformer (N_s > N_p, V_s > V_p), used at generating stations; (ii) step-down transformer (N_s < N_p, V_s < V_p), used for distribution to consumers; and by construction, core-type and shell-type transformers, and single-phase and three-phase transformers.

Exercise - Numerical Problems

Q1. A coil 200 turns linked by a flux of 40 mWb. If this flux is reversed in a time of 4 ms then determine the average induced e.m.f. in the coil. (Book answer: 2000 V) NOTE: the book's answer corresponds to a flux change of 40 mWb, not to a full reversal - see the solution.

Flux linked with the coil changes from +40 mWb to -40 mWb on reversal, so the change in flux is Delta phi = 40 - (-40) = 80 mWb = 80 x 10^-3 Wb = 0.08 Wb.

Time of change Delta t = 4 ms = 4 x 10^-3 s, and number of turns N = 200.

Average induced e.m.f. epsilon = N (Delta phi / Delta t) = 200 x (0.08 / 0.004) = 200 x 20 = 4000 V.

The average induced e.m.f. for a full reversal of the flux is 4000 V. The book's printed answer of 2000 V is obtained only if the flux

change is taken as 40 mWb (that is, if the flux merely falls to zero instead of reversing); since the question states the flux is reversed, the change is 80 mWb and the corrected value 4000 V is used here, with the book answer flagged.

Q2. At what rate would it be necessary for a single conductor loop to cut the flux in order that the current of 1.2 mA flows through it when 10 ohm resistor is connected across its ends? (Book answer: 1.2 x 10^-2 Wb s^-1)

The induced e.m.f. needed to drive the current through the resistor is epsilon = I R = (1.2 x 10^-3 A) x (10 ohm) = 1.2 x 10^-2 V.

For a single loop, Faraday's law gives epsilon = rate of change of flux = Delta phi / Delta t.

Hence the flux must be cut at the rate of 1.2 x 10^-2 Wb per second.

The loop must cut flux at the rate of 1.2 x 10^-2 Wb s^-1. This matches the book answer.

Q3. A 60 m long conductor is moving in a uniform magnetic field 0.9 T at a speed of 7 ms^-1. If the motion of the conductor is at right angle to the field then calculate the e.m.f. induced in the conductor. (Book answer: 378 V)

Motional e.m.f. for motion at right angles to the field is epsilon = vBl.

epsilon = 7 x 0.9 x 60 = 378 V.

The e.m.f. induced in the conductor is 378 V. This matches the book answer.

Q4. When the current in a coil A is increased uniformly from zero to 18 A in 0.3 s, an e.m.f. of 6 V is induced in a nearby coil B. What is the value of the mutual inductance between the coils A and B? (Book answer: 0.1 H)
Rate of change of current in coil A: Delta I / Delta t = 18 / 0.3 = 60 A/s.

Mutual induction gives epsilon = M (Delta I / Delta t), so M = epsilon / (Delta I / Delta t) = 6 / 60.

M = 0.1 H.

The mutual inductance between the coils is 0.1 H. This matches the book answer.

Q5. The current in a coil changes at the rate of 3 A in 60 ms. An e.m.f. of 10 V is induced in the coil. What is the self-inductance of the coil? (Book answer: 0.2 H)
Rate of change of current: Delta I / Delta t = 3 / (60 x 10^-3) = 3 / 0.06 = 50 A/s.

Self-induction gives epsilon = L (Delta I / Delta t), so L = epsilon / (Delta I / Delta t) = 10 / 50.

L = 0.2 H.

The self-inductance of the coil is 0.2 H. This matches the book answer.

Q6. An inductor of inductance 0.5 H carries a current of 6 A. Calculate the energy stored in the magnetic field of the inductor. (Book answer: 9 J)
Energy stored in an inductor: U = (1/2) L I^2.

U = 0.5 x 0.5 x (6)^2 = 0.25 x 36 = 9 J.

The energy stored in the magnetic field is 9 J. This matches the book answer.

Q7. The frequency of the coil of an A.C. generator is 60 Hz has 150 turns. If its area is 150 cm^2 and it induces maximum e.m.f. of 250 V during its rotation in a uniform magnetic field, then calculate the magnitude of such magnetic field. (Book answer: 0.3 mT) NOTE: the book's answer has a unit error - see the solution.
Maximum induced e.m.f. of a generator: epsilon_0 = N A B omega, where omega = 2 pi f = 2 x 3.14 x 60 = 377 rad/s.

Area A = 150 cm^2 = 150 x 10^-4 m^2 = 0.015 m^2.

Hence B = epsilon_0 / (N A omega) = 250 / (150 x 0.015 x 377) = 250 / 848.25 = 0.2947 T, about 0.3 T.

The magnetic field is about 0.3 T. The book's printed answer of 0.3 mT is a unit misprint - it is smaller by a factor of one thousand - so the corrected value 0.3 T is used here and the book answer is flagged.

Q8. An A.C. motor is rotated by applying the potential difference of 220 V. If the motor has a resistance of 4 ohm and a back e.m.f. of 50 V is induced in it during its rotation, then calculate (a) the current when the motor just start up (b) current when the motor is running at normal speed. (Book answers: 55 A, 42.5 A)
(a) At start-up the motor is stationary, so the back e.m.f. is zero and the current is I = V/R = 220/4 = 55 A.

(b) When running, the back e.m.f. of 50 V opposes the applied voltage, so the current is I = (V - epsilon_back)/R = (220 - 50)/4 = 170/4 = 42.5 A.

The current at start-up is 55 A and when running at normal speed it is 42.5 A. This matches the book answers.

Q9. A step-down ideal transformer operates on a 2.5 kV line and supplies a load with 80 A. The ratio of the primary winding to the secondary winding is 20:1. Determine (a) the secondary voltage (b) the primary current (c) the output power. (Book answers: 0.13 kV, 4 A, 10 kW)
(a) For a transformer V_s/V_p = N_s/N_p = 1/20, so V_s = V_p/20 = 2500/20 = 125 V = 0.125 kV, about 0.13 kV.

(b) For an ideal transformer $V_p I_p = V_s I_s$, so $I_p = (V_s I_s)/V_p = (125 \times 80)/2500 = 10000/2500 = 4$ A. (Equivalently $I_p = I_s/20 = 80/20 = 4$ A.)

(c) Output power $P_s = V_s I_s = 125 \times 80 = 10{,}000$ W = 10 kW.

The secondary voltage is about 0.13 kV, the primary current is 4 A and the output power is 10 kW. This matches the book answers.

Q10. A step-down transformer is used on 1 kV line to deliver 220 V. How many turns are on the primary winding if the secondary has 50 turns? (Book answer: 227 turns)

For a transformer $N_p/N_s = V_p/V_s$, so $N_p = N_s \times (V_p/V_s)$.

$N_p = 50 \times (1000/220) = 50 \times 4.545 = 227.27$ turns.

Since turns must be a whole number, the primary has 227 turns.

The primary winding has 227 turns. This matches the book answer.

Unit 15: Alternating Current

Exercise - Multiple Choice Questions (MCQs)

Q1. The mean value of current over one complete cycle of A.C. is

✓ Zero

One

I

I0

Q2. Root mean square value of alternating current is equal to

50% of I0

50.7% of I0

70% of I0

✓ 70.7% of I0

Q3. When the initial phase of A.C. is pi/2 then it will complete its one cycle at the phase of

pi

3pi/2

2pi

✓ 5pi/2

Q4. Average power dissipation in an A.C. circuit across the resistor is

VI

V0I0

✓ VrmsIrms

V^2I^2

Q5. When a capacitor is connected across the A.C. source then

V & I are in phase

I lagging behind V

✓ I leading by V

V leading by I

Q6. When the frequency of A.C. is increased then the reactance of the inductor will

Remain the same

Decreased

✓ Increased

Become negative

Q7. The power dissipation in LC-circuit is

V.I

V0I0

VrmsIrms

✓ Zero

Q8. A capacitor in an A.C. circuit is working at

Any frequency

Low frequency

✓ High frequency

Zero frequency

Q9. The unit of impedance is

- Ampere
- Volt
- Watt
- ✓ Ohm

Q10. According to the phasor diagram, the phase difference between VL and VC is

- pi/2
- 3pi/2
- ✓ pi
- 2pi

Q11. Power factor is equal to

- VI
- VI cos phi
- ✓ cos phi
- tan phi

Q12. In a choke and a resistor series circuit, the A.C. source delivers its maximum power when

- R = 0
- Z = 0
- ✓ R = Z
- R > Z

Q13. The A.C. source transfers its maximum power to load when

- Impedance of source is equal to 0

Impedance of source less than impedance of load

Impedance of source greater than impedance of load

✓ Impedance of source equals to impedance of load

Q14. In RLC-series circuit the amplitude of the current has its maximum value when

XC = 0

XL = 0

Both zero

✓ XL = XC

Q15. In RCL-parallel circuit, the resonance phenomenon will be observed when the net current in the circuit is

Increasing

Decreasing

Reverse

✓ Zero

Q16. The resonance frequency is

f = 2pi sqrt(LC)

f = 2pi/sqrt(LC)

✓ f = 1/(2pi sqrt(LC))

f = sqrt(LC)/(2pi)

Q17. The propagation of electromagnetic waves was predicted by

Hertz

✓ Maxwell

Faraday

Ampere

Q18. The phase difference between electric and magnetic line of forces in electromagnetic waves is

45 degrees

✓ 90 degrees

180 degrees

270 degrees

Q19. The speed of electromagnetic waves in free space is given by the equation

c = 1/sqrt(epsilon mu)

✓ c = 1/sqrt(epsilon0 mu0)

c = 1/sqrt(epsilon/mu)

c = 1/sqrt(epsilon0/mu0)

Q20. The electromagnetic waves do not transport

Energy

✓ Charges

Momentum

Information

Q21. Which one of the following waves has the shortest wavelength?

Radio wave

Microwave

Ultraviolet wave

✓ gamma-rays wave

Q22. A process in which audio signal superimpose on radio frequency wave is called

- Rectifier
- Amplifier
- ✓ Modulation
- Demodulation

Exercise - Short Questions

Q1. Why we use I^2 instead of I while taking the mean value of A.C. cycle?

Over one complete cycle the mean value of the alternating current I itself is zero, because the positive half cycle cancels the negative half cycle. The mean of I^2, however, is not zero, since the square of a negative value is positive. That is why the effective (r.m.s.) value of A.C. is defined through the mean of I^2, not through the mean of I.

Q2. Why 220V of A.C. is more effective than 220V of D.C.?

The 220 V marked on an A.C. supply is its r.m.s. (effective) value, not its peak value. Its peak value is sqrt(2) x 220 V, about 311 V. Because the voltage actually rises to 311 V in each half cycle, 220 V A.C. produces a stronger effect, and a stronger shock, than a steady 220 V D.C. supply.

Q3. At what value, the A.C. and the D.C. become equal?

An alternating current becomes equal in effect to a direct current at its r.m.s. (root mean square) value, also called the effective value.

An A.C. of r.m.s. value Irms produces the same heating effect in a resistor as a D.C. of the same value Irms.

Q4. What is the cause of the current leading by voltage in an A.C. circuit?

In a capacitive A.C. circuit the current leads the voltage by 90 degrees. The cause is the capacitor: current must flow first to build up charge on the plates, so the current reaches its peak a quarter cycle before the voltage across the capacitor reaches its peak.

Q5. What do you know about the impedance in RLC-circuit of A.C.?

The impedance Z of a series RLC circuit is the total opposition it offers to alternating current. It combines resistance R with the net reactance (XL - XC) and is given by Z = sqrt(R^2 + (XL - XC)^2). At resonance XL = XC, so Z = R, its minimum value.

Q6. What will happen if the frequency of A.C. across the inductor is increased?

The inductive reactance XL = omega L = 2 pi f L increases directly with frequency. If the frequency across an inductor is increased, its reactance increases, so for the same applied voltage the current through the inductor decreases.

Q7. What do you know about the power factor of A.C. circuit?

The power factor of an A.C. circuit is the ratio of the true power to the apparent power, and it is equal to cos phi, where phi is the phase angle between voltage and current. For a pure resistor it is 1, for a pure inductor or capacitor it is 0, and for a series RLC circuit it

is R/Z.

Q8. What is a phasor and phasor diagram?

A phasor is a rotating vector that represents a sinusoidally varying quantity such as alternating voltage or current; its length gives the peak value and its angle gives the phase. A phasor diagram is a diagram showing several such phasors together, so that the phase differences between voltage and current in a circuit can be read off at a glance.

Q9. Why a choke cannot control D.C.?

A choke coil controls A.C. because its inductive reactance XL = omega L opposes the changing current without wasting power as heat. For D.C., the frequency is zero, so omega = 0 and XL = 0; the choke then offers only the small resistance of its winding, so it cannot limit a direct current the way it limits an alternating current.

Q10. Under what condition a source transfers its maximum power to a load?

By the maximum power transfer theorem, an A.C. source transfers its maximum power to a load when the impedance of the load equals the impedance of the source. In the form used in this chapter, the source delivers maximum power when the load impedance matches the source impedance.

Q11. What is the condition of the resonance phenomenon in RLC-series circuit?

Resonance in a series RLC circuit occurs when the inductive

reactance equals the capacitive reactance, XL = XC. At that frequency the impedance is minimum and equal to R, and the current amplitude is maximum. The resonance frequency is f = 1/(2 pi sqrt(LC)).

Q12. How does the amplitude of the current becomes zero in LC parallel circuit?

In an ideal parallel LC circuit at resonance, the currents through the inductor and the capacitor are equal in magnitude but opposite in phase, so they cancel each other. The net current drawn from the source therefore becomes zero (in a real circuit it falls to a very small minimum value), and the parallel circuit behaves like a very high impedance.

Q13. What is the working principle of metal detector?

A metal detector works on a resonant LC circuit. Its search coil is part of a circuit tuned to resonance. When a metallic object comes near the coil, eddy currents induced in the metal change the inductance of the coil, which disturbs the resonance. This change is detected and signals the presence of metal.

Q14. How did Maxwell predict the electromagnetic waves?

Maxwell showed from his equations that a changing electric field produces a magnetic field and a changing magnetic field produces an electric field, so the two sustain each other and travel through space as a wave. He calculated the speed of such a wave as 1/sqrt(epsilon0 mu0), about 3 x 10^8 m/s, equal to the speed of light, and predicted that light itself is an electromagnetic wave.

Q15. How does transmitting antenna transmit the electromagnetic waves?

A transmitting antenna is fed with a rapidly oscillating current. The oscillating charges in the antenna produce changing electric and magnetic fields around it, and these fields detach from the antenna and propagate away through space as electromagnetic waves carrying the signal.

Q16. What do you know about the spectrum of electromagnetic waves?

The electromagnetic spectrum is the arrangement of all electromagnetic waves in order of wavelength (or frequency), from the longest radio waves, through microwaves, infrared, visible light, ultraviolet and X-rays, to the shortest gamma rays. All of them travel at the same speed in free space, 3 x 10^8 m/s, and differ only in wavelength and frequency.

Q17. How does the receiving antenna receive the electromagnetic waves?

When an electromagnetic wave reaches a receiving antenna, its oscillating electric field drives the free electrons in the antenna to oscillate with the same frequency. This produces an alternating current in the antenna, which is the received signal; tuning selects the wanted frequency from it.

Q18. How can you calculate the speed of electromagnetic waves?

The speed of electromagnetic waves in free space is calculated

from c = 1/sqrt(epsilon0 mu0), where epsilon0 is the permittivity and mu0 the permeability of free space. Substituting their values gives c = 3 x 10^8 m/s. It can also be found from c = f lambda, the product of frequency and wavelength.

Q19. Distinguish between the process of modulation and demodulation.

Modulation is the process of superimposing a low-frequency audio signal on a high-frequency radio (carrier) wave so that the signal can be transmitted over long distances. Demodulation is the reverse process, carried out in the receiver: the audio signal is separated from the carrier wave so that the original sound can be recovered.

Q20. Distinguish between electrocardiography and electrocardiogram.

Electrocardiography is the instrument (and the technique) that records the voltage pulses associated with the action of the heart. The electrocardiogram is the record itself, the heart pulses traced on paper, which a doctor reads to study the working of the heart.

Exercise - Comprehensive Questions

Q1. Explain alternating current and alternating voltage with their mathematical and graphical representation.

Alternating current (A.C.) is a current whose magnitude changes continuously and whose direction reverses periodically. Alternating voltage varies in the same way. Mathematically they are written as I = I0 sin(omega t) and V = V0 sin(omega t), where I0 and V0 are

the peak values and omega = 2 pi f is the angular frequency. Graphically, both are sine waves against time: starting from zero, rising to a positive peak, falling back through zero to an equal negative peak, and returning to zero in one complete cycle of period T = 1/f.

Q2. What do you know about the instantaneous value, root mean square value and peak value of alternating current and alternating voltage?

The instantaneous value is the value of current or voltage at a particular instant, I = I0 sin(omega t) or V = V0 sin(omega t). The peak value I0 (or V0) is the maximum value reached in a cycle. The root mean square (r.m.s.) value is the effective value: Irms = I0/sqrt(2) = 0.707 I0, and Vrms = V0/sqrt(2) = 0.707 V0. The r.m.s. value is the value of direct current or voltage that would produce the same heating effect in a resistor.

Q3. What is phase of A.C.? Explain phase lag and phase lead between alternating current and voltage with the help of phasor diagram.

The phase of an alternating quantity at any instant is the angle omega t (plus any initial phase) that fixes its position in the cycle; it tells how far the quantity has advanced through its sine wave. If the current reaches its peak after the voltage, the current lags behind the voltage, as in an inductive circuit where current lags by 90 degrees. If the current reaches its peak before the voltage, it leads the voltage, as in a capacitive circuit where current leads by 90 degrees. On a phasor diagram the voltage phasor is drawn as

reference and the current phasor is drawn behind it (lag) or ahead of it (lead) by the phase angle phi.

Q4. Study alternating current through a resistor and calculate power dissipation in the resistor.

When an A.C. voltage V = V0 sin(omega t) is applied across a resistor R, the current I = V/R = I0 sin(omega t) is in phase with the voltage. The instantaneous power is p = V I = V0 I0 sin^2(omega t). Its average over a complete cycle is P = (1/2) V0 I0 = Vrms Irms. Since Vrms = Irms R, this can also be written as P = Irms^2 R = Vrms^2/R. All the power is dissipated as heat in the resistor.

Q5. Discuss A.C. through a capacitor and power dissipation in it.

When an A.C. voltage is applied across a capacitor C, the current is I = I0 sin(omega t + pi/2): it leads the voltage by 90 degrees. The opposition offered is the capacitive reactance XC = 1/(omega C) = 1/(2 pi f C), and I0 = V0/XC. The instantaneous power alternates between positive and negative as energy is stored in the electric field of the capacitor and then returned to the source, so the average power over a complete cycle is zero. An ideal capacitor therefore dissipates no power.

Q6. What are the value of instantaneous current and voltage in an inductor when it is connected across the A.C. source?

When an ideal inductor L is connected across an A.C. source V = V0 sin(omega t), the current is I = I0 sin(omega t - pi/2), where I0 = V0/XL and XL = omega L = 2 pi f L is the inductive reactance. The instantaneous voltage across the inductor is v = V0 sin(omega t)

and the instantaneous current is i = (V0/XL) sin(omega t - pi/2): the current lags the voltage by 90 degrees, and the average power dissipated is zero.

Q7. What is impedance? Calculate the impedance of RCL-series circuit.

Impedance is the total opposition offered by an A.C. circuit to the flow of alternating current; it combines resistance and reactance and is measured in ohms. For a series RLC circuit the voltage across R is in phase with the current, the voltage across L leads the current by 90 degrees, and the voltage across C lags by 90 degrees. Adding these phasors gives the total voltage V = I sqrt(R^2 + (XL - XC)^2), so the impedance is Z = V/I = sqrt(R^2 + (XL - XC)^2), where XL = omega L and XC = 1/(omega C).

Q8. Calculate impedance and phase angle of RC-series circuit.

In a series RC circuit the resistance R and the capacitive reactance XC = 1/(omega C) act at right angles to each other in the phasor diagram. The impedance is therefore Z = sqrt(R^2 + XC^2). The phase angle phi by which the voltage lags the current is given by tan phi = XC/R, so phi = tan^-1(XC/R). The current leads the voltage by this angle phi.

Q9. Discuss the behaviour of A.C. through RL-series circuit.

In a series RL circuit the same current flows through R and L. The voltage across R is in phase with the current, while the voltage across L leads the current by 90 degrees. The total voltage is the phasor sum V = I sqrt(R^2 + XL^2), so the impedance is Z =

sqrt(R^2 + XL^2) with XL = omega L. The current lags the applied voltage by a phase angle phi given by tan phi = XL/R. As frequency increases, XL increases, so Z increases and the current decreases.

Q10. State and explain power in an A.C.-circuit and power factor.

The instantaneous power in an A.C. circuit is p = V I, and its average over a cycle is P = Vrms Irms cos phi, where phi is the phase angle between voltage and current. Vrms Irms is the apparent power, measured in volt-ampere, and P is the true (real) power, measured in watts. The power factor is cos phi = true power/apparent power = R/Z for a series circuit. It is 1 for a pure resistor and 0 for a pure inductor or capacitor, where no net power is consumed.

Q11. What do you know about choke coil, its function and application?

A choke coil is an inductor, usually wound on a laminated iron core, whose power dissipation is ideally zero. Its function is to control the current in an A.C. circuit: its reactance XL = omega L limits the current without wasting energy as heat, unlike a resistor. Its common application is in fluorescent tube circuits, where it limits the current through the tube, and in radio circuits for blocking high-frequency currents while passing low-frequency or direct currents.

Q12. State and explain resonance in RLC-series circuit with the resonance frequency.

Resonance in a series RLC circuit occurs when the inductive

reactance equals the capacitive reactance, XL = XC. At that point the impedance Z = sqrt(R^2 + (XL - XC)^2) becomes minimum and equal to R, so the current amplitude I0 = V0/R is maximum, and the circuit behaves like a pure resistor. Setting omega L = 1/(omega C) gives omega^2 = 1/(LC), so the resonance frequency is f = 1/(2 pi sqrt(LC)).

Q13. How can resonance be observed in LC-parallel circuit? Also calculate the resonance frequency in LC-parallel circuit.

In a parallel LC circuit the inductor and capacitor are connected across the same A.C. source. At resonance their currents are equal in magnitude but opposite in phase, so they cancel and the net current drawn from the source falls to a minimum (zero for an ideal circuit), while the impedance rises to a maximum. Resonance can therefore be observed by the minimum in the supply current, or the maximum in the impedance, as the frequency is varied. The resonance frequency is the same as for the series circuit, f = 1/(2 pi sqrt(LC)).

Q14. Explain metal detector, its working principle and function.

A metal detector is an electronic instrument used to detect metal and metallic objects, including buried metals. It works on a resonant LC circuit: its search coil forms part of a circuit tuned to resonance. When a metallic object comes near the coil, eddy currents induced in the metal change the effective inductance of the coil and disturb the resonance. The detector senses this change and gives an audible or visible signal, so its function is to reveal the presence of metal without digging.

Q15. What are electromagnetic waves? Discuss the process of generation, propagation and reception of electromagnetic waves.

Electromagnetic waves are composed of oscillating electric and magnetic fields at right angles to each other and to the direction of travel, and they do not require any medium for their propagation. They are generated when electric charges oscillate, as in a transmitting antenna fed with a high-frequency alternating current: the oscillating charges produce changing fields that detach and travel away. They propagate through space at c = 3 x 10^8 m/s, with the electric and magnetic fields sustaining each other. They are received when their oscillating fields drive electrons in a receiving antenna into oscillation, producing an alternating current that carries the transmitted signal.

Q16. What do you know about the spectrum of electromagnetic waves? State and explain all classes of electromagnetic waves.

The electromagnetic spectrum is the complete range of electromagnetic waves arranged in order of wavelength and frequency. From the longest wavelength to the shortest they are: radio waves, used in broadcasting and communication; microwaves, used in radar and cooking; infrared waves, felt as heat and used in remote controls; visible light, the small band the eye can see, from red to violet; ultraviolet waves, which cause tanning and are used for sterilisation; X-rays, which penetrate soft tissue and are used in medical imaging; and gamma rays, the shortest and most penetrating, emitted by radioactive nuclei and used in cancer treatment.

Q17. Explain electrocardiography with its working principle and its function.

Electrocardiography is the technique, and the instrument, that records the voltage pulses associated with the action of the heart. Its working principle is that each heartbeat is triggered by a small electrical pulse travelling through the heart muscle; electrodes placed on the skin pick up these pulses, and the instrument amplifies them and traces them on paper as the electrocardiogram. Its function is to help doctors study the rhythm and working of the heart and to detect heart disease.

Exercise - Numerical Problems

Q1. The r.m.s. value of alternating current is 5A and its frequency is 50Hz which flows in a circuit through a resistor. Calculate the peak current and the value of the current after 0.002s. (Book answers: 7A, 4.4A)

Peak current: I0 = sqrt(2) x Irms = 1.414 x 5 A = 7.07 A, about 7 A.

Angular frequency: omega = 2 pi f = 2 x 3.14 x 50 = 314 rad/s.

At t = 0.002 s, the phase is omega t = 314 x 0.002 = 0.628 rad, which is 36 degrees.

Instantaneous current: i = I0 sin(omega t) = 7 x sin 36 degrees = 7 x 0.588 = 4.1 A.

FLAG: the book's printed value 4.4 A comes from using sin(omega

t) = omega t = 0.628 instead of sin 36 degrees = 0.588. The correct value is 4.1 A (4.2 A with the unrounded peak 7.07 A).

The peak current is 7 A and the current after 0.002 s is 4.1 A.

Q2. An alternating current flows through a resistor of 10 ohm and produces heat at the rate of 360W. Calculate the effective value of current and voltage. (Book answers: 6A, 60V)

Power dissipated in the resistor: P = Irms^2 R.

So Irms = sqrt(P/R) = sqrt(360/10) = sqrt(36) = 6 A.

Effective voltage: Vrms = Irms R = 6 x 10 = 60 V.

The effective current is 6 A and the effective voltage is 60 V.

Q3. A capacitor of capacitance 6 microF is connected across the A.C. source of 220V. Calculate the current through the capacitor, if the frequency of the source is 50Hz. (Book answer: 0.4A)

Capacitive reactance: XC = 1/(2 pi f C) = 1/(2 x 3.14 x 50 x 6 x 10^-6).

XC = 1/(1.885 x 10^-3) = 530.5 ohm.

Current: I = V/XC = 220/530.5 = 0.415 A, about 0.4 A.

The current through the capacitor is 0.4 A.

Q4. An inductor of inductance 0.6H is connected across the A.C. source of 220V. Calculate the current through the inductor, if the frequency of the source is 50Hz. (Book answer: 1.2A)

Inductive reactance: XL = 2 pi f L = 2 x 3.14 x 50 x 0.6 = 188.5 ohm.

Current: I = V/XL = 220/188.5 = 1.167 A, about 1.2 A.

The current through the inductor is 1.2 A.

FLAG - SOURCE GAP: printed book page 196, the last page of Unit 15, is ABSENT from the source scan (the scan jumps from printed p195 at PDF page 196 straight to printed p197 at PDF page 197, which holds the displaced printed p251). Any Numerical Problems beyond Q4 that may have appeared on page 196 could not be verified from this scan and are therefore NOT included here; they must not be invented. If a complete scan is obtained, the missing items should be added. All other Unit 15 exercise pages (pp192-195) are present and solved above.

Unit 16: Physics of Solids

Exercise - Multiple Choice Questions (MCQs)

Q1. When the temperature increases, the distance between the molecules of a crystal is

- Decreased
- ✓ Increased
- Same
- Zero

Q2. Which one of the following has low melting point?

- Crystal
- Amorphous
- ✓ Polymer
- Glassy Solid

Q3. Which one of the following is called glassy solid?

- Crystal
- ✓ Amorphous
- Polymer
- Diamond

Q4. Which one of the following is a polymeric solid?

- ✓ Wool
- Glass
- Sodium chloride

Copper

Q5. The applied force which produces a change in a body is called

Elastic force

✓ Deforming force

Electric force

Magnetic force

Q6. The property of a body to attain its original position after the removal of force is called

Plasticity

Rigidity

✓ Elasticity

Resistivity

Q7. Which one of the following without any dimension and unit?

Stress

✓ Strain

Tangential Stress

Young Modulus

Q8. Modulus of elasticity is dimensionally equal to

Strain

Shearing Strain

✓ Stress

Surface tension

Q9. The Young's modulus 'Y' of an elastic body is due to the applied

stress 'S', the energy stored in the body per unit volume is

(1/2).(S/Y)

✓ (1/2).(S^2/Y)

(1/2).(Y/S)

(1/2).(Y^2/S)

Q10. If the applied stress is increased, then according to Hook's law the ratio of stress to strain will be

Increased

Decreased

✓ Constant

Zero

Q11. A substance which is permanently deformed by the applied stress is called

Elastic

✓ Plastic

Ductile

Brittle

Q12. A body which breaks down just crossing the elastic limit is known as

Elastic

Plastic

Ductile

✓ Brittle

Q13. Which one of the following substance is very strong in

compression?

Ductile

✓ Brittle

Monoatomic

Diatomic

Q14. Resistivity range of insulators is between

10^2 (ohm m)^-1 and 10^8 (ohm m)^-1

10^-2 (ohm m)^-1 and 10^-8 (ohm m)^-1

10^-11 (ohm m)^-1 and 10^-19 (ohm m)^-1

✓ 10^+11 (ohm m)^-1 and 10^+19 (ohm m)^-1

Q15. The order of the conductivity of the conductor's is

✓ 10^2 (ohm m)^-1 to 10^8 (ohm m)^-1

10^-2 (ohm m)^-1 to 10^-8 (ohm m)^-1

10^-11 (ohm m)^-1 to 10^-19 (ohm m)^-1

10^+11 (ohm m)^-1 to 10^+19 (ohm m)^-1

Q16. Which one of the following substance has partially filled valence band?

Insulator

✓ Conductor

Semiconductor

Superconductor

Q17. The forbidden energy gap between valance and conduction bands in semiconductor is upto

✓ 1eV

2eV

4eV

6eV

Q18. When the temperature is increased, the conduction in the semiconductor material is

Decrease

✓ Increase

Same

Zero

Q19. Which one of the following does not exhibit the superconductivity?

Mercury

✓ Copper

Aluminum

Zinc

Q20. Which one of the following is not a magnetic substance?

Iron

Nickel

✓ Brass

Cobalt

Q21. Which substance has relative permeability less than 1?

✓ Diamagnetic

Paramagnetic

Ferromagnetic

None of those

Q22. When a diamagnetic material is placed in an external magnetic field, the direction of field induced in the material is

Along the direction of external field

✓ Opposite to the direction external field

Perpendicular to direction of external field

None of these

Q23. The relative permeability of an iron is

0.9999

1.001

10^2

✓ 10^4

Q24. Which substance shows a strong magnetic moment due to the co-operation of its atoms to each other?

Diamagnetic

Paramagnetic

✓ Ferromagnetic

Bismuth

Q25. The temperature at which the ferromagnetic substance becomes paramagnetic substance is known as

Critical temperature

✓ Curie temperature

Absolute temperature

Normal temperature

Q26. When the area of hysteresis loop is small, energy dissipation is

- ✓ Small
- Large
- Uniform
- Zero

Exercise - Short Questions

Q1. Why solids have definite shape and volume?

Solids have a definite shape and volume because their atoms, molecules or ions are held together by strong interatomic forces in fixed positions. The particles cannot move freely past one another; they only vibrate about their mean positions, so the solid keeps both its shape and its volume.

Q2. How atoms, molecules or ions are arranged in a crystal?

In a crystal the atoms, molecules or ions are arranged in a regular, repeating three-dimensional pattern called a crystal lattice. Every particle occupies a fixed lattice site, and the same unit of arrangement repeats throughout the crystal, which is why crystals have flat faces and sharp melting points.

Q3. Why frozen amorphous has same property as that of a liquid?

An amorphous solid has no regular arrangement of its particles; its structure is disordered like that of a liquid. It can therefore be regarded as a liquid frozen in place, and like a liquid it has no sharp melting point but softens gradually over a range of temperature.

Q4. Why amorphous is also called glassy solids?

Amorphous solids are also called glassy solids because glass is the commonest example of an amorphous solid. Like glass, all amorphous solids lack a regular crystal structure and a sharp melting point, and they soften gradually when heated.

Q5. Why polymer is also called poor crystalline?

A polymer is also called poor crystalline because its long chain molecules are only partly arranged in an orderly way. Some regions of the chains line up in a regular pattern while other regions remain disordered, so a polymer is neither fully crystalline nor fully amorphous.

Q6. Is stress different from pressure?

Yes. Pressure is the normal force per unit area exerted by a fluid, and it acts equally in all directions. Stress is the internal restoring force per unit area set up inside a deformed solid, and it can be tensile, compressive or shearing, depending on the direction of the applied force relative to the area.

Q7. Differentiate among tensile, compressive and shear strains.

Tensile strain is the fractional increase in length when a body is pulled, equal to change in length over original length. Compressive strain is the fractional decrease in length (or volume) when a body is squeezed. Shear strain is the angular deformation, the angle through which a layer of the body is displaced sideways relative to the layer below it, when tangential forces are applied.

Q8. Define Hooke's law in terms of stress and strain.

Hooke's law states that within the elastic limit, the stress applied to a body is directly proportional to the strain produced in it. In symbols, stress = (constant) x strain, and the constant of proportionality is the modulus of elasticity of the material.

Q9. What do you know about the co-efficient of elasticity?

The co-efficient of elasticity, or modulus of elasticity, is the ratio of stress to strain within the elastic limit. For a change in length it is Young's modulus Y, for a change in volume it is the bulk modulus K, and for a change in shape it is the shear modulus G. A larger modulus means a stiffer material.

Q10. Differentiate between elastic and plastic limits.

The elastic limit is the maximum stress up to which a body regains its original shape and size completely when the deforming force is removed. Beyond the elastic limit, up to the plastic limit, the body is permanently deformed: it does not return to its original shape even after the force is removed.

Q11. Differentiate yield and breaking points of a solid.

The yield point is the stress beyond which a small increase in stress produces a large increase in strain, so the material begins to flow. The breaking point (fracture point) is the stress at which the material finally breaks apart. Between yield and breaking points the material undergoes permanent, plastic deformation.

Q12. What do you know about the ultimate strength of a solid?

The ultimate strength (ultimate tensile strength) of a solid is the maximum stress the material can bear before it begins to neck down and break. It is the highest point on the stress-strain curve, and it marks the greatest load per unit area the material can withstand.

Q13. Distinguish between brittle and ductile substances.

A brittle substance breaks down just after crossing its elastic limit, with almost no plastic deformation; glass and cast iron are examples. A ductile substance undergoes a large permanent deformation before breaking and can be drawn into wires; copper and aluminium are examples.

Q14. Which is more elastic rubber or iron?

Iron is more elastic than rubber. Elasticity is measured by the modulus of elasticity: a material is more elastic if a given stress produces a smaller strain. Iron has a Young's modulus of about 1.9 x 10^11 Pa, far larger than rubber, so iron regains its shape more strongly and is the more elastic material, even though rubber stretches further.

Q15. Differentiate between valence and conduction bands.

The valence band is the highest energy band that is filled (or partly filled) with electrons at ordinary temperatures; the electrons in it are bound to their atoms. The conduction band is the next higher band, normally empty in insulators and semiconductors; electrons that gain enough energy to reach it are free to move through the

solid and conduct electricity.

Q16. What do you know about the forbidden gap?

The forbidden gap (forbidden energy gap) is the range of energies between the top of the valence band and the bottom of the conduction band in which no electron states exist. An electron can only cross from the valence band to the conduction band if it gains energy at least equal to the gap. The gap is large in insulators, about 1 eV in semiconductors, and zero (overlapping bands) in conductors.

Q17. What is critical temperature?

The critical temperature (in this chapter, the Curie temperature for a ferromagnetic substance) is the temperature above which a ferromagnetic material loses its ferromagnetism and becomes paramagnetic. For superconductors the term is also used for the temperature below which the resistance drops to zero.

Q18. How spin and orbital motions of electrons cause of magnetic moment of an atom?

An electron carries a magnetic moment from two motions: its orbital motion around the nucleus acts like a tiny current loop, and its spin about its own axis gives it an intrinsic magnetic moment. The magnetic moment of an atom is the vector sum of the orbital and spin magnetic moments of all its electrons.

Q19. Why the magnetic moment of diamagnetic material is zero?

In a diamagnetic material the atoms have paired electrons whose

spin and orbital magnetic moments cancel each other, so the resultant magnetic moment of each atom is zero. The material is therefore only very weakly magnetised, in the direction opposite to an applied field, by the induced change in the orbital motion.

Q20. Distinguish between paramagnetic and ferromagnetic materials.

Paramagnetic materials have atoms with a small permanent magnetic moment, but the moments point in random directions, so the material shows only a weak magnetisation in an applied field, and loses it when the field is removed. Ferromagnetic materials have atomic moments that line up together in regions called domains, giving a very strong magnetisation that partly remains after the field is removed.

Q21. What is Curie temperature?

The Curie temperature is the temperature at which a ferromagnetic substance becomes paramagnetic. Above this temperature the thermal motion of the atoms overcomes the alignment of the magnetic domains, so the strong cooperative magnetism disappears and the material behaves like an ordinary paramagnetic substance.

Q22. What do you know about hysteresis and hysteresis loop?

Hysteresis is the lagging of the magnetic flux density B behind the magnetising force H when a ferromagnetic material is taken through a cycle of magnetisation and demagnetisation. The hysteresis loop is the closed curve obtained by plotting B against H

over one complete cycle; its shape shows how much magnetisation remains (retentivity) and how large a reverse field (coercive force) is needed to remove it.

Q23. What is hysteresis loss?

Hysteresis loss is the energy wasted as heat in a ferromagnetic material in each cycle of magnetisation, because the magnetic domains must be repeatedly turned against friction-like resistance. It is proportional to the area of the hysteresis loop, so materials with a small loop area, like soft iron, have a small hysteresis loss.

Q24. What do you know about domains?

Domains are small regions inside a ferromagnetic material within which the magnetic moments of the atoms are all aligned in the same direction. In an unmagnetised specimen the domains point in different directions and cancel out; applying a magnetic field makes favourably oriented domains grow at the expense of others, producing the strong overall magnetisation.

Q25. What is residual flux density?

Residual flux density (retentivity) is the magnetic flux density B that remains in a ferromagnetic material after the magnetising force H has been reduced to zero. It is the intercept of the hysteresis loop on the B axis, and it measures how strongly the material stays magnetised on its own.

Q26. What do you know about the coercive force?

The coercive force (coercivity) is the reverse magnetising force

needed to reduce the residual flux density of a magnetised ferromagnetic material to zero. It is the intercept of the hysteresis loop on the H axis. A large coercive force means a hard magnetic material, suitable for permanent magnets; a small one means a soft magnetic material.

Exercise - Comprehensive Questions

Q1. Discuss the three classes of solids: crystal, amorphous and polymeric with examples.

Solids are classified by the arrangement of their particles into three classes. Crystalline solids have a regular, repeating arrangement of atoms, molecules or ions; examples are sodium chloride, diamond and copper, and they have sharp melting points. Amorphous solids have a disordered arrangement, like a frozen liquid; examples are glass and pitch, and they soften over a range of temperature instead of melting sharply. Polymeric solids consist of long chain molecules, partly ordered and partly disordered; examples are wool, rubber and plastics, and they are poor crystalline solids with generally low melting points.

Q2. What do you know about the mechanical properties of solids? Explain it with examples.

The mechanical properties of solids describe how they respond to applied forces. Elasticity is the property of regaining the original shape when the force is removed, as in a steel spring. Plasticity is permanent deformation, as in a bent copper wire. Ductility is the ability to be drawn into wires, as in copper; brittleness is breaking without appreciable deformation, as in glass. Strength is the ability

to bear large loads, as in a steel column, and hardness is resistance to scratching, as in diamond. These properties follow from the strength of the bonds between the particles and their arrangement.

Q3. State and explain stress, strain and their kinds with suitable examples.

Stress is the internal restoring force per unit area set up in a body when it is deformed, measured in N/m^2 (pascal). Tensile stress acts when a body is pulled, as in a stretched wire; compressive stress when it is squeezed, as in a loaded column; shearing (tangential) stress when opposite tangential forces act on its faces, as in a rivet being cut. Strain is the fractional deformation produced, and it has no unit. Tensile strain is change in length over original length; compressive strain is the corresponding decrease; volumetric strain is change in volume over original volume; shear strain is the angle of shear, the sideways displacement of one layer relative to another.

Q4. Define Hooke's law in terms of stress and strain. Also discuss the three kinds of elastic moduli.

Hooke's law states that within the elastic limit, stress is directly proportional to strain: stress = modulus x strain. The three moduli are: Young's modulus Y = tensile stress / tensile strain, for a change in length; bulk modulus K = volumetric stress / volumetric strain, for a change in volume under uniform pressure; and shear modulus G = shearing stress / shear strain, for a change in shape. Each modulus is measured in pascal and is a property of the material, not of the size of the specimen.

Q5. Explain graphical representation of Hooke's law and define the terms: Elastic limit, plastic limit, yield strength and ultimate strength.

The stress-strain graph for a ductile material rises as a straight line from the origin (Hooke's law region) and then bends. The elastic limit is the greatest stress up to which the material returns fully to its original shape when unloaded. Beyond it lies the plastic region, where deformation is permanent; the plastic limit marks the end of fully recoverable behaviour. The yield strength is the stress at which a small increase in stress causes a large increase in strain, so the material begins to flow. The ultimate strength is the maximum stress on the curve, the greatest load per unit area the material can bear before necking and fracture.

Q6. What is meant by the strain energy? Derive the mathematical relation for strain energy.

Strain energy is the potential energy stored in a body when it is deformed within the elastic limit; it is released when the deforming force is removed. For a wire stretched by a force F producing an extension x, the average force during stretching is F/2, so the work done, and hence the strain energy, is (1/2) F x. Writing F = stress x area and x = strain x length gives strain energy = (1/2) x stress x strain x volume. The strain energy per unit volume is therefore (1/2) x stress x strain, which for a linear material equals (1/2) x stress^2 / Young's modulus.

Q7. What is band theory of solids? Explain insulator, conductor and

semiconductor by using band theory of solid.

Band theory says that when many identical atoms combine in a solid, their individual energy levels overlap and spread into allowed energy bands separated by forbidden gaps. In an insulator the valence band is full and the forbidden gap to the empty conduction band is large (several eV), so electrons cannot reach the conduction band and no current flows. In a conductor the valence band is partly filled, or it overlaps the conduction band, so electrons move freely and conduct. In a semiconductor the gap is small, about 1 eV, so at ordinary temperatures a few electrons gain enough thermal energy to cross into the conduction band, giving a conductivity between that of insulators and conductors, which increases with temperature.

Q8. Define superconductor and explain graphical representation of superconductivity.

A superconductor is a conductor whose resistance drops to zero when it is cooled below its critical temperature. Graphically, the resistance of an ordinary conductor falls gradually as temperature falls, but for a superconductor the resistance-versus-temperature curve drops suddenly to zero at the critical temperature and stays zero below it. Below this temperature the material also expels magnetic fields from its interior (the Meissner effect). Mercury was the first superconductor discovered, with a critical temperature of about 4.2 K.

Q9. Discuss magnetic properties of solids and explain the three classes of solids diamagnetic, paramagnetic and ferromagnetic.

The magnetic properties of solids come from the magnetic moments of their atoms, produced by electron spin and orbital motion. Diamagnetic solids have atoms whose resultant magnetic moment is zero, because paired electrons cancel; they are weakly repelled by a magnetic field, and their relative permeability is slightly less than 1, as in bismuth and copper. Paramagnetic solids have atoms with a small permanent moment, randomly oriented, so they are weakly attracted to a field, with relative permeability slightly greater than 1, as in aluminium and platinum. Ferromagnetic solids have atomic moments that co-operate in domains and align strongly with a field, giving a very large permeability and a permanent magnetisation, as in iron, nickel and cobalt.

Q10. State and explain hysteresis and hysteresis loop.

Hysteresis is the lagging of the magnetic flux density B behind the magnetising force H in a ferromagnetic material during a cycle of magnetisation. Starting from an unmagnetised state, B rises with H to saturation; when H is reduced to zero, B does not return to zero but keeps a residual value (retentivity); a reverse coercive force is needed to bring B to zero, and continuing the reverse field saturates the material in the opposite direction. Repeating the cycle traces a closed curve, the hysteresis loop. The area of the loop equals the energy lost as heat per unit volume per cycle (hysteresis loss).

Exercise - Numerical Problems

Q1. An iron rod 4m long and 0.5cm^2 in cross section area

stretches by 0.5mm when a mass of 115kg is hung from its lower end. Compute stress, strain and Young's modulus. (Book answers: 2.254 x 10^11 Pa, 1.25 x 10^-4, 1.8 x 10^11 Pa)
Force applied: F = mg = 115 x 9.8 = 1127 N.

Cross section area: A = 0.5 cm^2 = 0.5 x 10^-4 m^2 = 5 x 10^-5 m^2.

Stress = F/A = 1127/(5 x 10^-5) = 2.254 x 10^7 Pa.

FLAG: the book prints the stress as 2.254 x 10^11 Pa, which is 10^4 times too large; the value from its own data is 2.254 x 10^7 Pa, and the book's own Young's modulus below confirms this.

Strain = change in length / original length = 0.5 x 10^-3 / 4 = 1.25 x 10^-4.

Young's modulus Y = stress/strain = (2.254 x 10^7)/(1.25 x 10^-4) = 1.8 x 10^11 Pa.

The stress is 2.254 x 10^7 Pa, the strain is 1.25 x 10^-4 and Young's modulus is 1.8 x 10^11 Pa.

Q2. Calculate change in volume of a copper cube 40mm on each edge, when subjected to a pressure of 2MPa. The bulk modulus for copper is 125GPa. (Book answer: 1 x 10^-9 m^3 = 1 mm^3)
Original volume: V = (40 x 10^-3)^3 = 6.4 x 10^-5 m^3.

Bulk modulus K = volumetric stress / volumetric strain, so change in volume = V x pressure / K.

Change in volume = (6.4 x 10^-5 x 2 x 10^6)/(125 x 10^9) = (6.4 x 10^-5) x (1.6 x 10^-5) = 1.024 x 10^-9 m^3, about 1 x 10^-9 m^3.

Since 1 mm^3 = 10^-9 m^3, this is 1 mm^3.

The volume decreases by 1 x 10^-9 m^3, that is, by 1 mm^3.

Q3. A metallic wire of length 15m is stretched 6 x 10^-9 m by the applied stress of 5 x 10^8 Nm^-2. Calculate the strain energy per unit volume in the wire. (Book answer: 0.1J)
Strain = change in length / original length = (6 x 10^-9)/15 = 4 x 10^-10.

Strain energy per unit volume = (1/2) x stress x strain = 0.5 x (5 x 10^8) x (4 x 10^-10) = 0.1 J/m^3.

The book prints 0.1J; since the question asks per unit volume, the unit is J/m^3.

The strain energy per unit volume is 0.1 J/m^3.

Q4. A solid cylindrical steel column is 6m long and 10cm is diameter. What will be its increase in two length when carrying a load of 2 x 10^5 kg (Young Modulus for steel is 1.9 x 10^11 Pa)

(Book answer: 7.88mm)

Load: F = mg = 2 x 10^5 x 9.8 = 1.96 x 10^6 N.

Cross section area: A = pi r^2 = 3.14 x (0.05)^2 = 7.854 x 10^-3 m^2.

Increase in length = F L/(A Y) = (1.96 x 10^6 x 6)/(7.854 x 10^-3 x 1.9 x 10^11).

Increase in length = (1.176 x 10^7)/(1.4923 x 10^9) = 7.88 x 10^-3 m = 7.88 mm.

The column increases in length by 7.88 mm.

Q5. Two parallel and opposite forces, each 4000N are applied tangentially to the upper and lower faces of cubical metal block 25cm on a side. Find the angle of shear. (Shearing modulus is 80GPa) (Book answer: 8 x 10^-5 rad.)

Area of a face: A = 0.25 x 0.25 = 0.0625 m^2.

Shearing stress = F/A = 4000/0.0625 = 6.4 x 10^4 Pa.

Angle of shear = shearing stress / shearing modulus = (6.4 x 10^4)/(80 x 10^9) = 8 x 10^-7 rad.

FLAG: the book's printed answer 8 x 10^-5 rad is 100 times too large; the value from its own data is 8 x 10^-7 rad.

The angle of shear is 8 x 10^-7 rad.

Q6. Calculate change in volume of a solid cube 50mm on each edge, when subjected to a stress of 25MPa. (The bulk modulus for copper is 125GPa) (Book answer: 25 mm^3)

Original volume: V = (50 x 10^-3)^3 = 1.25 x 10^-4 m^3.

Change in volume = V x stress / K = (1.25 x 10^-4 x 25 x 10^6)/(125 x 10^9) = (1.25 x 10^-4) x (2 x 10^-4) = 2.5 x 10^-8 m^3.

Since 1 mm^3 = 10^-9 m^3, 2.5 x 10^-8 m^3 = 25 mm^3.

The volume changes by 25 mm^3.

Unit 17: Electronics

Exercise - Multiple Choice Questions (MCQs)

Q1. A germanium at 0K is

- Conductor
- Semiconductor
- ✓ Insulator
- P&N types semiconductor

Q2. By increasing the temperature, the conductivity of the semiconductor is

- Decreased
- ✓ Increased
- Constant
- Not affected

Q3. Which one of the following is an intrinsic semiconductor?

- Boron
- Copper
- Indium
- ✓ Silicon

Q4. In extrinsic semiconductor, the doping level should be in the ratio

- 1:10^-6
- ✓ 1:10^6

1:10^-8

1:10^8

Q5. In P-type semiconductor, the doping of Silicon is taken place with

Bivalent

✓ Trivalent

Tetravalent

Pentavalent

Q6. When P-type and N-type semiconductor are combined then we have a

Resistor

Capacitor

✓ Diode

Triode

Q7. The depletion region in PN-Junction carries

Positively charge

Negatively charge

Both positively and negatively charge

✓ No mobile charge carrier

Q8. The potential difference across the silicon PN-junction is

0.3V

0.5V

✓ 0.7V

0.9V

Q9. When P-type of the PN-Junction is connected with the negative terminal and N-type with positive terminal of the battery, the width of the depletion region

- decreases
- ✓ increases
- remain constant
- vanished

Q10. In forward biasing, the knee voltage for Germanium PN-junction is

- ✓ 0.3V
- 0.7V
- 3V
- 7V

Q11. One of the most important property of a diode

- ✓ Allow current in one direction
- Allow current in bidirection
- Use as an amplifier
- Use as an oscillator

Q12. The number of diodes used in a bridge rectifier is

- One
- Two
- Three
- ✓ Four

Q13. Transistor stands for

✓ Transfer of Resistance

Transfer of charge carriers

Transfer of power

Transfer of voltage

Q14. Which region of the transistor is most wide in terms of area?

Emitter

Base

✓ Collector

All have equal area

Q15. In biasing of a transistor, which region must be reverse biased?

Emitter-Base

✓ Collector-Base

Emitter-Collector

None of these

Q16. The thickness of the base region of a transistor is

10^-2 m

10^-4 m

✓ 10^-6 m

10^-8 m

Q17. Which one of the following relation is true for a transistor?

IE = IC = IB

IE = IB = IC

✓ alpha = IC/IE

beta = IB/IE

Q18. Which one of the following relations holds for cutoff mode of transistor?

IC = IB

IC = beta IB

IC = alpha IB

✓ IC = 0

Q19. The flow of charge carriers in a transistor is controlled by

Emitter

✓ Base

Collector

All of them

Q20. If the base current of a silicon transistor is 0.01mA, its collector current is

0.99mA

9.9 microA

0.1mA

✓ 1mA

Q21. VCE drops across the load resistor of a transistor when

IC = 0

IC = IB

IC = alpha IB

✓ IC = beta IB

Q22. When a transistor starts its working in its active region

- Before its saturation
- After its saturation
- ✓ After its cutoff
- At the end of its cutoff

Q23. A transistor is used as switch when it operates in its

- Normal mode
- Active mode
- Breakdown mode
- ✓ Saturation and cutoff mode

Exercise - Short Questions

Q1. Distinguish between intrinsic and extrinsic semiconductors?

An intrinsic semiconductor is an extremely pure semiconductor, such as pure silicon or germanium, in which the charge carriers are produced only by thermal breaking of covalent bonds, giving equal numbers of electrons and holes. An extrinsic semiconductor is a pure semiconductor to which a small, controlled amount of impurity (doping) has been added, greatly increasing either the number of free electrons (N-type) or the number of holes (P-type).

Q2. What are the advantages of extrinsic semiconductor over the intrinsic semiconductor?

Extrinsic semiconductors have a much higher conductivity than intrinsic semiconductors because doping supplies a large number of free charge carriers. Their conductivity can also be controlled

precisely by the amount and type of impurity added, which is what makes diodes, transistors and integrated circuits possible.

Q3. Under what conditions, the semiconductor materials behave as conductor and as insulator?

At very low temperature (0 K) a semiconductor behaves as an insulator, because no covalent bonds are broken and no free charge carriers are available. At higher temperatures, or when doped, more bonds break (or impurity carriers are supplied), so free electrons and holes become available and the material behaves as a conductor.

Q4. What is meant by the process of doping?

Doping is the process of adding a small, controlled amount of impurity atoms, such as trivalent or pentavalent atoms, to a pure (intrinsic) semiconductor crystal. The impurity atoms take the place of some semiconductor atoms in the crystal and supply extra free electrons or holes, greatly increasing the conductivity of the material.

Q5. What are the facts about holes?

A hole is the vacancy left in a covalent bond when an electron breaks free; it behaves like a positively charged particle. Holes move through the crystal in the direction opposite to the electrons, as neighbouring electrons successively fill the vacancy. In a P-type semiconductor holes are the majority charge carriers, and the current due to holes is in the same direction as the conventional current.

Q6. What do you know about the depletion region in PN-Junction?

The depletion region is the thin layer on both sides of a PN junction that is depleted of mobile charge carriers. When P-type and N-type materials are joined, electrons diffuse from the N side to the P side and holes diffuse the other way, leaving behind uncovered positive ions on the N side and negative ions on the P side. This layer of fixed ions carries no mobile charge and sets up a potential barrier across the junction.

Q7. What is the diffusion in the PN-Junction?

Diffusion in a PN junction is the movement of charge carriers from a region of higher concentration to a region of lower concentration: electrons diffuse from the N-type region, where they are plentiful, into the P-type region, and holes diffuse from the P-type region into the N-type region. This diffusion creates the depletion region and continues until the electric field of the uncovered ions balances it.

Q8. How can the conductivity of semiconductor be increased by increasing the temperature?

Increasing the temperature breaks more covalent bonds in the semiconductor crystal, producing more electron-hole pairs. With more free charge carriers available, the conductivity of the semiconductor increases. This is the opposite of metals, whose conductivity decreases with temperature.

Q9. How does the width of depletion region in PN-Junction increases and decreases?

In forward bias (P-type connected to the positive terminal, N-type to the negative), the applied field pushes carriers towards the junction, narrowing the depletion region and lowering the barrier. In reverse bias (P-type to the negative terminal, N-type to the positive), carriers are pulled away from the junction, widening the depletion region and raising the barrier, so almost no current flows.

Q10. How does a diode convert A.C. into D.C.?

A diode conducts current in one direction only: it offers very low resistance in forward bias and very high resistance in reverse bias. When A.C. is applied, the diode conducts during the half cycle that forward-biases it and blocks the half cycle that reverse-biases it, so the output is a pulsating direct current. This process is called rectification.

Q11. What is the rectification of a diode?

Rectification is the process of converting alternating current into direct current using a diode. Because a diode allows current to flow in one direction only, it passes one half of the A.C. wave and blocks the other, giving a unidirectional (pulsating D.C.) output.

Q12. Distinguish between half and full wave rectifications.

In half-wave rectification a single diode is used, so only one half cycle of the input A.C. appears at the output and the other half is lost; the output is pulsating D.C. with gaps. In full-wave rectification (two diodes with a centre-tapped transformer, or four diodes in a bridge) both half cycles appear at the output in the same direction, giving a smoother output with no gaps and twice the average

current.

Q13. What is the biasing of a diode?

Biasing of a diode means applying an external voltage across its PN junction. Forward biasing connects the P-type region to the positive terminal and the N-type region to the negative terminal, narrowing the depletion region and allowing current to flow. Reverse biasing connects them the other way, widening the depletion region so that almost no current flows.

Q14. Distinguish between forward and reverse biased characteristics of a diode.

In forward bias the depletion region narrows, the barrier falls, and once the applied voltage exceeds the knee voltage (0.3 V for germanium, 0.7 V for silicon) a large current flows, rising steeply with voltage. In reverse bias the depletion region widens, only a very small reverse saturation current flows, and the current stays almost constant until the breakdown voltage is reached, where it rises suddenly.

Q15. What is meaning of a transistor?

A transistor is an electronic device consisting of two PN junctions with three electrodes named emitter, base and collector. The name comes from 'transfer of resistance': a small change in the low-resistance input (emitter-base) circuit controls a large change in the high-resistance output (collector-base) circuit, so the device can amplify signals or act as a switch.

Q16. How does a transistor operate?

A transistor operates only when its emitter-base junction is forward biased and its collector-base junction is reverse biased. The forward-biased emitter injects a large number of carriers into the thin, lightly doped base; most of them cross the base and are collected by the reverse-biased collector, giving a collector current much larger than the small base current that controls it.

Q17. How does the biasing of a transistor take place?

A transistor is biased by connecting batteries so that the emitter-base junction is forward biased and the collector-base junction is reverse biased. For an NPN transistor the emitter is made negative with respect to the base, and the collector positive with respect to the base, so that electrons flow from emitter through the base to the collector.

Q18. Why is the base region of a transistor thin and lightly doped?

The base is made thin and lightly doped so that most of the charge carriers injected by the emitter pass straight through it into the collector instead of recombining with carriers in the base. If the base were thick or heavily doped, most carriers would recombine there, the base current would be large, and the transistor would give little current gain.

Q19. Why is the collector of a transistor anode wider than the emitter and the base?

The collector region is made wider (larger in area) than the emitter and the base because it has to collect all the charge carriers

coming through the base and to dissipate the heat produced there. The collector-base junction also has to withstand a larger reverse voltage, and the larger area helps it do so.

Q20. Why the input resistance of a transistor is low while its output resistance is high?

The input (emitter-base) circuit is forward biased, so a small change in input voltage produces a large change in input current: its resistance is low. The output (collector-base) circuit is reverse biased, so a large change in output voltage produces only a small change in collector current: its resistance is high. This transfer from low to high resistance is the basis of amplification.

Q21. Why the current gain alpha is less than beta?

The current gain alpha = IC/IE is always less than 1, because the collector current is slightly smaller than the emitter current (a small part becomes base current). The current gain beta = IC/IB is much larger, typically 100 or more, because the base current is very small compared with the collector current. Hence alpha is always less than beta.

Q22. Why a common emitter configuration transistor is mostly used?

The common emitter configuration is mostly used because it gives both a high current gain and a high voltage gain, so its power gain is the largest of the three configurations. It also has a moderate input resistance and a moderate output resistance, and it reverses the phase of the signal by 180 degrees, which suits most amplifier

circuits.

Q23. Under what conditions a transistor acts as an open and a closed switch?

A transistor acts as an open switch in the cutoff region: the emitter-base junction is not forward biased, so the base current and the collector current are both zero, and no current flows through the load. It acts as a closed switch in the saturation region: the emitter-base junction is forward biased so strongly that the collector current reaches its maximum, and current flows freely through the load.

Q24. What do you know about the saturation, cutoff and active regions of a transistor?

The cutoff region is the state in which both junctions are effectively off: base current and collector current are zero, and the transistor acts as an open switch. The saturation region is the state in which the transistor is fully on: the collector current is at its maximum and the transistor acts as a closed switch. The active region is the state in between, with the emitter-base junction forward biased and the collector-base junction reverse biased, in which the collector current is proportional to the base current and the transistor works as an amplifier.

Exercise - Comprehensive Questions

Q1. State and explain intrinsic and extrinsic semiconductors.

An intrinsic semiconductor is an extremely pure semiconductor, such as pure silicon or germanium. Its charge carriers come only

from the thermal breaking of covalent bonds, which produces equal numbers of free electrons and holes; its conductivity is low and increases with temperature. An extrinsic semiconductor is made by doping an intrinsic semiconductor with a small amount of impurity. Doping with pentavalent atoms gives an N-type semiconductor, in which electrons are the majority carriers; doping with trivalent atoms gives a P-type semiconductor, in which holes are the majority carriers. Extrinsic semiconductors have a much higher, controllable conductivity.

Q2. What do you know about the N-type and P-type semiconductors? Explain the development of N-type and P-type crystals.

An N-type semiconductor is formed by doping a pure silicon or germanium crystal with pentavalent impurity atoms, such as arsenic or phosphorus. Each impurity atom uses four of its five valence electrons in covalent bonds with neighbouring atoms, leaving one free electron, so electrons are the majority carriers. A P-type semiconductor is formed by doping with trivalent atoms, such as boron or indium. Each impurity atom can form only three complete bonds, leaving a vacancy or hole in the fourth bond, so holes are the majority carriers.

Q3. What is PN junction? How can you form a PN junction?

A PN junction is formed by joining N-type and P-type semiconductor materials in the form of a single crystal. It can be formed by growing a crystal so that one part is doped with pentavalent impurity and the adjoining part with trivalent impurity, or by

diffusing a trivalent impurity into one side of an N-type crystal. At the junction a depletion region and a potential barrier are set up, which give the junction its one-way conducting property.

Q4. Define biasing of PN junction and discuss forward biased and reverse biased of a PN junction.

Biasing of a PN junction means applying an external potential difference across it. In forward bias the P-type region is connected to the positive terminal and the N-type region to the negative terminal of the battery: the depletion region narrows, the barrier falls, and once the knee voltage is passed a large current flows. In reverse bias the connections are reversed: the depletion region widens, the barrier rises, and only a very small reverse saturation current flows until breakdown is reached.

Q5. What is rectification? How can a diode be used as a rectifier? Explain half-wave rectification.

Rectification is the conversion of alternating current into direct current. A diode can be used as a rectifier because it conducts in one direction only. In half-wave rectification a single diode is placed in series with the A.C. source and the load. During the half cycle that forward-biases the diode, current flows through the load; during the other half cycle the diode is reverse biased and blocks the current. The output is therefore a pulsating direct current consisting of one half cycle in every full cycle.

Q6. State and explain full wave bridge rectification by using four diodes.

In full-wave bridge rectification four diodes are connected in a bridge across the A.C. source, with the load connected across the other two corners of the bridge. In each half cycle of the input, a different pair of diodes is forward biased, and the current always flows through the load in the same direction. Both half cycles of the input therefore appear at the output, giving a continuous pulsating direct current with twice the average value of half-wave rectification and no gaps.

Q7. What is transistor? Explain the operation of NPN transistor?

A transistor is an electronic device which consists of two PN junctions with three electrodes named emitter, base and collector; its name comes from 'transfer of resistance'. In an NPN transistor the emitter-base junction is forward biased and the collector-base junction is reverse biased. The emitter injects electrons into the thin, lightly doped base; most of them cross the base without recombining and are pulled into the collector by the reverse-biased collector-base junction. A small base current thus controls a much larger collector current, giving current gain beta = IC/IB.

Q8. Discuss the three configurations of a transistor.

The three configurations are named by the terminal common to input and output. In the common base configuration the base is common; it gives a current gain alpha slightly less than 1, a low input resistance and a high output resistance. In the common emitter configuration the emitter is common; it gives a high current gain beta and a high voltage gain, and is the most used. In the common collector configuration the collector is common; it gives a

current gain near beta, a high input resistance and a low output resistance, and is used for impedance matching.

Q9. What is amplification? Explain the amplification of common emitter NPN transistor.

Amplification is the process by which a small input A.C. signal is converted into a large output A.C. signal using a transistor. In a common emitter NPN amplifier the input signal is applied to the forward-biased emitter-base circuit, where it produces a small change in base current. This produces a much larger change in collector current, beta times as large, which flows through the load resistor and produces a large output voltage across it. The voltage gain is beta times the ratio of output resistance to input resistance.

Q10. How can transistor be used as an automatic switch?

A transistor can be used as an automatic switch by operating it only in its cutoff and saturation regions. In cutoff the base-emitter junction is not forward biased, the base and collector currents are zero, and the transistor behaves as an open switch, so the load (for example a bulb) is OFF. In saturation the base-emitter junction is forward biased, the collector current is at its maximum, and the transistor behaves as a closed switch, so the load is ON. A small base signal can therefore switch a larger load current on and off automatically.

Exercise - Numerical Problems

Q1. Calculate the resistance across the PN-junction silicon diode that has 0.5A of current through it with a 0.8V drop across the two

terminals of the diode. (Book answer: 1.6 ohm)
Resistance = voltage / current = 0.8/0.5 = 1.6 ohm.

The resistance across the diode is 1.6 ohm.

Q2. How much is the emitter current with 100mA for the collector current and 800 microA for base current? (Book answer: 100.8mA)
Emitter current = collector current + base current: IE = IC + IB.

IB = 800 microA = 0.8 mA.

IE = 100 + 0.8 = 100.8 mA.

The emitter current is 100.8 mA.

Q3. A transistor has alpha = 0.995. If the emitter current is 80mA, what is (i) the collector current (ii) the base current (iii) current gain beta? (Book answers: (i) 79.6mA, (ii) 0.4mA, (iii) 199)
(i) Collector current: IC = alpha x IE = 0.995 x 80 = 79.6 mA.

(ii) Base current: IB = IE - IC = 80 - 79.6 = 0.4 mA.

(iii) Current gain: beta = alpha/(1 - alpha) = 0.995/(1 - 0.995) = 0.995/0.005 = 199.

(i) The collector current is 79.6 mA, (ii) the base current is 0.4 mA, (iii) the current gain beta is 199.

Q4. The current gain beta of a transistor is 200. If the base current is 0.2mA. Find (i) the collector current (ii) emitter current (iii) the value of alpha. (Book answers: (i) 40mA (ii) 40.2mA (iii) 0.995)

(i) Collector current: IC = beta x IB = 200 x 0.2 = 40 mA.

(ii) Emitter current: IE = IC + IB = 40 + 0.2 = 40.2 mA.

(iii) alpha = beta/(beta + 1) = 200/201 = 0.995.

(i) The collector current is 40 mA, (ii) the emitter current is 40.2 mA, (iii) alpha is 0.995.

Q5. The input resistance in the common emitter amplifier circuit of a given transistor is 2k ohm. The output resistance of this circuit is 8k ohm. What would be the output voltage corresponding to an input voltage 6mV if the current gain is 100? (Book answer: 2.4V)

Voltage gain = beta x (output resistance / input resistance) = 100 x (8/2) = 100 x 4 = 400.

Output voltage = voltage gain x input voltage = 400 x 6 mV = 2400 mV = 2.4 V.

The output voltage is 2.4 V.

Unit 18: Dawn of the Modern Physics

Exercise - Multiple Choice Questions (MCQs)

Q1. Inertial frame of reference is one which satisfies

✓ Newton's Theory

Einstein Theory

Hertz Theory

Special theory of relativity

Q2. Non-inertial frame of reference has

Zero acceleration

Zero velocity

Uniform velocity

✓ Variable velocity

Q3. If the source of light is moving towards the observer, then the speed of light received by the observer will be

Decreased

Increased

✓ Remain same

Maximum

Q4. The relativistic length of an moving object will be

Remain same

✓ Decreased

Increased

Doubled

Q5. If the rest mass of a particle is zero, then its speed is

✓ Equal to speed light

Less than speed of light

Greater than speed of light

Not comparable with speed of light

Q6. Blackbody radiation depends upon

Pressure

Volume

✓ Temperature

Density

Q7. If the temperature of black body is doubled then the emitted energy from it will be increased

Doubled

Four time

Eight time

✓ Sixteen time

Q8. The dimension of a Planck's constant is

[MLT^-1]

✓ [ML^2T^-1]

[ML^2T^-2]

[ML^2T^2]

Q9. Who did observe 1st time the photoelectric effect?

Maxwell

✓ Hertz

Einstein

Heisenberg

Q10. Photo electric effect depends upon the photon's

Pressure

Temperature

Intensity

✓ Frequency

Q11. A photon can transfer its energy into an electron, it was first explained by

Maxwell

Hertz

✓ Einstein

Bohr

Q12. The momentum of a photon is

mv

mc

hf

✓ hf/c

Q13. A change in energy of a photon occurs when it collides with an electron at rest is known as

Photoelectric effect

✓ Compton effect

Pair production

Annihilation

Q14. Which phenomenon does not verify the particle nature of light

Photoelectric effect

Compton effect

Pair production

✓ diffraction

Q15. The antiparticle of electron is

Neutron

Proton

Photon

✓ Positron

Q16. Davisson and Germer proved experimentally the wave nature of particle under the phenomenon of

Reflection

Refraction

Interference

✓ Diffraction

Q17. In the electron positron pair production, the speed of electron is

Zero

Less than speed of positron

✓ Equals to speed of positron

Greater than speed of positron

Q18. If the energy of the used light is high, then the momentum of the investigated particle has

- Less uncertainty
- ✓ High uncertainty
- High accuracy
- Equal uncertainty

Q19. For small uncertainty in the measurement of position of a particle, the wavelength of the incident light should be

- ✓ Small
- Large
- Average
- Does not depend of wavelength

Exercise - Short Questions

Q1. How inertial frame of reference is different from non-inertial frame of reference?

An inertial frame of reference is one in which Newton's laws of motion hold: a body free of external forces either stays at rest or moves with uniform velocity, and the frame itself has zero acceleration. A non-inertial frame is accelerated (its velocity changes), so Newton's laws do not hold in it without adding fictitious forces; a rotating or speeding-up frame is non-inertial.

Q2. State the postulates of special theory of relativity.

The special theory of relativity rests on two postulates. First, the laws of physics are the same in all inertial frames of reference.

Second, the speed of light in free space is the same for all observers, whatever the motion of the source or the observer; it is always c = 3 x 10^8 m/s.

Q3. Mention the important results of special theory of relativity.

The important results are: length contraction, a moving object appears shorter along its direction of motion; time dilation, a moving clock runs slower; mass increase, the mass of a moving body increases with speed; mass-energy equivalence, E = mc^2, so mass can be converted into energy; and the conclusion that no material body can reach or exceed the speed of light.

Q4. Under what condition, a particle can move with a speed of light?

A particle can move with the speed of light only if its rest mass is zero. From the mass formula m = m0/sqrt(1 - v^2/c^2), a particle with m0 = 0 can travel at v = c without its mass becoming infinite. The photon is such a particle; any particle with a non-zero rest mass would need infinite energy to reach c.

Q5. Give the formula to convert the mass of a particle into energy.

Mass is converted into energy by Einstein's relation E = mc^2, where m is the mass, c is the speed of light (3 x 10^8 m/s) and E is the energy. Even a small mass gives a very large energy because c^2 is so large; for a moving body the total energy is E = m c^2 with m = m0/sqrt(1 - v^2/c^2).

Q6. Does a perfect black body exist? If yes, then give an example.

No perfect black body exists in nature, because no real surface absorbs all the radiation falling on it and reflects none. The closest practical example is a hollow cavity with a very small hole in it: radiation entering the hole is trapped by repeated internal reflections and is almost completely absorbed, so the hole behaves like a perfect black body.

Q7. How can you construct a blackbody?

A blackbody can be constructed as a hollow enclosure (cavity) with a small hole in one wall, the inside walls being blackened. Radiation entering through the hole undergoes many reflections inside and is almost completely absorbed, so the hole acts as a perfect absorber and, when the enclosure is heated, as a perfect emitter of blackbody radiation.

Q8. How did Max Planck solve the dilemma of distribution of energy by blackbody?

Classical theory predicted that the energy radiated by a blackbody should keep increasing towards short wavelengths (the ultraviolet catastrophe), which experiments contradicted. Planck solved the dilemma by proposing that energy is not emitted continuously but in discrete packets, quanta, of energy E = hf, where h is Planck's constant and f the frequency. With this quantum hypothesis he derived the correct distribution of blackbody radiation over all wavelengths.

Q9. How can you calculate the K.E of photoelectrons?

The kinetic energy of photoelectrons is calculated from Einstein's

photoelectric equation: hf = phi + K.E.max, where hf is the photon energy and phi is the work function of the metal. So K.E.max = hf - phi. It can also be found from the stopping potential Vs by K.E.max = e Vs, where e is the charge of the electron.

Q10. What is the difference between work function and threshold energy?

The work function phi of a metal is the minimum energy needed to eject an electron from its surface. The threshold energy is the minimum photon energy that can just do this, and it equals the work function: threshold energy = phi = h f0, where f0 is the threshold frequency. In short, the work function is the metal's property (an energy), and the threshold energy is the same minimum energy expressed as the energy of the incident photon at the threshold frequency.

Q11. Name the metals which emit the photoelectrons for visible light.

The alkali metals, especially potassium, sodium and caesium, emit photoelectrons even for visible light, because their work functions are very low (about 2 eV). Visible-light photons, with energies of about 1.8 to 3.1 eV, therefore have enough energy to eject electrons from these metals, whereas most other metals need ultraviolet light.

Q12. What do you know about the Einstein's equation for photo electric effect?

Einstein's photoelectric equation is hf = phi + K.E.max. It states

that when a photon of energy hf strikes a metal surface, a part of its energy, the work function phi, is used to free an electron from the surface, and the remainder appears as the maximum kinetic energy of the emitted photoelectron. The equation explains why emission depends on frequency, not intensity, and why there is a threshold frequency below which no electrons are emitted.

Q13. What is the function of photocell?

A photocell is a device which converts light energy into electrical energy. When light of suitable frequency falls on its photosensitive surface, photoelectrons are emitted, and the resulting current varies with the intensity of the light. Photocells are used in automatic doors, burglar alarms, street-light controls and solar energy systems.

Q14. How does Compton effect verify the wave theory of light?

FLAG: the question as printed says 'wave theory', but the Compton effect actually verifies the PARTICLE nature of light, as Comprehensive Question 7 of this same chapter states correctly; 'wave' here appears to be a misprint for 'particle'. In the Compton effect an X-ray photon collides with an electron at rest and is scattered with a longer wavelength, having lost energy and momentum to the electron. Only a particle picture, in which the photon carries momentum p = hf/c and obeys conservation of energy and momentum in the collision, can explain the observed wavelength shift.

Q15. What is meant by the Compton shift in wavelength?

The Compton shift is the increase in the wavelength of an X-ray photon when it is scattered by an electron at rest. It is given by delta lambda = lambda' - lambda = (h/m0c)(1 - cos theta), where theta is the scattering angle and h/m0c = 2.43 x 10^-12 m is the Compton wavelength. The shift is greatest (twice the Compton wavelength) for back scattering at 180 degrees and zero for no scattering.

Q16. What should be the minimum value of energy of photon to induce the pair production?

Pair production needs a photon of at least 1.02 MeV. The photon must supply the rest energy of both the electron and the positron it creates, each 0.511 MeV, so the minimum photon energy is 2 x 0.511 = 1.02 MeV. Any extra energy appears as kinetic energy of the pair.

Q17. What is the difference between pair production and annihilation of matter?

Pair production is the creation of matter from energy: a high-energy gamma photon, near a nucleus, disappears and produces an electron-positron pair. Annihilation of matter is the reverse: a particle and its antiparticle, such as an electron and a positron, meet and disappear, releasing their total energy as two gamma-ray photons travelling in opposite directions.

Q18. What do you know about the de-Broglie wavelength?

The de-Broglie wavelength is the wavelength associated with a moving material particle. De Broglie proposed that, just as light has

both wave and particle properties, a particle of momentum p = mv has a wavelength lambda = h/p = h/(mv), where h is Planck's constant. The wavelength is significant for very light, fast particles like electrons, and negligible for ordinary objects.

Q19. At what angle the intensity of diffraction electrons is maximum?

In the Davisson and Germer experiment, electrons accelerated through 54 V were scattered by a nickel crystal, and the intensity of the scattered electrons was maximum at a scattering angle of 50 degrees from the incident beam (a colatitude angle giving a diffraction peak). This peak matched the wavelength predicted by de Broglie's relation, confirming the wave nature of electrons.

Q20. Why position and momentum of an electron cannot be measured simultaneously with perfect accuracy?

Because of Heisenberg's uncertainty principle: the act of measuring one quantity disturbs the other. To locate an electron accurately we must use light of short wavelength (high energy), but such photons give the electron a large, uncertain kick, disturbing its momentum. Conversely, a gentle measurement leaves the position uncertain. The product of the uncertainties obeys delta x . delta p >= h/(4 pi), so both can never be known exactly together.

Exercise - Comprehensive Questions

Q1. What do you know about the relative motion? Explain the relative motion with examples.

Relative motion is the motion of a body as observed from a

particular frame of reference; the same motion looks different to different observers. For example, a passenger sitting in a moving train is at rest relative to the train but in motion relative to a person standing on the platform. Similarly, two cars moving in the same direction at the same speed appear at rest relative to each other, while to a roadside observer both are moving. All motion is therefore relative, and there is no absolute rest or absolute motion.

Q2. Distinguish between inertial and non-inertial frame of reference.

An inertial frame is one in which Newton's laws hold without correction: it is either at rest or moving with uniform velocity, and a free body in it stays at rest or in uniform motion. A non-inertial frame is accelerated, so a free body in it appears to accelerate without any real force; to apply Newton's laws in it, fictitious forces (such as centrifugal force in a rotating frame) must be introduced. A train moving steadily is nearly inertial; a train speeding up or rounding a bend is non-inertial.

Q3. State and explain special theory of relativity with its consequences.

The special theory of relativity, proposed by Einstein in 1905, rests on two postulates: the laws of physics are the same in all inertial frames, and the speed of light in free space is the same for all observers. Its consequences are length contraction, L = L0 sqrt(1 - v^2/c^2); time dilation, T = T0/sqrt(1 - v^2/c^2); mass increase, m = m0/sqrt(1 - v^2/c^2); and mass-energy equivalence, E = mc^2. It also follows that the speed of light is the greatest possible

speed for any material body.

Q4. What is blackbody radiation? Explain the distribution of energy from the black body under various laws.

Blackbody radiation is the electromagnetic radiation emitted by a perfect absorber (blackbody) when heated; its spectrum depends only on temperature, not on the material. Its energy distribution is described by three laws: Wien's displacement law, lambda_max T = constant (2.9 x 10^-3 m K), so hotter bodies radiate most strongly at shorter wavelengths; the Stefan-Boltzmann law, total radiated energy per unit area E = sigma T^4, so energy rises steeply with temperature; and Planck's law, which gives the energy radiated at each wavelength and explains the whole distribution curve, including its fall at short wavelengths.

Q5. Define photoelectric effect and derive its fundamental equation.

The photoelectric effect is the emission of electrons from a metal surface when light of suitable frequency falls on it; the emitted electrons are called photoelectrons. Its fundamental equation is derived from the photon picture: a photon carries energy hf. When it strikes the surface, part of this energy, the work function phi, is spent in freeing an electron, and the rest becomes the electron's kinetic energy. By conservation of energy, hf = phi + K.E.max, which is Einstein's photoelectric equation.

Q6. State and explain Einstein's equation of photoelectric effect and different laws of photoelectric effect.

Einstein's equation is hf = phi + K.E.max, where hf is the incident photon energy, phi the work function, and K.E.max the maximum kinetic energy of the emitted photoelectrons. The laws of the photoelectric effect are: (i) electrons are emitted only if the frequency exceeds a threshold frequency f0 = phi/h, whatever the intensity; (ii) the number of emitted electrons (photocurrent) is proportional to the intensity of the incident light; (iii) the maximum kinetic energy depends on the frequency of the light, not on its intensity; (iv) emission is instantaneous, with no time lag.

Q7. What do you know about the Compton effect? How does it verify the particle nature of light by Compton theory.

The Compton effect is the increase in wavelength of X-rays scattered by electrons in a target such as carbon. Compton explained it by treating the photon as a particle of energy hf and momentum hf/c colliding elastically with an electron at rest. Conservation of energy and momentum in the collision gives the shift delta lambda = (h/m0c)(1 - cos theta). Because only a particle picture, with photons carrying momentum, can explain this shift, the Compton effect verifies the particle nature of light.

Q8. Describe pair production and annihilation of matter.

Pair production is the conversion of energy into matter: a gamma-ray photon of energy at least 1.02 MeV, passing near a nucleus, disappears and creates an electron-positron pair; any extra energy becomes their kinetic energy. Annihilation of matter is the reverse process: when a particle meets its antiparticle, for example an electron meeting a positron, both disappear and their

total energy is released as two gamma-ray photons of 0.511 MeV each, travelling in opposite directions to conserve momentum.

Q9. What is the de-Broglie hypothesis? How such hypothesis was verified experimentally by Davisson and Germer.

De Broglie's hypothesis (1924) states that every moving material particle has a wave associated with it, of wavelength lambda = h/p = h/(mv). Davisson and Germer verified it by scattering electrons from a nickel crystal: the electrons showed a diffraction pattern, with intensity maxima at angles matching Bragg's law for a wavelength equal to the de Broglie wavelength of the electrons. Since diffraction is a wave property, this proved that particles behave as waves.

Q10. State and explain electron microscope with its function and its working principle.

An electron microscope is a device which has a much higher magnification and resolving power than an optical microscope, and is used to obtain high-resolution images of extremely small specimens. Its working principle is the wave nature of electrons: a beam of electrons accelerated through a high voltage has a very short de Broglie wavelength (much shorter than visible light), so it can resolve far smaller details. Magnetic lenses focus the electron beam on the specimen, and the transmitted or scattered electrons form the magnified image.

Q11. What do you know about Heisenberg uncertainty principle? Express uncertainty principle under two mathematical relations.

Heisenberg's uncertainty principle states that it is impossible to determine simultaneously the position and the momentum of a particle with perfect accuracy. Its two mathematical relations are: for position and momentum, delta x . delta p >= h/(4 pi) (in the simplified form used in this chapter, delta x . delta p is of the order of h); and for energy and time, delta E . delta t >= h/(4 pi) (of the order of h). The principle sets a fundamental limit of nature, not a limit of our instruments.

Exercise - Numerical Problems

Q1. How fast a rocket has to go for its length to be contracted to 80% of its rest length? (Book answer: 1.8 x 10^8 m s^-1)

Length contraction: L = L0 sqrt(1 - v^2/c^2).

L/L0 = 0.80, so sqrt(1 - v^2/c^2) = 0.80.

Squaring: 1 - v^2/c^2 = 0.64, so v^2/c^2 = 0.36 and v = 0.6 c.

v = 0.6 x 3 x 10^8 = 1.8 x 10^8 m/s.

The rocket must travel at 1.8 x 10^8 m/s.

Q2. The period of a second pendulum is measured to be 2s in an inertial frame of reference of the pendulum. What is its period measured by an observer moving with a speed of 0.9c with respect to the pendulum's frame of reference? (Book answer: 4.6s)

Time dilation: T = T0/sqrt(1 - v^2/c^2).

v = 0.9c, so v^2/c^2 = 0.81 and sqrt(1 - 0.81) = sqrt(0.19) = 0.4359.

T = 2/0.4359 = 4.59 s, about 4.6 s.

The moving observer measures a period of 4.6 s.

Q3. Calculate the variation in the mass of a moving object with a speed of 0.85c. (Book answer: 1.9m0)
Mass increase: m = m0/sqrt(1 - v^2/c^2).

v = 0.85c, so v^2/c^2 = 0.7225 and sqrt(1 - 0.7225) = sqrt(0.2775) = 0.5268.

m = m0/0.5268 = 1.898 m0, about 1.9 m0.

The mass of the moving object becomes 1.9 times its rest mass.

Q4. What is the energy of a photon of a blue light of wavelength 450nm (in joules and in eV). (Book answers: 4.41 x 10^-19 J, 2.76eV)
Photon energy: E = hc/lambda = (6.626 x 10^-34 x 3 x 10^8)/(450 x 10^-9).

E = (1.9878 x 10^-25)/(4.5 x 10^-7) = 4.417 x 10^-19 J, about 4.41 x 10^-19 J.

In electron volts: E = (4.417 x 10^-19)/(1.602 x 10^-19) = 2.76 eV.

The photon energy is 4.41 x 10^-19 J, that is, 2.76 eV.

Q5. Calculate the wavelength of light in which the photons have an energy of 650eV. (Book answer: 1.9nm)

Wavelength: lambda = hc/E.

E = 650 x 1.602 x 10^-19 = 1.0413 x 10^-16 J.

lambda = (6.626 x 10^-34 x 3 x 10^8)/(1.0413 x 10^-16) = (1.9878 x 10^-25)/(1.0413 x 10^-16) = 1.909 x 10^-9 m = 1.9 nm.

The wavelength is 1.9 nm.

Q6. Determine the maximum K.E of photoelectrons ejected from a potassium surface by ultraviolet radiation of wavelength 200nm. If the work function of the potassium surface is 2.8eV, calculate the stopping potential. (Book answers: 3.4eV, 3.4V)

Photon energy: E = hc/lambda = (6.626 x 10^-34 x 3 x 10^8)/(200 x 10^-9) = 9.939 x 10^-19 J.

In eV: E = (9.939 x 10^-19)/(1.602 x 10^-19) = 6.2 eV.

Maximum kinetic energy: K.E.max = hf - phi = 6.2 - 2.8 = 3.4 eV.

Stopping potential: K.E.max = e Vs, so Vs = 3.4 V.

The maximum kinetic energy is 3.4 eV and the stopping potential is

3.4 V.

Q7. With what speed will the fastest photoelectrons be emitted from a surface whose work function is 2eV, when the surface is illuminated with a light of wavelength 4 x 10^-7 m? (Book answer: 6 x 10^5 ms^-1)

Photon energy: E = hc/lambda = (6.626 x 10^-34 x 3 x 10^8)/(4 x 10^-7) = 4.9695 x 10^-19 J = 3.10 eV.

Maximum kinetic energy: K.E.max = 3.10 - 2 = 1.10 eV = 1.10 x 1.602 x 10^-19 = 1.765 x 10^-19 J.

Speed: v = sqrt(2 x K.E./m) = sqrt((2 x 1.765 x 10^-19)/(9.11 x 10^-31)) = sqrt(3.875 x 10^11) = 6.2 x 10^5 m/s, about 6 x 10^5 m/s.

The fastest photoelectrons are emitted with a speed of 6 x 10^5 m/s.

Q8. Consider a photon that scatters from an electron at rest. If the Compton's wavelength shift is observed to be triple the wavelength of the incident photon and if the photon scatters at 60 degrees then calculate the wavelength of the incident photon. (Book answer: 4.05 x 10^-13 m)

Compton shift: delta lambda = (h/m0c)(1 - cos theta) = 2.43 x 10^-12 x (1 - cos 60 degrees).

cos 60 degrees = 0.5, so delta lambda = 2.43 x 10^-12 x 0.5 =

1.215 x 10^-12 m.

Given delta lambda = 3 lambda, so lambda = (1.215 x 10^-12)/3 = 4.05 x 10^-13 m.

The wavelength of the incident photon is 4.05 x 10^-13 m.

Q9. Determine the de-Broglie wavelength of an electron accelerated from rest through a potential difference of 3.3kV volt. (Book answer: 0.38 nm)
Kinetic energy gained: K.E. = eV = 1.602 x 10^-19 x 3300 = 5.287 x 10^-16 J.

Momentum: p = sqrt(2 m K.E.) = sqrt(2 x 9.11 x 10^-31 x 5.287 x 10^-16) = sqrt(9.632 x 10^-46) = 3.104 x 10^-23 kg m/s.

de-Broglie wavelength: lambda = h/p = (6.626 x 10^-34)/(3.104 x 10^-23) = 2.135 x 10^-11 m = 0.021 nm.

FLAG: the book's printed answer 0.38 nm does not follow from its own data (the shortcut lambda(nm) = 1.227/sqrt(V) gives 1.227/sqrt(3300) = 0.021 nm); the correct value is 0.021 nm.

The de-Broglie wavelength is 0.021 nm.

Q10. An electron-positron pair, each with a K.E of 220KeV is produced by a photon. Calculate the energy and wavelength of the photon. (Book answers: 1.46 MeV, 8.5 x 10^-13 m)

Energy needed: photon energy = rest energy of electron + rest energy of positron + kinetic energies = 0.511 + 0.511 + 0.220 + 0.220 = 1.462 MeV, about 1.46 MeV.

Wavelength: lambda = hc/E = (6.626 x 10^-34 x 3 x 10^8)/(1.462 x 1.602 x 10^-13) = (1.9878 x 10^-25)/(2.342 x 10^-13) = 8.49 x 10^-13 m, about 8.5 x 10^-13 m.

The photon energy is 1.46 MeV and its wavelength is 8.5 x 10^-13 m.

Q11. The speed of an electron is measured to be 4 x 10^4 ms^-1 to an accuracy of 0.002%. Find the minimum uncertainty in determining the position of this electron. (Book answer: 0.92mm)
Uncertainty in speed: delta v = 0.002% of 4 x 10^4 = 0.00002 x 4 x 10^4 = 0.8 m/s.

Uncertainty in momentum: delta p = m delta v = 9.11 x 10^-31 x 0.8 = 7.288 x 10^-31 kg m/s.

From delta x . delta p of the order of h: delta x = h/delta p = (6.626 x 10^-34)/(7.288 x 10^-31) = 9.09 x 10^-4 m = 0.91 mm, about 0.92 mm.

The minimum uncertainty in position is 0.92 mm.

Unit 19: Atomic Spectra

Exercise - Multiple Choice Questions (MCQs)

Q1. When light is passed through a gas at low pressure then we observe

- Emission spectra
- ✓ Absorption spectra
- Band spectra
- Molecular spectra

Q2. The spectra due to absorption or emission of radiation by atoms of a gas is known as

- ✓ Line spectra
- Band spectra
- Continuous spectra
- Molecular spectra

Q3. The band spectra is produced due to emission or absorption of radiation by

- Atoms
- ✓ Molecules
- Electrons
- Photons

Q4. Dark lines correspond to

- Emission spectra

✓ Absorption spectra

Band spectra

Continuous spectra

Q5. Name of the scientist who identified the four lines spectrum series for the first time

Lyman

✓ Balmer

Paschen

Brackett

Q6. Which one of the following spectrum series is found in the visible region?

Brackett

Pfund

✓ Balmer

Paschen

Q7. Lyman series lies in which region

Visible

Infrared

Violet

✓ Ultraviolet

Q8. The dimensions of Rydberg constant are

[M^0L^0T]

[M^0LT^0]

[M^0LT]

✓ [M^0L^-1T^0]

Q9. When an electron in hydrogen atom is raised from the ground state to the 2nd energy state, how many times its radius is greater than the radius of its ground state

- Same
- Half
- Twice
- ✓ Four time

Q10. When an electron in hydrogen atom is raised from the ground state to the excited state then its K.E. and P.E. will be

- K.E. increased and P.E. decreased
- ✓ K.E. decreased and P.E. increased
- Both K.E. and P.E. increased
- Both K.E. and P.E. decreased

Q11. The diameter of hydrogen atom is

- 0.35A^o
- 0.53A^o
- 0.70A^o
- ✓ 1.06A^o

Q12. The ionization potential of hydrogen atom is

- 6.13eV
- 6.13V
- ✓ 13.6V
- 13.6eV

Q13. Laser action cannot occur with the process of

- Spontaneous absorption
- Spontaneous emission
- ✓ Normal population
- Population inversion

Q14. Stimulated emission occurs only when the population inversion is in

- Ground state
- ✓ Metastable state
- 3rd energy state
- Any energy state

Q15. Lifetime of metastable state is about

- 10^-2 s
- ✓ 10^-3 s
- 10^-5 s
- 10^-8 s

Q16. Lifetime of higher excited states is

- Equal to the metastable state
- ✓ Shorter than the metastable state
- Longer than the metastable state
- Is not compare with metastable state

Exercise - Short Questions

Q1. How can you produce atomic spectra?

Atomic spectra are produced by exciting the atoms of a gas, for example by passing an electric discharge through the gas at low pressure or by heating it. The excited atoms emit light when their electrons fall back to lower energy levels, and this light, analysed by a spectroscope, shows the emission spectrum. If white light is passed through the cool gas, the gas absorbs its characteristic wavelengths and shows an absorption spectrum.

Q2. What is the difference between absorption spectra and emission spectra?

An emission spectrum consists of bright lines on a dark background, produced when excited atoms emit light of certain wavelengths as their electrons fall to lower energy levels. An absorption spectrum consists of dark lines on a bright continuous background, produced when white light passes through a cool gas and the atoms absorb exactly those wavelengths that would excite their electrons to higher levels.

Q3. How can you distinguish among line, band and continuous spectra?

A line spectrum consists of sharp, separate bright or dark lines, produced by atoms of a gas. A band spectrum consists of groups of closely spaced lines (bands) with sharp edges, produced by molecules. A continuous spectrum is an unbroken range of all wavelengths, with no gaps, produced by hot solids, liquids or gases at high pressure, such as the filament of a bulb.

Q4. What is the reason of dark lines in the line spectra?

Dark lines in a line spectrum are absorption lines. When white light passes through a cool gas, the atoms of the gas absorb photons whose energy exactly matches the difference between two of their energy levels, raising their electrons to higher levels. Those wavelengths are therefore missing from the transmitted light and appear as dark lines against the bright continuous background.

Q5. Calculate the value of Rydberg constant.

The Rydberg constant is R_H = E1/(hc), where E1 = 13.6 eV is the ground-state energy of hydrogen. Substituting E1 = 13.6 x 1.602 x 10^-19 J, h = 6.626 x 10^-34 J s and c = 3 x 10^8 m/s gives R_H = (13.6 x 1.602 x 10^-19)/(6.626 x 10^-34 x 3 x 10^8) = 1.0974 x 10^7 m^-1.

Q6. How the stability of an atom is explained by Bohr's postulates?

Bohr explained the stability of the atom with his first two postulates: an electron revolving in an allowed orbit does not radiate energy, even though classical theory says an accelerated charge must radiate; and only those orbits are allowed for which the electron's angular momentum is an integral multiple of h/(2 pi). Because the electron in an allowed orbit loses no energy, it does not spiral into the nucleus, so the atom is stable.

Q7. What do you know about the quantization of orbits?

Quantization of orbits means that an electron in an atom can only occupy certain discrete orbits, not any orbit. By Bohr's postulate, the allowed orbits are those for which the angular momentum is quantized: m v r = n h/(2 pi), where n = 1, 2, 3, ... is the quantum

number. Each allowed orbit has a definite radius r_n = n^2 x 0.53 angstrom and a definite energy E_n = -13.6/n^2 eV.

Q8. Under what condition an electron can emit energy?

An electron emits energy only when it jumps from a higher energy orbit to a lower energy orbit. The energy of the emitted photon equals the difference between the two energy levels, hf = E_higher - E_lower. While the electron stays in one allowed orbit it radiates nothing.

Q9. What do you know about the Bohr's radius?

Bohr's radius is the radius of the first (ground-state) orbit of the hydrogen atom. It is given by r1 = 0.53 angstrom (0.53 x 10^-10 m). The radii of the higher orbits are r_n = n^2 r1, so the second orbit has radius 4 x 0.53 = 2.12 angstrom, and so on.

Q10. What is the significance of negative energy of electron in an orbit?

The negative sign of the electron's energy in an orbit means the electron is bound to the nucleus: energy must be supplied from outside to free it. The ground-state energy is -13.6 eV, so 13.6 eV must be supplied to remove the electron completely from the atom (ionization). Zero energy corresponds to the electron free at rest, infinitely far from the nucleus.

Q11. What do you know about excitation energy and excitation potential?

The excitation energy is the amount of energy needed to raise an

electron from its normal (ground) state to a higher energy state. For example, raising hydrogen's electron from n = 1 to n = 2 needs 13.6 - 3.4 = 10.2 eV. The excitation potential is the potential difference through which an electron must be accelerated to gain that energy; for the first excitation of hydrogen it is 10.2 V.

Q12. Calculate ionization energy and ionization potential for hydrogen atom.

The ionization energy is the energy needed to remove the electron completely from the ground state, E_ionization = 0 - (-13.6) = 13.6 eV. The ionization potential is the potential difference that gives an electron this energy, V = E/e = 13.6 V.

Q13. What is the working principle of a laser?

A laser works on stimulated emission with population inversion. Atoms are first pumped to a metastable excited state so that more atoms are in the excited state than in the ground state (population inversion). A passing photon then stimulates an excited atom to drop to the lower state, emitting a second photon identical in frequency, phase and direction. Repeated stimulated emission, amplified between two mirrors, produces an intense, monochromatic, coherent beam.

Q14. Distinguish between spontaneous and stimulated emission.

In spontaneous emission an excited atom falls to a lower state on its own, after a short lifetime, emitting a photon in a random direction with random phase; this is the ordinary light of a bulb or flame. In stimulated emission an incident photon of the right

energy causes the excited atom to fall, emitting a second photon identical to the first in frequency, phase and direction; this is the process that produces laser light.

Q15. What is the difference between normal population and population inversion?

In a normal population most atoms are in the ground state and only a few are excited, as given by thermal equilibrium. In population inversion the situation is reversed: more atoms are in an excited (metastable) state than in the ground state. Population inversion is an unnatural condition created by pumping, and it is necessary for laser action, because only then does stimulated emission exceed absorption.

Q16. What do you know about the metastable state?

A metastable state is an excited energy state in which an atom can stay for a comparatively long time, about 10^-3 s, instead of the usual 10^-8 s for ordinary excited states. In this chapter the second energy state (E2) of the laser medium is the metastable state. Because atoms accumulate there, population inversion can be built up, which is essential for laser action.

Q17. Why population inversion in metastable state is necessary for the action of laser?

Population inversion in the metastable state is necessary because stimulated emission can only dominate over absorption when more atoms are excited than in the lower state. The metastable state's long lifetime (about 10^-3 s) lets atoms pile up there instead of

falling back at once, so a large excited population can be maintained. An incident photon then triggers many stimulated emissions, amplifying the light into a laser beam.

Exercise - Comprehensive Questions

Q1. State and explain atomic spectra with its different kinds.

An atomic spectrum is the set of wavelengths (or frequencies) emitted or absorbed by the atoms of an element. Its kinds are: line spectra, sharp separate lines produced by atoms of a gas; band spectra, groups of closely spaced lines produced by molecules; and continuous spectra, an unbroken range of wavelengths produced by hot solids or liquids. Line spectra are further divided into emission spectra (bright lines from excited atoms) and absorption spectra (dark lines when white light passes through a cool gas). Because each element has its own characteristic lines, spectra are used to identify elements.

Q2. What do you know about spectrum of hydrogen atom? Discuss the Balmer's series with empirical formula.

The spectrum of hydrogen consists of several series of lines, each series corresponding to transitions ending on a particular orbit. The Balmer series is the series in the visible region, produced when electrons fall to the second orbit (p = 2) from higher orbits (n = 3, 4, 5, ...). Its empirical formula, found by Balmer, is 1/lambda = R_H (1/2^2 - 1/n^2), with n = 3, 4, 5, ... The first four lines are red, blue-green, blue-violet and violet; the series limit (n to infinity) is 3646 angstrom.

Q3. State and explain Bohr's atomic model of the hydrogen atom with postulates. Also derive the empirical formulas for radii and energies of the quantized orbits of the atom.

Bohr's model rests on three postulates: (i) an electron in an allowed orbit does not radiate energy; (ii) only orbits with angular momentum m v r = n h/(2 pi) are allowed; (iii) an electron radiates (or absorbs) energy only when jumping between orbits, with hf = E_higher - E_lower. Balancing the Coulomb attraction with the centripetal force and using the quantization condition gives the radii r_n = n^2 h^2 epsilon0/(pi m e^2) = n^2 x 0.53 angstrom, and the energies E_n = -m e^4/(8 epsilon0^2 h^2 n^2) = -13.6/n^2 eV. The wavenumber formula 1/lambda = R_H (1/p^2 - 1/n^2) follows from the energy differences.

Q4. Discuss the energy level diagram and various series due to the different transition of electrons in the excited atoms.

The energy level diagram of hydrogen shows the allowed energies E_n = -13.6/n^2 eV as horizontal levels: -13.6 eV (n = 1), -3.4 eV (n = 2), -1.51 eV (n = 3), -0.85 eV (n = 4), -0.54 eV (n = 5) and 0 at infinity. Transitions ending on n = 1 form the Lyman series (ultraviolet); ending on n = 2, the Balmer series (visible); ending on n = 3, the Paschen series (infrared); ending on n = 4, the Brackett series (infrared); and ending on n = 5, the Pfund series (infrared). Within each series the longest wavelength comes from the nearest higher level and the shortest (series limit) from infinity.

Q5. What do you know about the excitation and ionization of an atom? Also discuss the potentials which are being used for

excitation and ionization of atoms?

Excitation is raising an atom's electron from the ground state to a higher energy level by supplying exactly the energy difference; the potential that gives an electron this energy is the excitation potential (for hydrogen n = 1 to n = 2, 10.2 V). Ionization is removing the electron completely from the atom, needing 13.6 eV for hydrogen; the corresponding ionization potential is 13.6 V. These potentials are measured by accelerating electrons through a known voltage and observing the energy at which excitation or ionization first occurs.

Q6. Explain characteristic x-rays due to the transition of electrons in the excited atoms. Also explain the production of x-rays.

Characteristic x-rays are produced when an inner-shell electron of a heavy atom is knocked out and an electron from a higher shell falls into the vacancy, emitting a photon whose energy equals the difference between the two shells. Because inner-shell energies are large, these photons are X-rays, and their energies are characteristic of the element. In an X-ray tube, fast electrons strike a metal target: some are suddenly slowed down, giving continuous (bremsstrahlung) X-rays, while others knock out inner-shell electrons, giving the characteristic lines superimposed on the continuous spectrum.

Q7. What is LASER action? Explain the various process which are related with laser, such as spontaneous and stimulated emission and population inversion.

LASER stands for Light Amplification by Stimulated Emission of

Radiation. Laser action needs three processes. Stimulated absorption raises atoms from the ground state to an excited state by absorbing photons. Spontaneous emission is the natural, random fall of an excited atom with emission of a photon. Stimulated emission occurs when an incident photon causes an excited atom to fall, emitting a second photon identical to the first. For amplification, population inversion must first be created by pumping: more atoms are put into a metastable excited state than remain in the ground state, so stimulated emission exceeds absorption and the light beam is amplified.

Q8. Define Helium-Neon laser. Also write down the various applications of laser.

A Helium-Neon laser is a gas laser in which a mixture of helium and neon is excited by an electric discharge; helium atoms transfer their energy to neon atoms, creating population inversion between neon levels, and stimulated emission gives a continuous red beam of wavelength 632.8 nm. Applications of lasers include: reading compact discs and DVDs, fibre-optic communication, laser printing, holography (three-dimensional photography), eye surgery and other medical treatment, cutting and welding in industry, barcode scanners, weapons guidance and submarine tracking, and research into controlled fusion.

Exercise - Numerical Problems

Q1. Calculate the shortest wavelength of the Balmer series. (Book answer: 3646A^o)

Shortest wavelength of the Balmer series corresponds to the

transition from n = infinity to p = 2.

1/lambda = R_H (1/p^2 - 1/n^2) = R_H (1/4 - 0) = R_H/4.

lambda = 4/R_H = 4/(1.0974 x 10^7) = 3.645 x 10^-7 m = 3645 angstrom, about 3646 angstrom.

The shortest wavelength of the Balmer series is 3646 angstrom.

Q2. Determine the longest wavelength in the Lyman series. (Book answer: 1216A^o)
Longest wavelength of the Lyman series corresponds to the transition from n = 2 to p = 1.

1/lambda = R_H (1/1^2 - 1/2^2) = R_H (1 - 1/4) = 3 R_H/4.

lambda = 4/(3 R_H) = 4/(3 x 1.0974 x 10^7) = 1.215 x 10^-7 m = 1215 angstrom, about 1216 angstrom.

The longest wavelength in the Lyman series is 1216 angstrom.

Q3. Calculate the speed of an electron of hydrogen atom in the second orbit and also find its kinetic energy also (Book answers: 1.23 x 10^12 ms^-1, 3.4eV)
Speed in the nth orbit: v_n = (2.19 x 10^6)/n m/s. For n = 2: v = (2.19 x 10^6)/2 = 1.09 x 10^6 m/s.

FLAG: the book's printed speed 1.23 x 10^12 m/s is impossible (it

exceeds the speed of light) and does not follow from Bohr's theory; the correct speed is 1.09 x 10^6 m/s.

Kinetic energy in the nth orbit: K.E. = 13.6/n^2 eV. For n = 2: K.E. = 13.6/4 = 3.4 eV, which matches the book.

The speed is 1.09 x 10^6 m/s and the kinetic energy is 3.4 eV.

Q4. Show that the Paschen series of spectral lines is entirely in the infrared region.

Paschen series: transitions ending on p = 3, from n = 4, 5, ... to infinity.

Longest wavelength (n = 4): 1/lambda = R_H (1/9 - 1/16) = R_H x 7/144, so lambda = 144/(7 R_H) = 144/(7 x 1.0974 x 10^7) = 1.875 x 10^-6 m = 1875 nm.

Shortest wavelength (n = infinity): 1/lambda = R_H/9, so lambda = 9/R_H = 8.2 x 10^-7 m = 820 nm.

The whole series therefore lies between 820 nm and 1875 nm. Since visible light ends at about 700 nm, every Paschen line is beyond the red end of the visible spectrum, that is, entirely in the infrared region.

The Paschen series lies between 820 nm and 1875 nm, entirely in the infrared region.

Q5. Determine the time period of the first Bohr's orbit in the hydrogen atom. (Book answer: 1.53 x 10^-16 s)

Radius of first orbit: r1 = 0.53 x 10^-10 m. Speed in first orbit: v1 = 2.19 x 10^6 m/s.

Time period: T = 2 pi r1/v1 = (2 x 3.14 x 0.53 x 10^-10)/(2.19 x 10^6).

T = (3.33 x 10^-10)/(2.19 x 10^6) = 1.52 x 10^-16 s, about 1.53 x 10^-16 s.

The time period of the first Bohr orbit is 1.53 x 10^-16 s.

Q6. An electron jumps from a level Ei = -2.2eV to Ef = -7.5eV. What is the wavelength of the emitted radiation? (Book answer: 234nm)

Energy of emitted photon: E = Ei - Ef = -2.2 - (-7.5) = 5.3 eV.

Wavelength: lambda = hc/E = 1240/5.3 = 234 nm (using hc = 1240 eV nm).

The wavelength of the emitted radiation is 234 nm.

Q7. The ionization energy of hydrogen atom is 13.6eV. Calculate the wavelength of the 1st line of the Lyman series. (Book answer: 1212A^o)

First line of the Lyman series: transition from n = 2 to n = 1.

Photon energy: E = 13.6 x (1 - 1/4) = 13.6 x 0.75 = 10.2 eV.

Wavelength: lambda = hc/E = 1240/10.2 = 121.6 nm = 1216 angstrom.

Note: the book prints 1212 angstrom; the difference of about 0.3% comes from rounding of the constants used.

The wavelength of the first Lyman line is 1216 angstrom.

Q8. The wavelength of K X-ray from copper is 1.377 x 10^-10 m. What is the energy difference between the two levels from which this transition occurs? (Book answer: 9.025keV)
Energy difference: delta E = hc/lambda = (6.626 x 10^-34 x 3 x 10^8)/(1.377 x 10^-10).

delta E = (1.9878 x 10^-25)/(1.377 x 10^-10) = 1.4436 x 10^-15 J.

In electron volts: delta E = (1.4436 x 10^-15)/(1.602 x 10^-19) = 9011 eV = 9.01 keV, about the book's 9.025 keV (difference from rounding of constants).

The energy difference between the two levels is 9.0 keV.

Unit 20: Nuclear Physics

Exercise - Multiple Choice Question.

Q1. Proton was discovered by

- J.J Thomson
- Chad wick
- ✓ Ruther ford
- Neil Bohr

Q2. Which of the following particle was discovered by Chadwick?

- Electron
- Proton
- ✓ Neutron
- Photon

Q3. Which of the following particle has the greatest mass?

- Electron
- Positron
- ✓ Proton
- Photon

Q4. The energy equivalent of one atomic mass unit (a.m.u) is

- 1.6x10^-19J
- 1.6x10^19J
- ✓ 931 MeV
- 9.31 MeV

Q5. The number of neutrons in the nucleus of 235_92U is

92

✓ 143

235

327

Q6. The Nuclei 22_11Na and 22_10Ne are known as

Isotopes

✓ Isobars

Isotones

Isomers

Q7. Which of the following nucleus has the highest value of binding energy?

4_2He

✓ 56_26Fe

141_56Ba

235_92U

Q8. The binding energy per nucleon remains constant at

6 MeV per nucleon

7 MeV per nucleon

✓ 8 MeV per nucleon

9 MeV per nucleon

Q9. The nuclei have maximum binding energy per nucleon whose mass number lies between

- ✓ 50-60
- 100-150
- 150-200
- 200-235

Q10. Curie is the unit of

- Energy
- Intensity
- ✓ Radioactivity
- Half life

Q11. The rate of decay depends upon

- Pressure
- Temperature
- Intensity
- ✓ Number of nuclei

Q12. Which rays have highest ionization power?

- ✓ alpha - rays
- beta - rays
- gamma - rays
- X - rays

Q13. A radioactive element will decay when its atomic number is

- Equal to 82
- Less than 82
- ✓ Greater than 82
- does not depend upon number

Q14. When a radioactive element emits a beta-particle the mass number of the atom will be

- increased by 1
- decreased by 1
- increased by 2
- ✓ remain the same

Q15. The unit of decay constant 'lambda' is

- m
- m^-1
- s
- ✓ s^-1

Q16. The energy released per fission reaction is

- 0.85 MeV
- 28 MeV
- 150 MeV
- ✓ 200 MeV

Q17. Moderator used in a nuclear reactor is

- Ice
- ✓ heavy water
- boron rods
- cadmium rods

Q18. The energy released in a fusion reaction is

- 0.85 MeV

✓ 28 MeV

200 MeV

400 MeV

Q19. On average, the number of neutrons emitted per fission is

2

✓ 2.5

3

3.5

Q20. Which force has the negligible effect on elementary particles?

Strong nuclear force

Weak nuclear force

electromagnetic force

✓ gravitational force

Q21. The particles with half integral spin are known as

✓ Fermion

Photon

Bosons

Meson

Q22. Which of the following particles have the highest value of mass?

Photons

leptons

mesons

✓ Baryons

Exercise - Short Questions

Q1. Distinguish between atomic number and mass number.

The atomic number Z is the number of protons in the nucleus, and it fixes which element the atom is; in a neutral atom it also equals the number of electrons. The mass number A is the total number of protons and neutrons (nucleons) in the nucleus. A nucleus is written as ^A_Z X, so the number of neutrons is A - Z.

Q2. What do you know about the atomic mass unit?

The atomic mass unit (a.m.u, symbol u) is the unit used for atomic and nuclear masses. One atomic mass unit is defined as one-twelfth of the mass of a carbon-12 atom, and equals 1.6605 x 10^-27 kg. Its energy equivalent is 931 MeV, which makes it convenient for nuclear energy calculations: a mass difference of 1 u corresponds to 931 MeV.

Q3. Why carbon atom is being used as a reference for atomic mass unit?

Carbon-12 is used as the reference because it is stable, abundant and easy to handle, and its atoms can be compared very accurately in a mass spectrograph. Defining 1 u as exactly one-twelfth of the mass of a carbon-12 atom gives a fixed, reproducible standard against which the masses of all other atoms and nuclei are measured.

Q4. Differentiate between (a) Isotopes and Isobars (b) Isotones and Isomers.

(a) Isotopes are nuclei of the same element with the same atomic number Z but different mass numbers A, such as hydrogen-1, hydrogen-2 and hydrogen-3. Isobars are nuclei of different elements with the same mass number A but different atomic numbers Z, such as sodium-22 and neon-22. (b) Isotones are nuclei of different elements with the same number of neutrons (A - Z), but different Z and A. Isomers are nuclei with the same Z and the same A but in different energy states, one being an excited (metastable) state of the other.

Q5. Why the mass of the nucleus is always less than the sum of masses of its nucleons?

The mass of a nucleus is always less than the sum of the masses of its separate protons and neutrons because, when the nucleons bind together, part of their mass is converted into the energy that holds them together, the binding energy, and this energy is released. The missing mass is the mass defect, delta m, and the binding energy is delta m c^2.

Q6. Describe briefly binding energy.

Binding energy is the energy that holds the nucleons together in a nucleus, or equivalently the energy that must be supplied to break the nucleus into its separate protons and neutrons. It equals the mass defect times c^2: B.E. = delta m c^2. The binding energy per nucleon, B.E./A, measures the stability of the nucleus; it is greatest (about 8.8 MeV) for nuclei near iron-56.

Q7. How does stability of a nucleus depend upon the binding

energy per nucleon?

A nucleus is more stable when its binding energy per nucleon is larger, because more energy per nucleon would be needed to break it apart. The binding energy per nucleon rises with mass number to a maximum around iron-56 and then falls for heavier nuclei, which is why medium-mass nuclei are the most stable, while very heavy nuclei can release energy by fission and very light nuclei by fusion.

Q8. Express the relation between Curie and Becquerel.

The Becquerel (Bq) is the SI unit of radioactivity: 1 Bq = 1 disintegration per second. The Curie (Ci) is the older unit: 1 Ci = 3.7 x 10^10 disintegrations per second, the activity of 1 gram of radium. The relation is therefore 1 Ci = 3.7 x 10^10 Bq.

Q9. What do you know about the radioactive elements?

Radioactive elements are elements whose atomic numbers are greater than 82 and whose nuclei are unstable. They spontaneously emit radiations (alpha, beta and gamma rays) and change into other nuclei, a process called radioactivity. Well-known examples are uranium, radium and polonium; their decay is unaffected by temperature, pressure or chemical combination.

Q10. Explain the conversion of parent nucleus into the daughter nucleus.

When a radioactive parent nucleus decays, it changes into a different nucleus called the daughter nucleus. In alpha decay the parent loses 2 protons and 2 neutrons, so Z falls by 2 and A by 4. In beta decay a neutron changes into a proton, so Z rises by 1 while A

stays the same. In gamma decay the nucleus only loses excess energy, so Z and A stay the same and the daughter is the same nucleus in a lower energy state.

Q11. Why alpha-particles, beta-particles and gamma-rays are not emitted simultaneously?

Alpha, beta and gamma radiations are not emitted simultaneously because they come from different kinds of nuclear change. A nucleus in a particular unstable state decays by one mode at a time: an alpha decay changes the nucleus to a new element, a beta decay changes a neutron into a proton, and gamma emission only removes excess energy from an already formed excited nucleus. Each decay is a separate event with its own probability, so the three radiations appear in sequence as the nucleus passes through successive states, not all at once.

Q12. What is the decay constant?

The decay constant lambda is the probability per unit time that a given radioactive nucleus will decay. It is a property of the radioactive substance, independent of temperature, pressure and the amount present. It is related to the half-life by lambda = 0.693/T_half, and its unit is s^-1. The activity (rate of decay) is A = lambda N, where N is the number of undecayed nuclei.

Q13. What is meant by nucleus transmutation?

Nucleus transmutation is the conversion of the nucleus of one element into the nucleus of another element. It happens naturally in radioactive decay, and it can also be produced artificially by

bombarding a nucleus with a nuclear particle, as in the first artificial transmutation by Rutherford: nitrogen-14 bombarded with alpha particles changes into oxygen-17 and a proton.

Q14. What is the role of moderator in the nuclear reactor?

The moderator in a nuclear reactor slows down the fast neutrons produced in fission to thermal (slow) speeds, because slow neutrons are far more likely to cause further fission of uranium-235 and so sustain the chain reaction. Common moderators are heavy water, ordinary water and graphite. (Control rods of boron or cadmium, by contrast, absorb neutrons and control the rate of the reaction.)

Q15. Why fusion reaction releases more energy per nucleon than fission reaction?

Fusion releases more energy per nucleon than fission because the increase in binding energy per nucleon when very light nuclei fuse is larger than the increase when a heavy nucleus splits. Fusing hydrogen into helium raises the binding energy per nucleon from about 1 MeV to about 7 MeV, a gain of roughly 6 MeV per nucleon, whereas fission of uranium-235 gains only about 0.9 MeV per nucleon (about 200 MeV shared among 235 nucleons).

Q16. Why a fast reactor does not require moderator?

A fast reactor does not require a moderator because it is designed to sustain its chain reaction with fast neutrons, not slow ones. It uses highly enriched uranium or plutonium fuel, in which fast neutrons can still cause enough fissions, and it avoids a moderator

so that the neutrons keep their high energy, which also allows it to breed new fuel from uranium-238.

Q17. Why fusion reaction is more difficult than fission reaction?

Fusion is more difficult than fission because the positively charged nuclei that must fuse repel each other strongly, and they can only be brought close enough to fuse if they are given very high kinetic energies, which means temperatures of millions of degrees. Confining such hot material without it touching and cooling on the walls is extremely difficult, whereas fission only needs a slow neutron to strike a heavy nucleus, which happens readily at ordinary temperatures.

Q18. What is process by which the Sun generates its energy?

The Sun generates its energy by nuclear fusion in its core. In the proton-proton chain, four hydrogen nuclei (protons) fuse in steps to form one helium nucleus, two positrons and neutrinos, releasing about 28 MeV of energy per helium nucleus formed. The energy slowly reaches the surface and is radiated away as sunlight.

Q19. How many forces of nature are there?

There are four fundamental forces of nature: the gravitational force, the electromagnetic force, the strong nuclear force and the weak nuclear force. The strong force holds nucleons together, the weak force is responsible for beta decay, the electromagnetic force acts between charges, and gravity acts between masses; on the scale of elementary particles gravity is negligibly small.

Q20. Distinguish between elementary and sub atomic particles.

Subatomic particles are all the particles smaller than an atom, including protons, neutrons and electrons. Elementary particles are the truly fundamental subatomic particles, which are not made of anything smaller and have no internal structure: the quarks and the leptons. Protons and neutrons are subatomic but not elementary, because they are composed of quarks; electrons are both subatomic and elementary.

Q21. What is the difference between Fermions and Bosons?

Fermions are particles with half-integral spin (1/2, 3/2, ...), such as quarks and leptons; they obey the Pauli exclusion principle, so no two identical fermions can occupy the same state. Bosons are particles with integral spin (0, 1, ...), such as photons, mesons and the Higgs boson; they do not obey the exclusion principle and any number can occupy the same state. Fermions are the building blocks of matter, while bosons carry forces.

Q22. How many hadrons are there?

Hadrons are the composite particles made of quarks, and they fall into two classes: baryons, made of three quarks (such as protons and neutrons), and mesons, made of a quark and an antiquark (such as pions and kaons). A great many hadrons are known, well over a hundred, including the proton, neutron, lambda, sigma, xi and omega baryons and the pi, K, rho and many other mesons; new ones continue to be found in high-energy experiments.

Exercise - Comprehensive Questions

Q1. State and explain the composition of a nucleus of an atom with all its properties.

The nucleus of an atom is composed of protons and neutrons, collectively called nucleons. The number of protons is the atomic number Z, which fixes the element; the total number of nucleons is the mass number A; the number of neutrons is A - Z. The nucleus is tiny (about 10^-14 m) compared with the atom (10^-10 m) but contains almost all its mass. Protons carry positive charge and repel each other electrically, but the short-range strong nuclear force overcomes this repulsion and binds the nucleons together. Nuclei are written as ^A_Z X.

Q2. What do you know about the isotopes, isobars, isotone and isomers? Explain all them with examples.

Isotopes are nuclei of the same element (same Z) with different mass numbers, such as hydrogen-1, hydrogen-2 (deuterium) and hydrogen-3 (tritium), or carbon-12 and carbon-14. Isobars are nuclei of different elements with the same mass number but different atomic numbers, such as sodium-22 and neon-22. Isotones are nuclei of different elements with the same number of neutrons, such as carbon-13 and nitrogen-14 (both have 7 neutrons). Isomers are nuclei with the same Z and A but different energy states, one being an excited metastable state of the other, such as the metastable and ground states of technetium-99.

Q3. Define mass spectrograph. How can we separate the isotopes of an element by this method?

A mass spectrograph is a device in which the isotopes of an

element are separated by their masses. Atoms of the element are ionized and accelerated into a magnetic field, where the magnetic force bends their paths into circles whose radii depend on their mass-to-charge ratio. Lighter isotopes are bent more and heavier ones less, so the isotopes strike a photographic plate (or detector) at different positions, forming separate lines whose positions give the masses and whose intensities give the relative abundances.

Q4. What do you know about mass defect and binding energy? Discuss graphically, the relation between mass defect and binding energy.

The mass defect is the difference between the sum of the masses of the separate nucleons and the actual mass of the nucleus: delta m = (Z m_p + (A - Z) m_n) - M_nucleus. The binding energy is B.E. = delta m c^2, the energy released when the nucleus is formed. Graphically, the binding energy per nucleon plotted against mass number rises steeply for light nuclei, reaches a broad maximum of about 8.8 MeV near iron-56, and then slowly falls for heavy nuclei. This curve explains why energy is released both by fusion of light nuclei and by fission of heavy nuclei.

Q5. State and explain radioactivity with all its properties.

Radioactivity is the spontaneous emission of radiations (alpha, beta and gamma rays) from the nuclei of unstable elements, those with atomic number greater than 82. Its properties are: it is a nuclear process, unaffected by temperature, pressure or chemical state; it changes the parent nucleus into a daughter nucleus; the rate of decay is proportional to the number of undecayed nuclei (A =

lambda N); and the radiations have different penetrating and ionizing powers: alpha rays are the most ionizing but least penetrating, gamma rays the least ionizing but most penetrating.

Q6. Discuss various laws of radioactivity. Also derive equation for decay of a radioactive element.

The laws of radioactivity are: (i) the rate of decay is proportional to the number of undecayed nuclei present; (ii) the decay constant is characteristic of the substance and independent of external conditions; (iii) the number of nuclei falls exponentially with time. Derivation: dN/dt = -lambda N, so dN/N = -lambda dt. Integrating from N0 at t = 0 to N at time t gives ln(N/N0) = -lambda t, or N = N0 e^(-lambda t). The half-life follows by setting N = N0/2, giving T_half = 0.693/lambda.

Q7. State and explain the functions and working principles of the following two radiations detectors. 1) GM-counter 2) Solid state detector.

(1) The Geiger-Muller (GM) counter detects and counts ionizing particles. It is a gas-filled tube with a central wire anode at high positive voltage. An entering particle ionizes the gas, and the ions are accelerated so strongly that they cause an avalanche of further ionization, giving an electrical pulse that is counted; each pulse corresponds to one particle. (2) A solid-state detector uses a semiconductor crystal (such as silicon) instead of gas. An ionizing particle creates electron-hole pairs in the crystal, and the applied voltage sweeps them to the electrodes, producing a pulse whose size is proportional to the particle's energy, so it both counts and

measures the energy of the radiation.

Q8. Discuss nuclear reaction with all its kinds.

A nuclear reaction is a process in which a change occurs in a nucleus when it is struck by a nuclear particle. Its kinds include: elastic scattering, in which the particle bounces off unchanged; inelastic scattering, in which the nucleus is left excited; transmutation, in which the nucleus changes into another element; fission, in which a heavy nucleus splits into two lighter nuclei with release of energy; fusion, in which two light nuclei join into a heavier nucleus with release of energy; and radioactive decay, the spontaneous breakdown of an unstable nucleus.

Q9. State and explain nuclear fission and chain reaction. Also discuss the various parts of the nuclear reactor.

Nuclear fission is the splitting of a heavy nucleus, such as uranium-235, into two lighter nuclei when struck by a slow neutron, releasing about 200 MeV of energy and two or three fresh neutrons. If these neutrons cause further fissions, a self-sustaining chain reaction results. A nuclear reactor controls this chain reaction with: fuel (uranium-235) in fuel rods; a moderator (heavy water or graphite) that slows the neutrons; control rods (boron or cadmium) that absorb neutrons and regulate the rate; a coolant that carries away the heat; and a thick concrete shield that absorbs escaping radiations.

Q10. What is nucleus fusion reaction? Explain nuclear fusion reaction in the Sun.

A nucleus fusion reaction is one in which two light nuclei join to form a heavier nucleus, releasing energy because the product has a larger binding energy per nucleon. In the Sun, the proton-proton chain operates: two protons fuse to deuterium, deuterium fuses with a proton to helium-3, and two helium-3 nuclei fuse to helium-4 plus two protons. Overall four protons become one helium nucleus, releasing about 28 MeV, which is the source of the Sun's energy.

Q11. State and explain radiation exposure and various methods used for its measurement.

Radiation exposure is the amount of ionizing radiation absorbed by a body, measured by the charge it produces in air or by the energy deposited per unit mass. It is measured with: a dosimeter (film badge or thermoluminescent dosimeter) worn by workers, which records their accumulated dose; a Geiger-Muller counter, which counts individual particles; and an ionization chamber, which measures the ionization current produced by the radiation. Units are the roentgen for exposure, the rad (or gray) for absorbed dose, and the rem (or sievert) for dose equivalent.

Q12. How can we diagnose and treat the fatal diseases like cancer by using radioactive radiations?

Radioactive radiations are used in diagnosis through tracers: a small amount of a radioactive isotope (such as iodine-131 for the thyroid or technetium-99m for general scanning) is introduced into the body, and its gamma rays are detected from outside to image the organ and its working. In treatment (radiotherapy), cancerous tissue is exposed to carefully aimed gamma rays (for example from

cobalt-60) or to implanted radioactive sources, which damage the DNA of the rapidly dividing cancer cells and destroy them, while the dose to healthy tissue is kept as low as possible.

Q13. What do you know about the basic forces of nature? Explain their interacti.

There are four basic forces of nature. The gravitational force acts between all masses; it is always attractive and, though the weakest, it governs the large-scale structure of the universe. The electromagnetic force acts between electric charges and is responsible for chemistry and for light. The strong nuclear force acts between quarks and holds nucleons together; it is the strongest but acts only over about 10^-15 m. The weak nuclear force acts over even shorter distances and is responsible for beta decay. Their relative strengths are roughly strong 1, electromagnetic 10^-2, weak 10^-13 and gravitational 10^-39.

Q14. Discuss the elementary particles and their classifications.

Elementary particles are the fundamental particles with no internal structure, and they are classified by spin into two groups. Fermions have half-integral spin and are the building blocks of matter: the six quarks (up, down, charm, strange, top, bottom) and the six leptons (electron, muon, tau and their three neutrinos). Bosons have integral spin and carry forces: the photon (electromagnetic), the W and Z bosons (weak), the gluons (strong) and the Higgs boson, which gives other particles their mass. Hadrons, such as protons, neutrons and mesons, are not elementary: they are composed of quarks.

Exercise - Numerical Problems

Q1. Calculate mass defect, binding energy and binding energy per nucleon of 14_7N. The mass of 14_7N nucleus is 14.003074u. (Book answers: 0.108513u, 101MeV and 7.2MeV)

Nitrogen-14 has Z = 7 protons and A - Z = 7 neutrons.

Sum of nucleon masses: 7 x 1.007276 + 7 x 1.008665 = 7.050932 + 7.060655 = 14.111587 u.

Mass defect: delta m = 14.111587 - 14.003074 = 0.108513 u.

Binding energy: B.E. = delta m x 931 = 0.108513 x 931 = 101.03 MeV, about 101 MeV.

Binding energy per nucleon: 101/14 = 7.21 MeV, about 7.2 MeV.

The mass defect is 0.108513 u, the binding energy is 101 MeV and the binding energy per nucleon is 7.2 MeV.

Q2. The half-life of 92_36Kr is 3.16 minutes. Find its decay constant. (Book answer: 3.66x10^-3 s^-1)

Half-life in seconds: T_half = 3.16 x 60 = 189.6 s.

Decay constant: lambda = 0.693/T_half = 0.693/189.6 = 3.655 x 10^-3 s^-1, about 3.66 x 10^-3 s^-1.

The decay constant is 3.66 x 10^-3 s^-1.

Q3. The half-life of radium is 1600 years. How many radium atoms decay in 1s in a 1g sample of radium? (Mass number of radium is 220 kg/k mol) (Book answer: Practically no atom of radium-220 will decay in 1s)

Number of atoms in 1 g: N = (1/220) x 6.02 x 10^23 = 2.74 x 10^21 atoms.

Decay constant: lambda = 0.693/T_half = 0.693/(1600 x 365 x 24 x 3600) = 0.693/(5.05 x 10^10) = 1.37 x 10^-11 s^-1.

Atoms decaying in 1 s: lambda x N x 1 = 1.37 x 10^-11 x 2.74 x 10^21 = 3.75 x 10^10 atoms.

FLAG: the book's remark that 'practically no atom will decay' understates the result. The FRACTION decaying in one second is indeed tiny (1.37 x 10^-11 of the sample), but because the sample contains 2.74 x 10^21 atoms, the absolute number decaying per second is about 3.75 x 10^10.

About 3.75 x 10^10 radium atoms decay per second in a 1 g sample.

Q4. The half-life of 14_6C is 5700 years. What fraction of a sample of 14C will remain unchanged after a period of five half-lives? (Book answer: 0.0315)

After one half-life, 1/2 remains; after n half-lives, (1/2)^n remains.

After five half-lives: (1/2)^5 = 1/32 = 0.03125, about the book's 0.0315.

The fraction remaining unchanged is 1/32, about 0.031.

Q5. Determine the energy associated with the following nuclear reaction. 14_7N + 4_2He -> 17_8O + 1_1H. The relevant nuclear masses are: m(14_7N) = 14.003074u, m(4_2He) = 4.002603u, m(17_8O) = 16.999131u and m(1_1H) = 1.007825u (Book answer: -1.191MeV)
Mass of reactants: 14.003074 + 4.002603 = 18.005677 u.

Mass of products: 16.999131 + 1.007825 = 18.006956 u.

Mass change: delta m = reactants - products = 18.005677 - 18.006956 = -0.001279 u.

Energy: Q = delta m x 931 = -0.001279 x 931 = -1.191 MeV. The negative sign means energy is absorbed, not released.

The energy associated with the reaction is -1.191 MeV (energy is absorbed).

Q6. Calculate the energy associated with the given nuclear fusion reaction. 2_1H + 3_1H -> 4_2He + 1_0n (Book answer: 17.6 MeV)
Relevant masses: m(2_1H) = 2.014102u, m(3_1H) = 3.016049u, m(4_2He) = 4.002603u, m(1_0n) = 1.008665u.

Mass of reactants: 2.014102 + 3.016049 = 5.030151 u.

Mass of products: 4.002603 + 1.008665 = 5.011268 u.

Mass change: delta m = 5.030151 - 5.011268 = 0.018883 u.

Energy released: Q = 0.018883 x 931 = 17.58 MeV, about 17.6 MeV.

The energy released in the fusion reaction is 17.6 MeV.

Q7. Calculate the amount of energy released in the nuclear fission reaction. 1_0n + 235_92U -> 141_56Ba + 92_36Kr + 3 1_0n + Q. The relevant masses are: m(1_0n) = 1.008665u, m(235_92U) = 235.043924u, m(141_56Ba) = 140.913740u, and m(92_36Kr) = 91.926270u. (Book answer: 174 MeV)

Mass of reactants: 1.008665 + 235.043924 = 236.052589 u.

Mass of products: 140.913740 + 91.926270 + 3 x 1.008665 = 235.866005 u.

Mass change: delta m = 236.052589 - 235.866005 = 0.186584 u.

Energy released: Q = 0.186584 x 931 = 173.7 MeV, about 174 MeV.

Note: the book's answer 174 MeV follows only if three neutrons are emitted in the reaction, as written here.

The energy released in the fission reaction is 174 MeV.

Q8. How much energy is released during following reaction? 1_1H + 7_3Li -> 4_2He + 4_2He + Q. The relevant nuclear masses are: m(1_1H) = 1.007825u, m(7_3Li) = 7.016005u, and m(4_2He) = 4.002603u (Book answer: 17.3 MeV)

Mass of reactants: 1.007825 + 7.016005 = 8.023830 u.

Mass of products: 2 x 4.002603 = 8.005206 u.

Mass change: delta m = 8.023830 - 8.005206 = 0.018624 u.

Energy released: Q = 0.018624 x 931 = 17.34 MeV, about 17.3 MeV.

The energy released is 17.3 MeV.

END MATTER: Appendix A (book p355, some useful fundamental constants), Appendix B (book p356, particles of atomic physics), the Glossary (book pp357-366) and the Index (book pp367-372) contain no student exercises. They are reference tables and lists, so no exercises from them are solved here; the chapter summaries above already define the key terms.